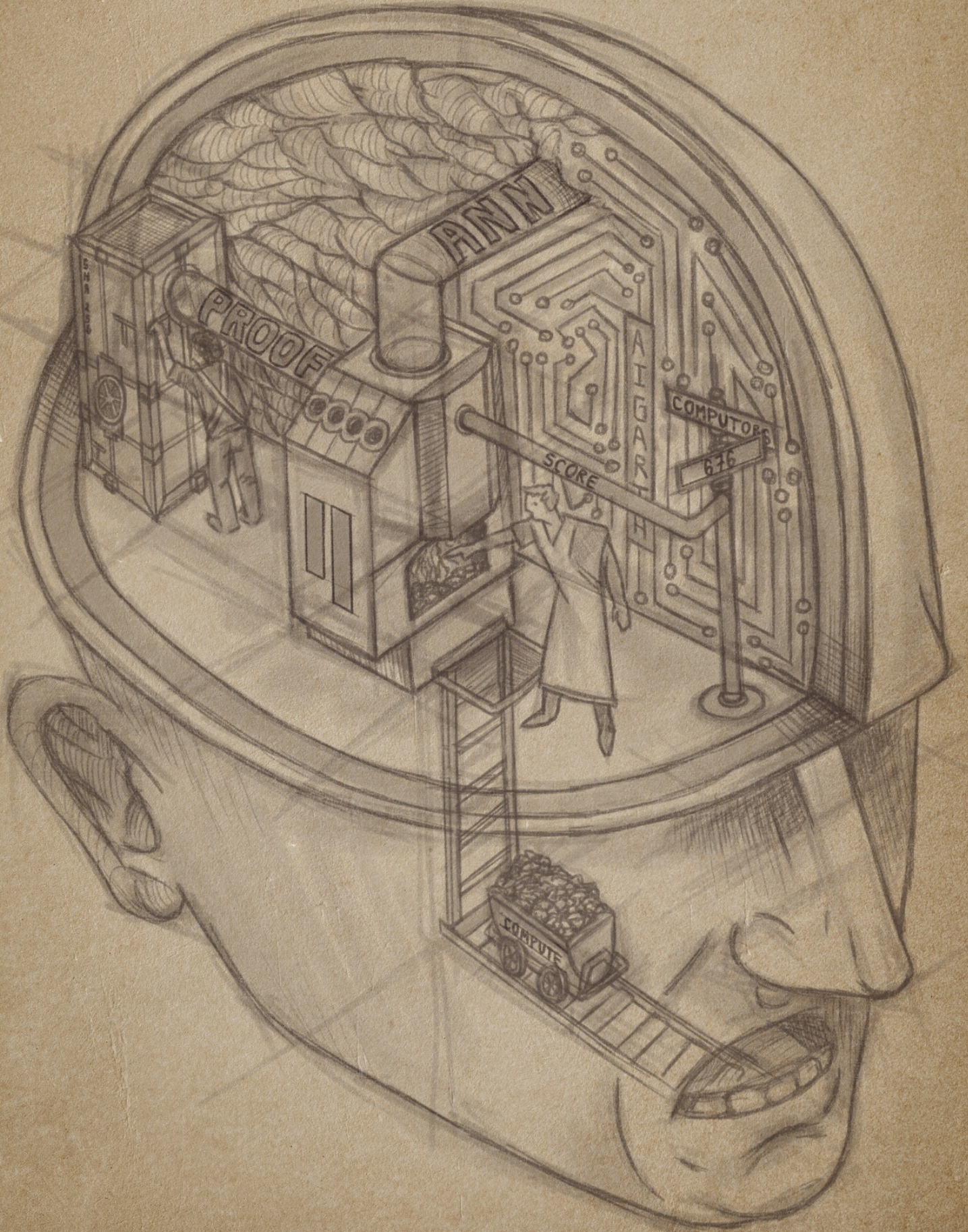


QUBIC

THE LONG VERSION



Q U B I C

The Long Version

QUBIC CHURCH • 2026

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Preface



“Qubic is doomed to succeed. Aigarth? Depends on miners.” — Come-from-Beyond, February 7, 2026

The story of Qubic begins long before Bitcoin and contains many vivid stages. To study the path that the visionary Come-from-Beyond lays beneath his own feet is genuinely interesting.

My father spent a significant amount of time studying this subject and, being a master of religious studies, looked at Qubic on a somewhat larger scale than is customary.

The main challenge for the ordinary person who today takes an interest in Qubic is to grasp the scale, the structure, the functions, and the meaning of every detail of a large mechanism. Believe me, this is a whole universe with a rich history and many secrets we’ll be uncovering over time. Come-from-Beyond, like an ancient artist, is painting a large picture. Elements that seem unconnected gradually begin to fuse and take shape.

This book does one simple thing: it gathers what’s been scattered into a single fresco, structures it, and presents it in plain language.

Who should read this? Anyone who’s even a little connected to Qubic will find something here for themselves. A newcomer will learn the basic principles of how it works and see connections where they didn’t see them before. An experienced holder can enjoy the history and the beautiful allegories my father’s speech is so full of. The professional and the master can take a look at the overall picture from the outside and, perhaps, see something they used to miss.

All Cfb quotes used in this book are taken verbatim from open primary sources — the official Qubic Discord, his publications on X, the team’s scientific papers. All of them appear in our book. If you spot an inaccuracy — write in. The book is alive; the next edition will incorporate corrections. A caveat is in order: Come-from-Beyond’s style is a wild ride on a motorcycle off-road. In the Qubic universe, you rarely find information laid out in a structured way. That’s why we believe this book will be a pleasant addition to every Qubic holder’s library. The book is distributed absolutely freely. The best thanks for us would be your appraisal, your criticism, your remarks. If, by some special and uncontrollable urge, you wish to pay for the book — you can voluntarily use the

donate button on the website qubic.church or our qubic wallet: BDRFCOFWHRTEHHM-QQUIYJBXEOLNARADAGFUSBFGJFABYZBZLQNWJIPPFRTXO

The book was originally written in Russian, but for convenience it has been translated into English.

Thank you, Father. Your contribution is invaluable.

PART I

The Problem

CHAPTER 1

Murdoch's Gas



Useful waste.

There's a kind of thing you learn to see one day — and then you wonder how you didn't notice it before. One of those things is the difference between work and its waste.

Take any furnace. Something burns inside it, something gets heated, and the desired product comes out. But besides the product, there's always something else left over. Smoke. Steam. Smell. Noise. Heat going off into the walls. For centuries people treated those leftovers as garbage — the inevitable companion of useful work, something that had to be disposed of somehow or simply released into the air.

Sometimes a person comes along who looks at this garbage more attentively than the rest. And asks a strange question: what if we don't release it?

This whole book is about a question of that kind, asked with respect to one particular kind of garbage. But to understand why this question matters, we first have to tell two old stories — about coal, and about the man who figured out how to light cities with it.

Both happened in England.

1709. Coalbrookdale

In the west of England, in the county of Shropshire, there is a valley with the river Severn flowing along its floor. The place is green, hilly, not particularly remarkable. In the early eighteenth century, a man named Abraham Darby moved there — he was in his thirty-second year, he was a Quaker, and he already had a notable record in industry behind him.

Quaker — that's an important detail, even if at first glance not the flashiest. The Society of "Friends" (as the Quakers called themselves) in the England of that time was a community whose members did not wear military uniforms, did not swear oaths in court, avoided ostentatious luxury, and took honesty in their dealings very seriously. Quaker dynasties of craftsmen were famous for the fact that their products weren't counterfeited and their prices weren't inflated. Darby was part of such a dynasty — and

grew up inside an ethic in which reputation was worth more than momentary profit. This detail is worth remembering: we'll come back to it when we talk about Bitcoin.

Darby had a problem, and it was common to all of English metallurgy at the time. Cast iron was smelted using charcoal — specially charred wood that burns hotter than ordinary firewood. But by 1700 England had cut down most of its forests. Charcoal was getting more expensive, there was less and less of it, and foundries were closing one after another. A country that depended on iron for everything — from plows to cannons — found itself facing what today would be called a resource crisis.

Coal in England did exist. There was plenty of it. But cast iron couldn't be smelted with it — sulfur and other impurities ruined the metal. Iron made on coal came out brittle, unfit for serious work.

Here it's worth saying how Darby arrived at his solution, because this wasn't the feat of a lone genius. Darby had spent several years before that working in the copper industry in Bristol, and before that he had apprenticed with a Quaker who made malt kilns for brewing. In both copper smelting and malt production, the technique of heating coal without access to air had already been used — out came coke, a fuel without sulfur, which burned cleanly and steadily. In copper smelting it worked well. In iron smelting, no one had tried it systematically before Darby.

Darby did two things. The first — he transferred a known technological technique from one industry to another. The second — he came to Coalbrookdale, where at hand he had at the same time low-sulfur coal, deposits of iron ore not far below the surface, and a fast river to power the bellows. By January 1709 he produced his first commercial cast iron, smelted on coke.

No one wrote down Darby's exact words when he first saw the finished casting from the coke furnace. But the consequences of that moment rewrote the next two hundred years of world history. Bridges. Railways. Steamships. The frames of skyscrapers. Tanks. Engines. All the iron of industrial civilization — from the first cast-iron beam to the last rail of the Trans-Siberian — traces back to the fact that an experienced Quaker copper-industrialist guessed that a technique that works in one production can be transferred to another, where no one had yet tried it.

That's the first story.

1792. Cornwall

More than eighty years after Darby, on the other side of England, an engineer by the name of William Murdoch was at work — a Scotsman, the son of a small farmer, by then thirty-seven years old. He served at the company Boulton & Watt — the very one that built James Watt's steam engines. Murdoch had been sent to Cornwall to look after the engines installed at the tin mines.

He lived in the small town of Redruth, in a small house. And here's what Murdoch noticed one day.

When you make coke — that same coke that Darby learned to produce eighty years earlier — a great deal comes out of the coal during heating. The coke stays in the furnace. And outside, into the air, a thick combustible gas escapes. It smells of sulfur and burning, looks like a dirty cloud, and everyone who worked with coke furnaces fought with this gas: it's poisonous, sometimes it explodes, and it stinks up the whole neighborhood. People tried to dissipate it, lead it away, so it wouldn't get in the way.

Murdoch collected this gas into a pipe.

In 1792 he began his experiments. By 1794 he had a working system: in an iron retort in his back yard he heated coal, the gas went through a pipe into the house, and at the end of the pipe stood a simple burner. The first pipe for the gas line was an old gun barrel. He lit it — and got a clean bright flame, without smoke, without soot, much brighter than candles or oil lamps. On an evening in 1794, Murdoch sat in his living room in Redruth by the light of what everyone else considered garbage.

What happens next is a curious thing that comes up in the history of technology more often than people tend to think.

Murdoch invented the system. But he was not the one who commercialized it.

In 1802, when the Peace of Amiens was being celebrated, Murdoch managed to convince the management of Boulton & Watt to put on a public demonstration — and that evening the facade of the Soho Foundry in Birmingham was lit by gas. The town came running to look. It was a triumph. But Murdoch, as his contemporaries wrote, “by character kept to the shadows,” and his employers were in no hurry to patent the discovery. Meanwhile another man, Frederick Albert Winsor — a German by origin who had moved to London — had read a paper by the Frenchman Lebon on combustible gas, saw the potential, and began giving lectures at the Lyceum Theatre in London. Winsor was a brilliant promoter and a poor engineer. Contemporaries wrote that “he knew almost nothing of chemistry and was so incompetent in mechanics that he could not give instructions for the building of apparatus.” But he knew how to convince investors and how to gather a crowd.

In January 1807 it was Winsor, not Murdoch, who lit the world's first gas streetlamps — on Pall Mall in London. By the fourth of June that same year, on the birthday of King George III, he put on a whole show: along the wall between Mall street and the gardens of Carlton House, thirty-two gas burners lit up, connected by a thin tin tube. The crowd stood there until midnight.

The first industrial factory lit entirely by gas — the Phillips & Lee mill in Manchester — was put into service by Murdoch's hand only in 1805–1806. By that time Winsor was already drafting an act to create the world's first gas company. By the 1820s, forty thousand gas streetlamps were burning in London. By the middle of the century, gas lighting stood across the entire Western world.

Murdoch did not patent his discovery and died in 1839 a respected engineer, known in narrow circles. The name “father of gas lighting” stayed with him in the history of technology, but the commercial empire was built by a man who didn't invent the tech-

nology — he sold it. This split between the inventor and the one who carries the invention to a mass market is a pattern we'll meet in this book more than once.

That's the second story.

What they have in common

Pay attention to the structure.

Darby didn't invent coal. Coal had been mined in England for centuries before him. Darby didn't invent the technique of coking either — it was already in use in copper smelting and in brewing. Darby transferred a known technology from one industry to another, where no one had tried it systematically before. Before him, coal in metallurgy was useless. After him, coal became the foundation of industry.

Murdoch didn't invent combustible gas. Gas had been coming out of coke furnaces from the very moment Darby first lit a furnace in 1709. For eighty-three years the gas went up into the sky over England, and nobody paid any attention. Murdoch didn't create a new resource — he saw value in what everyone else was throwing away.

And one more detail, which may seem unimportant for now but later turns out to be key. Neither Darby nor Murdoch destroyed what came before them. Charcoal kept being produced and used long after Darby. Oil lamps and candles kept burning long after gas lighting became available. The new technology didn't replace the old by decree from above. It simply began to work in parallel — and people gradually crossed over when they saw it was more advantageous for them.

In parallel, not instead.

Remember that phrase.

How this relates to Bitcoin

Now let's look at our own time.

In 2008 a person (or group of people) under the pseudonym Satoshi Nakamoto published a nine-page document describing a way to create digital money without a bank in the middle. A few months later, in January 2009, he launched a system called Bitcoin. The fundamental difficulty Satoshi solved was known in computer science as “the double-spending problem”: how to make sure that a digital file — money — can't be copied and spent twice, without a central arbitrator watching the accounts.

The solution was elegant. All participants in the network jointly maintain a ledger in which transactions are recorded. Each page of this ledger is protected by a mathematical problem that's very expensive to solve — you have to spend a colossal amount of electricity and computation. Whoever solves the problem first is rewarded with new bitcoins. And since the solution takes enormous resources, rewriting pages that have already been recorded becomes practically impossible: you'd have to outspend the entire network. This mechanism is called Proof-of-Work.

This was a discovery of the Darby class. Before Satoshi, it was thought that digital money without an intermediary was in principle impossible. After Satoshi, it exists, has been working for its sixteenth year, is more strongly protected than any bank vault in the world, and has a market capitalization larger than the GDP of most countries.

But this solution has one feature that few people pay attention to. The mathematical problems that Bitcoin's miners solve are, in themselves, meaningless.

A miner takes a list of transactions, adds a random number to it, runs it through a cryptographic function, and looks at whether the result has enough zeros at the beginning. If not — he adds another random number, runs it through again, checks again. And so on, billions of times per second, until he gets lucky. All the energy that gets spent in that moment — and as of 2026 that's about 170 terawatt-hours a year, roughly as much as Poland or Ukraine consumes — goes into one thing: picking a random number that meets the condition.

There's no by-product benefit from this picking. No one uses the results of these computations. No one needs them, except the network itself.

Bitcoin is Darby. The first great discovery of its class.

And what comes out into the air around Bitcoin — billions of computational cycles that nobody needs — is Murdoch's gas of 2009, which no one has yet collected into a pipe.

What exactly comes out into the air

It's worth saying more concretely what kind of "gas" is being talked about.

The modern Bitcoin network consists of hundreds of thousands of specialized devices — ASIC miners. These are computers sharpened for one single operation: hashing using the SHA-256 algorithm. They can do nothing else, and nothing else is required of them. Each such device makes trillions of attempts per second to pick the necessary number.

If you add up all the work of all the miners in the world, you get computing power comparable to the combined power of all the supercomputers in the planet's research centers, multiplied by thousands. This is one of the largest computational resources humanity has ever assembled into a single structure.

And this entire resource is spent on one action: picking random numbers to fit a hash condition.

Imagine you have a million highly educated specialists — scientists, engineers, doctors, architects — and you make them spend the whole day flipping a coin and writing down whether it came up heads or tails. They handle the task excellently, the coin really is being flipped, the results really are being written down. But all their education, their experience, their abilities — go unused. Each of them could, in the same time, write a scientific paper, design a bridge, treat ten patients, solve a hard engineering problem. Instead — the coin.

Roughly that's what's happening in Bitcoin. Computing power that could be solving real problems — training AI models, modeling the climate, searching for new medicines — is directed at picking random numbers.

Does it protect the network? Yes. Does it work? Yes. Is there a structural flaw in this? Absolutely. And this flaw is not Satoshi's mistake, just as the abundant gas from the coke furnaces wasn't Darby's mistake. It's simply a consequence of the fact that the first great discovery rarely notices its own by-products.

Under the hood. GHash.IO, summer 2014

There's one episode from the early history of Bitcoin that's useful to break down — because it foreshadows much of what will happen later in this book.

In 2013–2014, the largest mining pool in Bitcoin became GHash.IO. A pool is an association of miners who put their computations together and split the reward in proportion to contribution. GHash.IO grew quickly. In January 2014 its share approached 50 percent, the community panicked, some of the miners voluntarily left, and the share temporarily decreased. But in summer everything repeated in a far more serious form.

From eleven o'clock in the morning of June twelfth to ten o'clock in the morning of June thirteenth, 2014 — for nearly twenty-four hours straight — GHash.IO held approximately 55 percent of Bitcoin's total hashrate. Not 51. Fifty-five.

For Bitcoin, this is a catastrophic figure. If a single participant controls more than half of the network's computing power, he can theoretically rewrite the history of transactions in his own interests, double-spend the same coins, block someone's transfers. The network ceases to be decentralized at the moment majority control appears.

People expected an attack. They expected exploitation. They expected the end of the decentralized era.

Nothing happened.

Here's what happened. The pool's owner, the exchange CEX.IO, published a statement saying it had no intention of attacking the network. The key Bitcoin developers, academic researchers, and large holders simultaneously began a pressure campaign: miners were urged to leave GHash.IO for other pools, CEX.IO's reputation was publicly dissected in the press, the price of bitcoin sank by about five percent over the course of a few hours. On July ninth, 2014, a roundtable was held in London with representatives of all the major mining pools and equipment manufacturers. At that roundtable it was proposed that GHash.IO voluntarily limit its share with a ceiling of 39.99 percent. On July sixteenth the pool publicly agreed to this restriction and called on the other pools to do the same. GHash.IO's share began to decline. By the end of summer it had fallen to thirty-five percent. By 2016 the pool had closed.

Why didn't a pool that held the technical ability to destroy Bitcoin in its hands use that ability?

The official answer: reputation, trust, long-term interests. Breaking the network you yourself earn from is irrational.

That's the right answer. But it's not the full one. Because "breaking the network you yourself earn from is irrational" is not a self-evident truth. It's the consequence of very specific structural conditions. What exactly has to be set up so that such behavior becomes irrational? Why does it work in some systems and not in others? Why did community pressure in July 2014 work — while in other cases, with other networks, pressure doesn't work and attacks happen?

These questions were answered in 1980 by a political scientist from Michigan. His name was Robert Axelrod, and what he discovered explains not only the behavior of GHash.IO but almost everything this book is about.

But that's for the next chapter.

The promise we begin with

Before we close the chapter, allow me to formulate what awaits us further.

The story of coal, which I began to tell, is in fact longer than two scenes — Darby and Murdoch. It consists of five phases, each of which was opened by its own person, and each of which made what had previously been considered waste into the main product of the next industry.

At first, coal was simply burned for heat — and was considered a poor fuel, because it stinks.

Then Darby transferred the coking technique from copper smelting to iron — and everything that came out during the production of coke was considered garbage.

Then Murdoch collected the gas — and everything that was left over from the production of gas was considered garbage.

Then in 1856 an eighteen-year-old chemist named William Perkin, trying to synthesize a malaria drug, accidentally discovered that synthetic dyes could be made out of this "garbage" — and on those grew the entire chemical industry of the twentieth century. Aspirin, the first plastics, the first antibiotics, the first polymers — all out of what had been considered the waste of gas production.

And in the end, by the middle of the twentieth century, coal stopped being called fuel. People began to call it feedstock — raw material. From one and the same coal they learned to extract simultaneously coke, gas, tar, sulfur, ammonia, and dozens of chemical compounds. Each output went into its own industry. The heat from burning the leftovers became the cheapest and least valuable result.

What happened to coal is what's now beginning to happen, with us, to computation. Bitcoin in this story plays the role of Darby's coke — the first structural breakthrough that showed that compute can be turned into something valuable (the protection of digital money). It was a discovery of historic scale, and we'll be unpacking its consequences in this book more than once. But just like coke in 1709, Bitcoin uses its re-

source one-dimensionally: compute turns into hashes, hashes turn into security, all the rest goes off into the air. For three hundred years humanity has been learning to see that around the main operation — coke, iron, hashes — there are streams that are considered garbage only until the person comes along who looks at them.

The Murdoch step — when people start collecting what until now they’ve been throwing away — falls, in digital history, on 2022. That is Qubic. And it’s one of those engineering ideas about which people later write “how did no one do this earlier” — it’s that logical in hindsight and that radical at the moment of appearance. Because Qubic doesn’t try to compete with Bitcoin on Bitcoin’s own field. It makes a move that didn’t exist on this field at all. What exactly Qubic “collects into a pipe,” how it’s set up technically, and why it works — is the main subject of this book. We’ll get to it gradually, chapter by chapter, because explaining it fast is impossible: the idea looks simple only after all the layers under it have been broken down.

And then — if everything goes as planned by the people who are building Qubic — what awaits us is the third step. The same one that happened to coal in the twentieth century, when it ceased to be “fuel” and came to be called feedstock, raw material for dozens of industries at once. If Qubic delivers what it promises, computation will pass into the same kind of state: out of one and the same compute cycle, security, artificial intelligence, economic value, and several more things we don’t yet know how to formulate today will be extracted simultaneously. This is an ambition of roughly the same scale as Darby’s in 1709 — and how exactly it’s implemented, we’ll see in this book down to the last screw.

The main thing — this step has not yet been made. It’s planned for April 2027. And we’ll honestly see whether it works.

This book is a map between the phases. It will not move linearly, because the story of Qubic itself didn’t move linearly: it was made by different people, in different places, solving different tasks, and only afterwards did it become visible how those tasks add up into one big task.

But before we move on to the Coalbrookdale of the digital age, we have to deal with one simple truth. This truth Murdoch knew intuitively in 1792, and Abraham Darby — did not. Without it, gas lighting would never have become an industry. Without it, Bitcoin would have stopped at hashes. Without it, Qubic wouldn’t work for a single day.

This truth is formulated in four lines of code.

It was proven in 1980, in a tournament that fourteen programs took part in.

CHAPTER 2

Axelrod's Tournament



In June 2014, one person (more precisely, one exchange) had in its hands, for twenty-four hours, the technical capacity to destroy Bitcoin. He didn't destroy it. The standard explanation — “it's irrational to break the network you earn from” — is correct, but it doesn't explain why it's irrational. Because there are networks where breaking is profitable. There are situations where people destroy what they earn from. There are countries where elites kill the country they rule. There are relationships where partners methodically annihilate each other. Irrational? Yes. Happens all the time.

So the question isn't why GHash.IO didn't attack. The question is: under what conditions does honest behavior become rational and cheating — not.

This turned out to be a concrete, technical, verifiable question. And the answer to it was first found not by a philosopher and not by an economist. It was found by a thirty-eight-year-old political scientist who in 1977 was sitting in his office at the University of Michigan, watching computer programs play chess.

This chapter is about what he eventually figured out. But to understand why his discovery matters for Qubic, Bitcoin, and everything we'll be talking about later, we have to first break down one very simple game.

The Prisoner's Dilemma

Imagine that you and your accomplice have been caught. There isn't much evidence on either of you — they can solidly prove only a minor crime, for which each of you will get a year. But the investigators suspect you pulled off something big together, and they want to prove it.

You've been split up into separate cells. An investigator comes to each of you and offers a deal.

“Listen. If your accomplice keeps quiet, and you give testimony against him — he goes down for five years, and you walk free. If you both give testimony — you both go down, but you get not five but three years each. If you both keep quiet — we won't be able to prove the major crime, and each of you only gets a year for the minor one.”

You sit in your cell and think.

If my accomplice keeps quiet — it pays for me to give him up. I walk, he goes down. If my accomplice gives me up — it also pays for me to give him up. I'll get three years instead of five.

Any variant of his behavior leads to one conclusion: it pays me to give him up.

The accomplice in the next cell over is thinking the same thing. And he too arrives at the conclusion: it pays for him to give me up.

In the end, you both give each other up and you each get three years. If you'd both kept quiet, you'd each have gotten a year.

That's the Prisoner's Dilemma. In its pure form, it looks like this: there are two participants, each can choose to "cooperate" (keep quiet) or "defect" (give up). The payoffs are arranged so that the individually rational choice is always to defect, but when both do it, both lose.

The game isn't complicated. They came up with it in 1950 at the RAND Corporation — two mathematicians, Merrill Flood and Melvin Dresher, were trying to come up with an example that would stump the rational-behavior theories of the day. Their colleague Albert Tucker formalized it and gave it the story about the two prisoners, in which we know it.

Since then — for seventy years — this game has become one of the most studied objects in the humanities and social sciences. Because it turned out to be everywhere.

Arms race — Prisoner's Dilemma. It pays each country to build more weapons than the neighbor. But when everyone builds — everyone is worse off.

Tax evasion — Prisoner's Dilemma. It pays each person not to pay. But when nobody pays — the state stops working.

Pollution of the environment — Prisoner's Dilemma. It pays each factory to dump its waste in the river. But when everyone dumps — there's no river left for anyone.

Corruption, cartel collusion, overfishing, traffic jams, harvesting on common fields — everywhere the same structure. And everywhere — the same result: each person's rational choice leads to a bad result for all.

John Nash, a mathematician from Princeton, in 1950 proved one unpleasant thing about this game and got a Nobel Prize for it. He showed that "both defect" is an equilibrium. Meaning: if you're both in this state, neither of you benefits from a unilateral change of strategy. If my accomplice gives me up, and I decide to keep quiet — I lose. If I give him up, and my accomplice keeps quiet — he loses. Both giving up — stable. Not good. But stable.

The Nash equilibrium. The system stuck in a bad but durable state.

That was the diagnosis. The trap was described. Whether you could get out of it — for a long time there was no answer to that question.

Michigan, March 1980

Robert Axelrod was the name of the man I mentioned at the beginning of the chapter. By 1977 he had been teaching at the University of Michigan for six years, had defended his PhD at Yale in political science in 1969, specialized in conflicts between states. Forty years old, son of Lithuanian Jews, he had spent his whole life on the question of how people manage to come to agreement at all.

Axelrod had a whole closet's worth of papers on the Prisoner's Dilemma. He'd read them all and was unhappy with them. The papers were mostly trying to prove one of two things: either that people in real life are "irrational" and cooperate contrary to theory, or that you have to change the payoff matrix — add a tax on defection, an outside arbitrator, something like that. Neither explanation satisfied him.

He was interested in a concrete question. The game was run once — the rational move is to defect. Fine. But what if you run it many times? If two participants know they'll be playing each other not for one round but for two hundred? Will that change anything?

Intuition suggested it should. If I defect now, and we're going to play another one hundred and ninety-nine rounds, my accomplice will have many opportunities to take revenge on me. He'll stop cooperating too. We'll both get little. Most likely it'll be more profitable to cooperate from the very start.

But how exactly? Which strategy in the iterated Prisoner's Dilemma is the most profitable? The intuition wouldn't formalize. Different theorists proposed different strategies — some complex, others simple — and no one could convincingly show whose was best.

Axelrod was at the time following computer chess tournaments. By the late seventies, chess programs had begun outplaying each other — and you could see that victory belonged to whoever had written the most complex, deep, multi-tiered algorithms. The programs were getting longer, smarter, more cunning.

And then a thought occurred to him.

What if you set up a tournament like that for the Prisoner's Dilemma?

Send out invitations to all the well-known specialists who had published on this game. Ask each of them to send in a program. A program in Fortran — a simple one, that takes as input what the opponent has done in past rounds, and returns a decision: cooperate or defect. Bring all the programs together in a single tournament, each with each, two hundred rounds. Tally the points. See who wins.

In 1977–78 Axelrod sent out the invitations. His assistant Jeff Pynnonen wrote, in Fortran, the environment in which the programs would play. By the start of 1980, fourteen programs had been received.

The senders were world-class specialists. Economists. Mathematicians. Political scientists. Psychologists. One sociologist. They had all been publishing papers on the Prisoner's Dilemma for years. Each was convinced that his strategy was the best.

Axelrod opened the envelopes. He read the code. And launched the tournament.

What was in the envelopes

I won't retell all fourteen strategies. But it's worth breaking down a few — to understand against what background the main thing happened.

One program always defected. The author reasoned this way: if the theory says the only equilibrium is "defect," let's just do that. The program got a guaranteed bad result against other defectors and an excellent one — if it ran into someone who always cooperates.

Another program always cooperated. The author reasoned: if everyone does that, everyone will be well off. The program played excellently with itself and against other cooperators, but constantly lost when running into defectors.

Some programs were cunning. One — a program by the name of JOSS — cooperated in ninety percent of cases after the opponent's cooperation, but in ten percent of cases it randomly defected "as a test." The author thought this would let it pick up a bonus from naive partners.

Another program — DOWNING — tried to model the opponent's behavior. It assessed what was the probability that the opponent would cooperate after its defection, and what — after cooperation. If the opponent was predictable, DOWNING tried to exploit him. If not — it adapted.

Some programs were openly complex. One used regression analysis. Another — Markov chains. One program was seventy-seven lines long in Fortran — the author was proud of its multi-level logic.

And somewhere in the middle of an envelope from Toronto lay a program four lines long.

The four-line winner

The program from Toronto worked like this:

First move — cooperate. Every following move — do what the opponent did in the previous move.

If the opponent cooperated — cooperate. If the opponent defected — defect. Next time, look again at what he did in response, and mirror.

That's all of it. No modeling. No regression. No "when to exploit, when to forgive" logic. A mirror. Literally a mirror.

The author called it TIT FOR TAT — an everyday English expression. Though a more accurate name for what the program actually does would be: "mirror." Axelrod saw this

program, read the code — and decided someone had sent it as a joke. Then he looked at the sender's name. This person didn't like jokes.

His name was Anatol Rapoport.

Axelrod launched the tournament. Fourteen programs played each other, every one with every one, two hundred rounds each. Plus each played against itself. Plus one neutral program was added as a benchmark. All the points were counted.

When the last round ended, the results were such that Axelrod rewrote his entire research program.

Tit-for-Tat won. By a wide margin.

Everything complicated lost. The long seventy-seven-line program took a middling place. JOSS with its random defections — below average. DOWNING with its modeling of the opponent — mediocre. Programs that always defected — at the bottom. Programs that always cooperated — also at the bottom, because they got exploited.

Four lines of code from Toronto beat fourteen years of work by experts.

The man from Lozovaya

It's worth pausing for a minute and telling who the man was who wrote the winning program. Because this isn't just some random mathematician.

Anatol Borisovich Rapoport was born on May 22, 1911, in Lozovaya, in the Kharkov Governorate of the Russian Empire. Today Lozovaya is in eastern Ukraine, in the Kharkiv region. In December 1921, when Anatoly was ten years old, his family fled from Soviet Ukraine to Poland. Across the frozen border river Zbruch. On skates.

To be precise, he was the one fleeing on skates — a ten-year-old boy who, dressing warmly and buying himself new skates, walked up to the river, crouched by a fence, watched the carefree Polish children skating on the Polish side of the dam, waited for the moment when the border guards got distracted, and rolled out. Suddenly, from somewhere above, a Red Army soldier spotted him and started shouting. Anatol picked up speed, raced over to the Polish children, wedged himself into their circle — and calmly skated off with them, away from the Soviet border. Many years later he would describe this moment in his memoirs.

A year later the family made it to the United States and settled in Chicago. Anatol was sent to study music. At sixteen he was already a promising pianist. At eighteen he entered the Vienna Conservatory. Until 1934 he studied there as a composer, pianist, and conductor. He gave concerts in Austria, Poland, Hungary, Italy. He spoke four languages.

Then the Nazis came to power in Germany. The career of a Jewish concert pianist in central Europe ended before it had begun. Rapoport returned to the States and discovered that in America too, no one needed yet another pianist from Chicago. At the age of twenty-three he enrolled at the University of Chicago. Seven years later he defended his dissertation in mathematics.

Then for fifteen years he worked on mathematical biology — modeling the spread of epidemics, the behavior of parasites and symbionts. In 1955 he came to the University of Michigan as one of the founders of the Mental Health Research Institute. There he began studying the behavior of people in conflict situations — through laboratory games, including the Prisoner’s Dilemma. It was Rapoport who introduced this experiment into broad scientific use, in his 1965 book *Prisoner’s Dilemma: A Study in Conflict and Cooperation*.

In 1970, when the Vietnam War entered its dirtiest phase, Rapoport left the United States for Canada. He wrote later that he wanted to live “in a country not committed to a messianic role.” He took a position in Toronto, where by 1980 he was already a professor emeritus on a pension — a seventy-year-old elder, the author of twenty books and three hundred articles, who had spent his whole life on one question: how to get people to stop killing each other.

By this moment Rapoport was one of the world’s most well-known researchers in game theory and conflict. And one of the most consistent pacifists. He founded Science for Peace. He wrote that war is “a contagious disease with a fatal outcome.”

When the invitation came from Axelrod, Rapoport sat down and wrote four lines. Sent it.

Many years later he confessed to Axelrod that he himself didn’t much like Tit-for-Tat. Too harsh, he said. Too ready to take revenge. Not really suitable as a moral position for a peace researcher. But he wanted to see how it would play.

It won.

Why the mirror won

Axelrod spent the next two years trying to understand why.

After analyzing all the played matches, he singled out four properties of the winning strategy.

First: niceness. Tit-for-Tat never defects first. On the first move it always cooperates. On any move after the opponent’s cooperation — it cooperates. This property divides all fourteen programs into “nice” (never defect first) and “mean” (defect at some point without provocation). Eight of the fourteen programs were “nice.” In the final results table, all eight of the nice programs took the top places. All six mean ones — the bottom.

Second: retaliation. Tit-for-Tat answers a defection immediately. Not after a delay, not in reserve, not with some probability — exactly on the next move, exactly once. This protects it from exploitation. The “nice” programs that didn’t know how to retaliate (always-cooperate) got mercilessly exploited and in the end lost.

Third: forgiveness. Tit-for-Tat doesn’t hold grudges. If the opponent cooperated on the last move — it forgets everything that came before and cooperates in response. Pro-

grams that piled up grudges and started taking revenge forever after a single defection got stuck in eternal war and lost points. Forgiveness restores cooperation.

Fourth: clarity. Tit-for-Tat is predictable. It's easy for the opponent to understand what it's doing and why. That means cooperation can quickly be established with it — all it takes is one try at cooperating, and it'll cooperate too. Programs that tried to be cunning and unpredictable (like JOSS with its random defections) confused their partners and destroyed cooperation that could otherwise have been established.

Niceness, retaliation, forgiveness, clarity. All in four lines of code.

Axelrod published the results in March 1980 in the *Journal of Conflict Resolution*. The article was called "Effective Choice in the Prisoner's Dilemma." It made an impression, but the scientific community received it with the usual healthy skepticism: well, one program won one tournament. Maybe it was a fluke. Maybe the others didn't try. Maybe if you ran the tournament differently, somebody else would win.

Axelrod read the criticism and decided to run a second tournament.

The second round

For the second tournament, Axelrod did three things.

First: he published a full report on the first tournament. All the programs, all the matches, all the results, a full analysis of why Tft won. All future participants got this report. That is, now no one could say they hadn't known the weaknesses of their own strategy or the strengths of their opponent.

Second: he ran an announcement in computer magazines — like *Creative Computing* and *Personal Computing*. He invited everyone, not just academic experts. In the end he received sixty-two programs from all over the world — from professors to teenage hobbyists. In the first tournament there were fourteen programs. In the second — almost five times as many.

Third: he changed the rule for ending the game. In the first tournament there were exactly two hundred rounds. This created the "last move" problem — on the last round there's no point in cooperating, so you should defect; but then there's no point on the second-to-last either; and so on all the way back to the start. This is called "backward induction" and theoretically destroys the whole construction. In real life backward induction doesn't work, because nobody knows the exact moment of the end. Axelrod modeled this: in the second tournament, after each move, with probability 0.00346 the game ended. On average this gave about 200 moves, but no one knew in advance which move would be the last.

The participants knew everything. They knew Tft had won. They knew which properties make a program successful. They knew that Axelrod had even specifically suggested a strategy that might beat Tft — Tit-for-Two-Tats, a softer version that forgives two defections in a row before retaliating.

Out of sixty-two programs, many tried to beat Tft directly. Some were more aggressive. Others — more cunning. Still others used complex models of the opponent.

One person looked at all this commotion, sat down, and once again wrote four lines.

Anatol Rapoport submitted Tit-for-Tat again.

The second tournament Tft won too.

Axelrod published the results in September 1980. And he added something else I haven't mentioned. He ran six variants of the second tournament — with different ecological conditions, where certain types of programs were represented more or less. In five out of six variants Tft won again. In the sixth it took second place.

And one more thing: Axelrod ran an evolutionary simulation. He imagined that each program produces offspring in proportion to its success. The less successful die out, the more successful reproduce. After a thousand generations Tft dominated. After several thousand — it occupied most of the population. Cooperation, which had seemed like a naive alternative to rational selfishness, turned out to be a stable evolutionary attractor.

The picture was clear. Cooperation is stable. Cooperation is profitable. Cooperation wins in repeated interactions — provided that the strategy that implements it has four properties: niceness, retaliation, forgiveness, clarity.

Three conditions

Axelrod went further. Over the next several years, he and his colleagues worked on understanding under which conditions exactly the picture described above works. Because it's one thing to show that Tft wins in a specific tournament. It's another thing to understand why.

In the 1984 book *The Evolution of Cooperation*, and even more sharply — in articles from 1985–1987 with the biologist William Hamilton, Axelrod formulated the answer. It turned out that for cooperation to be able to arise and be sustained through something like Tft, three conditions are needed. All three. Remove any one of them — cooperation collapses.

Condition one: iteration. The game has to repeat. If I know I'll meet you only once in life, the optimal strategy is to defect. Take and run. You won't find me anyway. In the tournament each pair of programs had two hundred rounds. In real life iteration arises when people live in the same village, work at the same company, trade in the same market for many years. When I know I'll see you tomorrow, I behave differently than if I know I'll never see you again.

Destroy iteration — and you get the behavior of a tourist in an unfamiliar city. You can leave no tip. You can be rude. No consequences.

Condition two: memory. Participants have to remember what the opponent did in past rounds. Without memory Tft doesn't work — you can't "answer in the mirror" to something you don't remember. In the tournament memory was built into the mechan-

ics: each program got, as input, the full history of interaction. In real life memory arises when I recognize you by face, remember our past conversations, have an idea of how you've behaved with others.

Destroy memory — and you get the behavior of a big anonymous crowd. Nobody remembers anybody. Every interaction is the first and the last. To deceive is safe.

Condition three: clarity. Or reputation, or observability — it's formulated in different ways. The essence is one: behavior has to be visible. Not only to the one I'm interacting with, but also to other participants. Because if I cheated you and no one besides the two of us knows about it, I can go and cheat the next person. But if news of my cheating spreads through the community — they'll stop letting me through to the next ones.

Destroy clarity — and you get the behavior of a world where you can buy a new name and move to another town. Whatever I do, the trail doesn't follow.

Three conditions. Iteration, memory, clarity. Remove any one — and the optimal strategy shifts from Tft toward pure selfishness.

How this connects to the Nash equilibrium

Remember the Nash equilibrium from the start of the chapter? In the Prisoner's Dilemma it was "both defect." Stable but bad.

Axelrod found another equilibrium. When the three conditions are satisfied — when the game is repeated, the participants remember, and behavior is visible — a new equilibrium appears: both cooperate by Tft. If one deviates and starts to defect, the other mirrors — and both become worse off. If one returns to cooperation — the other does too. Stable. And good.

This is fundamental. Nash described the trap: the system gets stuck in a bad state because unilateral movement out of it isn't profitable. Axelrod showed the way out: if there's the right structure around the game (three conditions), a second equilibrium appears in which everyone cooperates. And this equilibrium is also stable — unilateral deviation from cooperation is automatically punished.

The task, then, isn't to convince people to "be better." The task is to create a structure where selfishness leads to cooperation. Not to change people. To change the rules of the game.

Five paths to one idea

For millennia before Axelrod, a similar idea had independently been found five times.

Confucius, around 500 BCE, taught: do not do unto others what you do not wish for yourself. This is formulated in the Analects, in a conversation between the disciple Zigong and Confucius. Zigong asked: is there one word by which one might live one's whole life? Confucius answered: perhaps, "reciprocity."

Rabbi Hillel the Elder in Jerusalem around 30 BCE, when a gentile asked him to lay out the whole essence of the Torah while standing on one foot, answered: what is hateful to you, do not do to your neighbor — that is the whole Torah, the rest is commentary.

Jesus of Nazareth, around the year 30 CE, said something similar but in a positive formulation: as you would have people treat you, so treat them. Later this would be called the Golden Rule.

Immanuel Kant in 1785 proved this logically, as the categorical imperative: act only according to that maxim by which you can at the same time will that it should become a universal law. If you can't will that everyone lie — don't lie.

Anatol Rapoport in 1980 proved it mathematically. Four lines of code in Fortran.

Five independent paths. Two and a half thousand years. Different cultures, religions, eras, instruments. One formula.

You can say this confirms a deep moral truth — that the Golden Rule has always existed and is found everywhere. You can also say it differently: this confirms a structural truth. In repeated interactions with memory and clarity, Tit-for-Tat is the optimal strategy. That is exactly why any culture, any philosophy that has seriously thought about the question “how to live among others” finds it. That is exactly why a mathematical proof gives the same answer as moral intuition.

The problem, as Rapoport himself formulated it, is not in the rule. The problem is in the architecture of its application.

The three conditions have to be satisfied. In a family, between friends, in a small business — they're often satisfied of themselves. We interact iteratively, we remember each other, we see behavior. Tft works intuitively, without mathematics. In big anonymous systems — the state, the global economy, the crowd — the conditions break. And the Golden Rule stops working not because people have gotten worse. But because the architecture is broken.

Axelrod and Rapoport showed what makes the architecture right. And, by the same token, showed what makes it wrong.

Return to Bitcoin

Now we can return to the question we started the chapter with.

Why didn't GHash.IO attack Bitcoin in July 2014?

Because in Bitcoin all three of Axelrod's conditions are satisfied.

Iteration. Each block is a new round of the game. A block comes out every ten minutes. Twenty-four hours is one hundred and forty-four rounds. A year is fifty-two thousand rounds. A mining pool that wants to earn over five years will play this game a

quarter of a million times. This isn't a one-off situation. The shadow of the future is longer than any possible one-time gain.

Memory. The blockchain is literally the network's shared memory. Every transaction, every block, every miner — all of it is on the record. If GHash.IO had tried to rewrite transaction history in July 2014, that attempt would have stayed in the blockchain forever. Forever — in the cryptographic sense of the word. To erase is impossible.

Clarity. The hashrate of every mining pool is publicly visible. This isn't secret information. Any researcher, any journalist, any community member can look in real time at what share which pool occupies. When GHash.IO crossed 50 percent in June 2014, this was known within hours. Not days, not months — hours.

Three conditions satisfied. Which means TftT should work. And it did work: the community mirrored the pool's actions — miners left, the reputation collapsed, the price of bitcoin fell. The economic pressure was a retaliatory response to the threat of breaking cooperation. When GHash.IO publicly promised to limit itself to 40 percent — forgiveness restored cooperation. After a few months the situation returned to normal.

That was TftT at the level of the global mining community. Without a coordination center. Without a formal contract. Just four properties working through three conditions.

Now it becomes clear why in other situations pressure doesn't work. If a network is small and not visible from the outside — clarity is broken. If the miners are anonymous and there's no reputation — memory is broken. If the game is played once (say, if the network is on the verge of collapse and won't survive the next year) — iteration is broken. Anywhere even one of the three conditions is broken, Tit-for-Tat stops being the optimal strategy, and defection becomes rational.

So if we're building a distributed system and we want cooperation without a central arbitrator to work in it — we have to design it so that all three conditions are satisfied. Iteration, memory, clarity.

This looks like a recipe. And it almost is a recipe. But there's one nuance which none of Axelrod, Rapoport, or anyone who read their work in the 1980s could have predicted.

What was left unclear

If everything is so simple — three conditions, four properties, an optimal strategy — then why did we say at the end of Chapter 1 that Bitcoin has a structural flaw? After all, in it the three conditions are satisfied.

The answer: Axelrod's three conditions are a necessary condition for cooperation. But not a sufficient one.

Bitcoin satisfies Axelrod. There is iteration in it, and memory, and clarity, and Tit-for-Tat works there. That explains why the network survived in 2014.

But there is one more factor that Axelrod didn't write about, because in 1980 it was impossible to see. This factor is connected not to the conditions of the game, but to what exactly is being won in this game. To the prize itself.

In Axelrod's tournament the prize was abstract — points. The programs couldn't exchange those points for something else, asymmetric. They couldn't, for example, use their points to buy votes in determining the future rules of the tournament. They couldn't buy their reputation.

In Bitcoin the prize is real money. And this money can be used, among other things, to influence the system itself. To buy more equipment. To capture a larger share. To push through certain changes in the protocol. To direct the industry in a particular direction.

This creates a different problem. Not the problem of the absence of the three conditions — the three conditions are satisfied. But the problem that the very prize in the game contains, inside itself, an asymmetry that was not in the 1980 tournament. The prize gives the ability to change the rules of the game. And that is a class of attack which Axelrod did not model.

It's exactly this flaw — structural, subtle, almost invisible — that is the reason Bitcoin, for all its remarkable properties, remains the Darby phase, not the Murdoch phase. It solves the problem of cooperation between participants within the framework of the current game. But it does not solve the problem that the money earned in this game can be used to change the game itself.

But before moving further, it's worth pausing for a minute. Because we've derived something important.

Cooperation without an arbitrator is possible. This is not a utopia and not mysticism. It is a mathematical fact, proven in 1980 in a tournament that fourteen programs took part in. It requires three conditions — iteration, memory, clarity — and one strategy made of four lines of code.

And if we want to build systems in which millions of people, without coercion, act in common interests, we now know what to build from.

That's the first piece of good news in this book.

The next chapter is about the bad news.

CHAPTER 3

Why Bitcoin Can't Fix Itself



A Gene You Can't Repair From Inside the Cell

Imagine an organism. It has DNA. The DNA codes for proteins. The proteins assemble cells, the cells assemble organs, the organs make the organism. If there's an error in the DNA — say, one letter is wrong — that error is reproduced in every cell, because every cell copies the same DNA. The consequences can range from imperceptible to catastrophic, but one thing is clear: no cell inside that organism can fix the DNA error. For that you need someone outside, with different tools.

That's exactly Bitcoin's situation right now. It has its original construction — Proof-of-Work with anonymous miners and an open protocol. That construction solved Satoshi's specific problem — to create digital money with no bank in the middle. And it solved it brilliantly. But it has one side effect: anything earned inside the system can be used outside it, and the system has no mechanism to defend itself against how that earning changes the system itself.

I'll try to show this on three different layers. First — on the layer of computational power, where money earned from mining buys still more mining and concentrates in a few hands. Then — on the layer of ownership, where the same money, having passed through exchanges and funds, comes back as large institutional positions. And finally — on the layer of development, where the same money pays the engineers who write the protocol itself.

Three layers. The same mechanism. And in each of them — a structural contradiction between what Bitcoin promises in its original construction and what actually comes out.

Layer One: Who Counts

Let's start with mining.

When Satoshi launched Bitcoin in 2009, he seems to have sincerely thought that mining would be done by thousands of people on home computers, each one independently. "One CPU — one vote," he wrote in the white paper. One processor, one vote. Every participant in the network defends it with their small bit of power, and no one

has an advantage over anyone else. In 2009–2010 that’s how it actually worked — Satoshi himself mined on an ordinary laptop, taking the whole block reward.

That picture died very quickly. In 2010 someone figured out that graphics cards could be used for mining — they hash dozens of times faster than processors. In 2011 the first FPGA miners appeared, several times faster again. In 2013 Bitmain released the first ASIC — an integrated circuit honed for one single operation, SHA-256 hashing. The ASIC was a thousand times more efficient than a GPU. Home computers were out of the game. So were graphics cards. Only ASIC farms remained.

By 2016 most mining had moved to China, where electricity was cheap and local authorities weren’t interfering yet. By 2021 China had banned mining, and the industry relocated to Texas, Kazakhstan, Russia, Iran, Iceland. By 2024 — after another “halving,” when the block reward was cut in half from 6.25 to 3.125 BTC — the small operators were squeezed out for good. Only the large industrial farms survived: the ones with long-term contracts for wholesale electricity, their own cooling infrastructure, and teams of engineers.

As of April 2026 the picture looks like this.

At the very top — mining pools. These are associations of miners that combine their computations and split the reward. Of all the world’s miners working on Bitcoin, Foundry USA serves about 29 percent. AntPool (owned by Bitmain, the largest ASIC manufacturer) serves another 21 percent or so. ViaBTC — about 11. F2Pool — about 10. SpiderPool — about 7. MARA Pool and SecPool — 6 each.

The first two pools together control roughly half of all Bitcoin hashrate. The first four — about seventy percent.

This is a fundamentally important number, and it’s worth pausing on it for a minute.

When a pool controls a certain share of the hashrate, it essentially decides which transactions go into the block. The pool doesn’t actually mine the blocks itself — that’s done by the miners connected to it. But the pool builds the “block template” — the list of transactions the miner then tries to hash. If the miner happens to find the right hash, the block is published with the transactions the pool selected.

In other words: four organizations — Foundry, AntPool, ViaBTC, F2Pool — decide which transactions get into seventy percent of all Bitcoin blocks. This isn’t a threat to the network in the sense of “they could destroy it.” It’s much more subtle: they can slowly, transaction by transaction, decide who pays more for the right to enter a block, whose transactions get processed first, and whose get stuck. They can (and sometimes do) exclude transactions from addresses under sanctions — for example, addresses on the U.S. Treasury blacklist. They can (and this has already happened) prioritize certain kinds of transactions that bring in higher fees.

And here’s the thing — they can do all of this without breaking a single protocol rule. They’re simply taking advantage of the fact that they have a lot of hashrate. And they have a lot of hashrate because they earned it on previous blocks and reinvested into more equipment, cheaper electricity, and better engineers.

The game rewards winners with money. Money buys more equipment. More equipment gives more hashrate. More hashrate — more wins. The cycle is closed.

Notice that this cycle violates none of Axelrod's conditions. The three conditions — iteration, memory, transparency — are still met. New blocks come out every ten minutes (iteration). Everything is recorded in the blockchain forever (memory). The hashrate of every pool is publicly visible (transparency). Everything's in order.

But inside this perfectly functioning architecture a process of concentration is going on that isn't accounted for by the rules of Tit-for-Tat. Because Tft is a strategy for equal players. It says nothing about what to do when one of the players is twenty times bigger than another. In Axelrod's tournament all programs had equal starting conditions. In real Bitcoin — they don't.

Layer Two: Who Owns

Now let's look at another level.

Bitcoin isn't only miners — it's holders. People who simply own coins and store them on their own wallets or on exchanges. In theory, since every bitcoin is identical, distribution should be uniform. In practice — it isn't.

In January 2024 the U.S. Securities and Exchange Commission (SEC) approved spot Bitcoin exchange-traded funds (ETFs) for the first time. This was the culmination of years of pressure from large asset managers — BlackRock, Fidelity, Grayscale — who wanted to be able to sell bitcoin to their institutional clients through ordinary brokerage infrastructure, without having to deal with crypto exchanges and cold wallets.

BlackRock is the world's largest asset management company; it has on the order of 11 trillion dollars under management. That's more than the GDP of any country except the United States and China. When BlackRock launched its Bitcoin ETF under the ticker IBIT on January 11, 2024, money poured in like a river.

As of April 2026 — two years later — IBIT holds roughly 800,000 bitcoins, which corresponds to about 63 billion dollars at the current price. That's about 49 percent of all assets in all U.S. spot Bitcoin ETFs combined. In the first quarter of 2026 alone, IBIT took in net inflows of 8.4 billion dollars.

But IBIT isn't the only large holder. The company Strategy (formerly MicroStrategy, a business-analytics firm that then reinvented itself as a pure Bitcoin treasury) as of April 17, 2026 owns 815,000 bitcoins with a total acquisition cost of 61.56 billion dollars. That's about four percent of all the Bitcoin that has ever existed. No single entity in history has held that kind of share — except, possibly, Satoshi himself, whose coins haven't moved since 2010.

If you add up everything held by U.S. spot ETFs, you get about 1.31 million bitcoins — roughly 6.2 percent of total supply. Add Strategy — you get more than ten percent. Add the large exchanges (Coinbase, Binance), corporate treasury positions, large private

holdings — and it turns out that several dozen entities account for a quarter of all Bitcoin.

What does this mean in practice?

It means that if BlackRock decides one fine day, on a regulator's order, to freeze IBIT's assets — eight percent of all Bitcoin instantly becomes illiquid. That's a market shock of a magnitude Bitcoin hasn't seen yet. It means that if Strategy faces a margin call on its loans (and it has been borrowing to buy bitcoin), it could be forced to liquidate 815,000 coins — which would crash the price several times over in an instant. It means that if the U.S. government decides to regulate Bitcoin with new legislation, the main channel of influence is no longer decentralized miners in Kazakhstan but BlackRock in New York, where one phone call to the right office can change the behavior of trillions of dollars.

The paradox is that Bitcoin was conceived as a network resistant to centralized pressure. Decentralization was not a side effect, it was a central requirement. And yet sixteen years later we have a construction in which a few large institutional players control such a share of supply that they are structurally able to influence price, liquidity, and the regulatory agenda more than the rest of the market combined.

How did this happen?

The same way it happened with mining. Each individual movement of money was voluntary and individually rational. A pension fund wanted exposure to Bitcoin but couldn't hold the coins directly because of regulatory restrictions — bought IBIT. A family office wanted diversification — bought IBIT. A corporation wanted a hedge against inflation — bought IBIT. No one was coerced, no one was manipulated. Each individual decision looked sensible.

But the aggregate result of millions of such individually rational decisions is gigantic concentration. Which now itself becomes a factor in the equation. Because the more bitcoin BlackRock has, the more voice BlackRock has in any public discussion of what Bitcoin should be in the future. And that voice isn't the same as an ordinary holder's. It's a voice that gets listened to at the SEC, in Congress, at the Federal Reserve.

Layer Three: Who Writes

The third layer is the most invisible, but the most important.

Bitcoin is code. That code is written by people. There aren't very many of them. According to a 2024 BitMEX Research report, on all of Bitcoin — which at that point was worth about a trillion and a half dollars — 41 developers were actively working: those

who made at least five commits to the main Bitcoin Core repository over the year. Forty-one people, on a trillion and a half.

Of those forty-one, about two-thirds live in the U.S. and Europe. Three developers — in Latin America (Argentina, Brazil, El Salvador). Four — in the rest of the world. That's Tier 1 statistics, drawn from public commit data.

Who funds them?

Bitcoin has no foundation. No central organization. No proverbial treasury that pays salaries from. That was all by Satoshi's design — so that nobody could halt development by pulling funding. But programmers have to work, and they get paid through various sources.

Blockstream. A company founded in 2014. Among its founders — Gregory Maxwell, Pieter Wuille, Matt Corallo, Jorge Timón, Mark Friedenbach (all active Bitcoin Core developers), as well as Adam Back (the inventor of hashcash, on which Satoshi built Bitcoin's PoW) and venture investor Austin Hill. Since its founding, Blockstream has raised more than half a billion dollars in investment in total — \$76 million by 2017, \$210 million in a Series B in 2021, another \$210 million in convertible notes in 2024. The company clearly has several key Core developers on its payroll. Adam Back, its president, explains it this way: the company has an interest in seeing Bitcoin supported, developed, and scaled.

Chaincode Labs. A research center in New York that funds several more key developers.

MIT Digital Currency Initiative. An academic cover at the MIT Media Lab; it started up around 2015 in response to the funding crisis among Core developers after the collapse of the Bitcoin Foundation.

Brink, OpenSats, Spiral, Human Rights Foundation. Nonprofit grant programs.

Large exchanges — Coinbase, OKCoin, OKX. Periodically issue grants.

The total annual funding for all Bitcoin Core development is estimated at 150–200 million dollars a year — for a network worth more than a trillion. By comparison, the Ethereum Foundation in 2024 spent about 50 million on development, with a market cap ten times smaller than Bitcoin's. Polkadot — 16 million, with a market cap a hundred times smaller. Per dollar of capitalization defended, Bitcoin underfunds development by an order of magnitude.

And — here's where it gets subtle — a disproportionately large share of the modest funding that does exist flows through Blockstream. A company that has its own commercial products: Liquid (a sidechain for financial institutions), the Lightning Network (an L2 for fast payments), mining hardware, institutional services. The interests of these products sometimes coincide with the interests of an ordinary Bitcoin user, and sometimes they don't.

A classic example is the 2023–2024 debates over the Ordinals protocol, which made it possible to write arbitrary data (images, NFT collections, even books) onto the Bitcoin blockchain. Technically this was done through Taproot — a protocol upgrade that Core developers shipped in 2021 for other purposes, but which Casey Rodarmor used in early 2023 in an unexpected way. Ordinals quickly took up a significant share of block space, driving up fees and changing the network's economics.

One of the senior Core developers, Luke Dashjr, submitted a pull request that would have technically restricted Ordinals. Another maintainer, Ava Chow (a former Blockstream employee), closed the PR, citing “lack of consensus and creating noise.” The community split. Some considered Ordinals a creative use of the network. Others — parasitism on block space meant for monetary transactions. The decision was, formally, made by one person. That decision could not be reversed by anyone except the maintainer himself or another maintainer with equal rights.

Ordinals didn't break a single Bitcoin rule. But they radically changed how the blockchain is used. And that change happened not through a vote of miners, not through a referendum of holders, but through the individual decision of a few developers with commit access. Who get paid by specific companies. Who have their own business interests.

Formally the network remains decentralized — it has no CEO, no foundation, no board of directors. In fact, a narrow circle of people, financially tied to a limited set of companies, makes decisions that affect trillions of dollars of capital and millions of users.

And again — this is no one's evil intent. It's just a consequence of how the development economy is set up. Bitcoin doesn't pay developers out of its blocks. They are paid by external companies. The companies have interests. The interests concentrate. The developers fed by those companies cluster around the tasks those companies consider priorities. Regardless of how well those tasks line up with Bitcoin's tasks as a public good.

The Three Layers Together

If you look at the three layers at once, a common structure is visible.

On the mining layer — money earned in previous blocks buys more hashrate. More hashrate gives more influence over which transactions go into blocks.

On the ownership layer — money earned from mining or from price appreciation buys more coins through institutional instruments. More coins gives more voice in the public discussion of Bitcoin's future.

On the development layer — money earned at exchanges, at ETFs, at mining infrastructure, funds developers. The developers who receive that money decide where the protocol moves.

On each of the three layers, the same mechanism is at work: an internal win is converted into external influence. And that influence, in turn, can be used to change the conditions of the next rounds of the game in one's own favor.

This is where Axelrod's model breaks. Not because Tit-for-Tat is a bad strategy. Tft is still optimal for the situation it describes. But it describes a situation in which points cannot be converted into the right to change the rules. Bitcoin is a situation in which they can.

Axelrod in 1980 couldn't have foreseen this, because in his tournament there was no money. There were abstract points that couldn't be used for anything except tallying up the winner. Whereas in a real economic system, the winnings can always be converted. The only question is into what, and how much.

Under the Hood. GHash.IO Once More

In the first chapter I told the story of GHash.IO in the summer of 2014 — a pool that for a day crossed the 55-percent threshold of Bitcoin hashrate and didn't catastrophically attack the network. In the second chapter I used that story as an example of how Tit-for-Tat works in reality — the community mirrored the threat, the pool retreated, everything ended well.

Now — the whole truth.

GHash.IO did not destroy Bitcoin. But it did use its majority position for a specific attack on a specific service. In the autumn of 2013, when GHash.IO's share was already significant but not yet 51 percent, the pool conducted a series of double-spending attacks against the online casino BetCoin Dice. The discussion of this on the BitcoinTalk forum, in thread number 327767, is still publicly available. The idea is simple: GHash.IO accepted losing bets at its addresses, and when it lost — it mined the same transaction with its own (winning) address into a different block, turning the loss into a win. The digital version of what in a physical casino is called "replaying the round when the ball didn't stop where it should have."

BetCoin Dice took real losses. The investigation established that. GHash.IO didn't formally acknowledge any public contrition, but there were reputational consequences: the pool stopped accepting deposits at its addresses, and the community started watching it more closely. This story is one of the reasons that, six months later, when GHash.IO once again grew toward the critical share, the pressure was so quick and so strong. People remembered.

This is an important correction to the picture usually painted in the Bitcoin community. People often say: "Bitcoin has never had a 51-percent attack." That's true only at a certain level of strictness. If you understand a 51-percent attack as "a complete re-writing of blockchain history months back, with double spending of all major transactions" — yes, that hasn't happened. If you understand it as "use of a majority position to

extract gain at the expense of other participants in the network” — it has. And it has been documented.

This is a critical detail for our subject. Because it shows: the mechanism of “converting winnings into the right to change the rules” is not a theoretical threat. It has already gone off. It just went off on a limited scale, because the pool, understanding that a full breach would kill the entire network (and its profit along with it), stopped at local exploitation.

Local exploitation is bad news in another sense. It means that a system in which the holder of a majority of computation can exploit the minority, even when not destroying the whole construction, is structurally unstable even when Axelrod’s three conditions are met. Because Tft equilibrium presupposes that the participants play more or less on equal footing. The moment a majority player appears who can carry out a “pinpoint attack” — take something from someone but not break the system as a whole — they will do it. Rationally. In keeping with their incentives. In full accordance with game theory.

And there’s no defense against this inside Bitcoin’s mechanics.

Under the Hood. Monero, August 2025

The second example is a very recent one. In August 2025 — that is, less than a year before these lines are being written — something happened that confirmed what I was just talking about, but on a far larger scale. And it was done not by one of the major exchanges and not by a hacker group, but by the project this book is, in principle, written about — Qubic.

To retell the whole story here is not the place — that’s the topic of chapter fifteen. But the brief fact is this. Monero is a network similar to Bitcoin, also Proof-of-Work, but with a different algorithm (RandomX, optimized for ordinary processors rather than for specialized ASICs). Monero’s market cap at that moment was about 6 billion dollars. Qubic’s market cap — about 300 million. A ratio of 1 to 20.

Over three months — from May to August 2025 — Qubic, through a mechanism we’ll talk about in detail in chapter 8, started paying Monero miners three times more than they were getting by mining Monero directly. No hack. No hacking. Just a different economics: “connect to our pool, we’ll route your computational work in such a way that you earn more.”

The miners looked at the numbers. Reconnected their equipment. On August 11, 2025, in a window of 122 blocks, Qubic owned 63 blocks out of 122 — 51.6 percent of the hashrate. The Monero network was under the control of a network twenty times smaller in market cap.

What happened next is a separate story. Qubic did not destroy Monero. But the technical possibility of destruction was there, and in a public statement on August 3, 2025 the Qubic developers spelled out exactly what they could have done if they’d

wanted to: double-spending against exchange deposits, rewriting block history, censoring transactions, crashing the price to zero. They didn't. They decided to stop. But they could have.

For our chapter — and specifically, for the question of why Bitcoin can't fix itself — this is the key episode. Because it proves empirically, not theoretically: any PoW network can be attacked by offering miners a better economics. The Qubic developers, summing up the experiment, formulated this as one of three proven theses:

“Incentives Dictate Consensus: Any Proof of Work blockchain can be attacked and controlled by providing superior economic incentives to its miners.”

This is a critical statement. It isn't saying “Bitcoin is weak” — Bitcoin is strong, which is precisely why it's more expensive to attack. It's saying “no PoW network is structurally protected.” The protection is just money. More money — more protection. Someone with bigger money can buy that protection out. And money, let me remind you, is exactly what accumulates with those who have already won the game.

The circle is closed.

What This Means for Our Story

Let me bring it all together.

The first chapter ended on the point that Bitcoin assembled a revolution — but in doing so left “Murdoch's gas” in the air, computational cycles that nobody needs. That was bad news about the use of a resource.

The second chapter gave good news: cooperation without an arbitrator is possible if Axelrod's three conditions are met, and Bitcoin on the whole meets those conditions.

The third chapter — that is, this one — says: the bad news is deeper than we thought in the first chapter. Bitcoin not only uses its resource inefficiently. It is also structurally incapable of fixing itself, because the money earned inside is constantly being converted into power outside, and that power comes back inside as the ability to change the rules. Concentration of mining. Concentration of ownership. Concentration of development. Three layers, the same dynamic.

You could say that all of this is just the general problem of capitalism, and Bitcoin isn't unique in it. That would be true if Bitcoin hadn't promised the opposite. And it did. It promised that every processor would be one vote. That mining is decentralized. That nobody controls the network. That the money it produces is neutral and resistant to any politics. Sixteen years on, all of that is either already untrue or becoming untrue right in front of us.

And not because Satoshi was wrong. He wasn't wrong — he solved a specific problem, and he solved it. It's just that his solution is the Darby phase. The first great breakthrough. It opens doors that were locked before. But it has a structural limit you can't step over from inside. The Murdoch phase is already a different engineering problem, with different conditions.

And here we come to the main question of this chapter.

The Fourth Condition

Axelrod in 1980 described three conditions under which cooperation is possible: iteration, memory, transparency. He didn't write about a fourth, because in his tournament that condition was satisfied automatically — the programs played in a closed system in which winnings could not be converted into anything external. In the real world this condition may or may not hold — and that is the defining difference between systems that stay cooperative and systems that drift toward oligopoly.

Let me try to state this condition explicitly:

The system has to be set up so that winnings inside the game cannot be converted into the right to change the game itself.

Or shorter: the boundary between “the prize” and “the rules” must be impossible to cross from inside.

In Axelrod's tournament that boundary was absolute — points and rules lived in different worlds. In Bitcoin that boundary is a permeable membrane. Internal money constantly seeps outward, acquires external influence, and comes back in already as pressure on the rules. Gradually. Year after year.

This fourth condition isn't an improvement on Axelrod. It's an addition. The three original conditions determine whether cooperation will be the equilibrium between players. The fourth determines whether that equilibrium structure will be preserved in time, or whether it gradually drifts toward a new equilibrium in which a few play and many submit. It's about the durability of durability itself.

If we want to build a system that satisfies not only Axelrod's three conditions but also the fourth, we need an architecture built differently. Not “an improved Bitcoin.” Not “Bitcoin with a patch against concentration.” A fundamentally different construction, in which the very mechanism of winning is built so as not to produce concentration.

This is a non-trivial engineering problem. The naive solution — to forbid the sale of bitcoin to large investors — doesn't work, because it destroys the first property (a free market). The naive solution — to make the developers receive a salary from the protocol — works partially, but runs into the question of who decides whom to pay. The naive solution — to forbid mining pools — doesn't work, because it destroys the ability of small miners to participate in the network.

What's needed is an architecture in which the answer to all of these questions is built into the very mechanism by which the network operates, rather than being decided by a vote or by an authoritative directive. Where the mining process itself produces decentralization rather than concentration. Where the ownership process itself prevents oligopoly among large holders. Where the development process itself depends on the entire community rather than on a narrow group of sponsors.

Such an architecture exists. It's called Qubic. And how it works is the subject of this book from chapter six to the end.

But before we talk about how it works, we need to talk about the man who made it. Not because the project's history is incomplete without the man's history — but because this man saw Bitcoin's structural defect not in 2025, when it became obvious to everyone, and not in 2020, when it began to show up in the data. He saw it in the early 2010s, when Bitcoin was still young and nobody doubted that it would remain decentralized forever. He saw it — and that's why he was able to build two crypto networks that came out at the very top of the all-time ICO ranking: the first, in first place; the second, in second. More than a million percent ROI on one, four hundred thousand on the other. And from those two networks he took away an understanding of exactly how crypto projects die from inside.

His name is Come-from-Beyond, and he really did come from beyond.

PART II

The Man

CHAPTER 4

Come-from-Beyond Before Bitcoin



This story may differ in micro-details from how its protagonist himself would tell it. Some things he has never publicly confirmed, some he has confirmed years later and in passing, some he has deliberately left in the air. I assembled it from forum archives, late interviews, throwaway remarks, technical traces in the code of three networks, and the few facts I could verify myself. Somewhere I'm probably wrong. If he ever reads this and says "not quite," then it's not quite. But the general outline is this.

There's a peculiarity about people who later turn out to be right: they see something earlier than others, and at the moment they see it, nobody believes them. Galileo, looking through his telescope at the moons of Jupiter. Semmelweis, demanding that surgeons wash their hands. Turing, asserting that one day a machine would play chess better than a human.

The hero of this chapter is no Galileo and no Turing. He didn't discover any laws of nature. He was simply one of the thousands of programmers reading the BitcoinTalk forum in 2010 and trying to figure out what strange thing this notorious Satoshi Nakamoto had just launched. The difference is that most of those thousand programmers saw a miracle. He — a year or two later — saw a crack.

That's an unusual ability. To see a defect in something everyone else admires, and not say "but it works, so everything's fine." To say "yes, it works, but right here there's going to be a problem — not tomorrow, but in ten years." And to start building something else while everyone around you is still celebrating.

I named this person at the end of the previous chapter: Come-from-Beyond. A handle taken back in the early 2000s. As with Satoshi, the handle here isn't a way to hide — it's a way to sign. That's a separate and curious parallel we'll come back to.

This chapter is about what this person was doing before Bitcoin entered his field of view. What his background was. Why he ended up in the right place at the right time with the right set of tools to see Bitcoin's flaws sooner than anyone else in the industry. And why by 2013 — when he first stepped on the stage himself — he already had everything he needed to build something fundamentally different.

A Programmer From the Nineties

If you trace the footprints backward, the earliest ones Come-from-Beyond left go back to the late 1990s. An Eastern European teenage programmer working on a large game project that never shipped. In one of his remarks many years later he'll mention that this project was twenty-five years ago — that is, somewhere around 1999. "It was written for one of my first big games which never released. ~25 years ago." A big game, group work, didn't make it to release. The classic fate of early 3D games of that era.

To understand the context, it's worth saying a couple of words about what programming looked like in Eastern Europe at the turn of the millennium. By the end of the 1990s, in the countries of the former socialist bloc, a generation had appeared with a computer at home, more or less stable internet, and access to any textbook or source code you could find. At the same time, this generation had almost no institutional career infrastructure — no Silicon Valley nearby, no venture capital, no internship programs at big companies. People taught themselves, worked remotely for Western clients or for local employers, and invented their own problems.

This turned out to be a very productive environment — in the sense that it produced people accustomed to solving problems that nobody pays for, without external validation, for their own interest. A habit rare in more settled ecosystems.

Come-from-Beyond was one of these. He played Ultima Online — one of the first big mass online games, in which hundreds of players simultaneously roamed a shared virtual world — and quickly discovered that some of the in-game tasks (gathering ore, fishing, hunting hides) lent themselves to automation. In 2024 he'll mention in passing that "My first farming bot in UO was doing it that way." Which is to say that as a teenager he was already writing programs that played the game for him.

To a modern eye this sounds like a small thing. But it's worth thinking about what this person was doing to make such a bot work. UO had no open API. The bot had to emulate mouse and keyboard input, recognize images on screen, react to changes in game state in real time. That's a real engineering problem — and in 1999–2000 it was nontrivial even for adult professionals.

The next footprint left behind is more formal. In 2003, in one of the regional technical publications (back then there were such things — computer portals in local languages, with or without a paper supplement) an article was published called "Methods for Detecting Plagiarism in Programming." The author — him, then still under his ordinary name, without the crypto handle. The article worked through the technical side of the question — how, looking at source code, you can determine that someone copied it from someone else, even if the variables were renamed and the comments rephrased. A topic relevant to educational institutions and to the protection of intellectual property in the software industry.

This article matters to us for two reasons.

The first — it proves that his first published work goes back to the early 2000s. By the time Bitcoin appears on the scene, he has ten years of programming practice behind him and at least one printed work on a professional topic.

The second reason is the subject of the article. Plagiarism in programming. How do you know that this code is yours and not someone else's stolen from you. What structural traces remain in a program that give away its author. The question of how to identify an anonymous creator by the style of the code is one of those questions that, when an anonymous Satoshi Nakamoto appears on the scene in 2008, will unexpectedly take on extreme relevance. And, amusingly, many years later he himself will refer back to this 2003 article of his in one of the most enigmatic phrases in this whole story. But it's too early for that now.

In parallel with this, by his own later admissions, in the very early 2000s he started thinking about something else, much more ambitious than games and plagiarism. In one of his early exchanges he'll say that Aigarth — a project he has been working on, with varying intensity, since the very beginning of the 2000s, for about twenty years by 2023. What exactly Aigarth is we'll work through later. For now it's enough to say: the task he set himself at twenty was bigger than crypto, bigger than Bitcoin, bigger than the entire industry that came afterward. Crypto would be infrastructure for Aigarth, not the other way around.

This is an important angle. He didn't come to crypto "to make his own coin." He came to crypto with a much older and bigger problem, for which he needed infrastructure that couldn't be built without crypto.

Meeting Bitcoin. 2010

Satoshi published the white paper in October 2008. In January 2009 he launched the network. In December 2010 he disappeared — leaving only a million mined coins and not a word about who he was.

Somewhere in that interval, Come-from-Beyond first opened BitcoinTalk.

He doesn't name an exact date, but many years later, in 2025, in a conversation with someone in the community, he'll mention that he had used the technique of finding sockpuppets (fake accounts of one and the same person under different names) on BitcoinTalk "like 15 years ago." Fifteen years from 2025 is 2010. That is, by the time Satoshi was still active and posting in the forum, he was already reading that forum closely enough to be doing account forensics.

How many bitcoins he had at this point and how they appeared — he has never publicly disclosed. Possibly he mined some himself in the early days, when CPU mining at home still made sense and one block reward was fifty coins. Possibly he bought some later. Possibly both. Someone hunting for sockpuppets on BitcoinTalk in 2010 is unlikely not to have tried running on his machine the thing that was being discussed there. But

let that stay open — it's one of those details he left in the air, and I'm not going to expose him.

Formal registration on the forum came later. In October 2011, Litecoin appeared — the first major variation on the Bitcoin theme, created by ex-Google engineer Charlie Lee. Litecoin had poor technical documentation, and Come-from-Beyond, wanting to dig into the details and not finding answers in open sources, was forced to register on BitcoinTalk to ask a question. He himself will later describe this as “an act of bravery for a lazy programmer.” His account got the number 46556; the profile is registered before November 22, 2011, which is publicly verifiable to this day — the profile page is open, anyone can go and check.

This single event — a technical registration to ask a question about Litecoin — will eventually grow into fifteen thousand messages over the years that follow. The BitcoinTalk archive holds them all. If anyone has the patience, they can reconstruct one specific programmer's evolving views on the crypto industry literally week by week.

By the end of 2011 he is already deeply involved. Not a public figure, not an author, not an activist. Just a very attentive observer who already has his own view of where this industry is going — and where its problems are.

What He Saw

The previous chapter ended on the point that this person saw Bitcoin's problems in the early 2010s — at the moment when nobody in the industry doubted its eternal decentralization. What exactly did he see?

Let's break it down step by step.

By 2011 Bitcoin had been running for two years. The network held, transactions went through, miners were getting more numerous. The price was volatile, but the trend was upward. On the BitcoinTalk forum, optimism reigned: a solution had been found to a problem that had been considered unsolvable. Digital money without a bank. Freedom from state issuance. Decentralization for the ages.

Come-from-Beyond, on top of the problems described in chapter 3, saw the most interesting thing of all:

Bitcoin uses its computational power irrationally. In a world where distributed computation could be solving real problems — training neural networks, modeling the climate, verifying cryptographic constructions — Bitcoin's miners are computing useless hashes. This isn't a bug — it's by design. Satoshi chose a solution that worked, and it worked brilliantly. But the solution was single-purpose. Compute turned into protection and nothing else.

Come-from-Beyond, working on Aigarth — a project that needs distributed computational substrate at industrial scale — looked at those hundred and seventy terawatt-hours a year (the figure for 2026; in 2011 it was counted in gigawatt-hours) and saw, literally, his infrastructure being burned on guessing random numbers.

For him this wasn't an abstract injustice. It was personal. He needed a computational substrate for the AI project he had been thinking about for ten years. Bitcoin was taking that substrate and burning it for nothing.

So: four problems, seen at once. Any one of them, on its own — noticeable. All together — no longer just a defect, but a structural diagnosis.

Most people, on seeing such a diagnosis, do one of two things. Either they close their eyes and keep using what's there, because it works. Or they write a long forum post, take a wave of downvotes from the enthusiasts, and walk away disappointed.

Come-from-Beyond did a third thing. He silently kept accumulating early coins, to have capital and independence, and started thinking about how to build it differently.

The Satoshi Question

This is a fitting place to stop for a minute and say a word or two about a question that any attentive reader will inevitably raise.

Is Come-from-Beyond Satoshi Nakamoto himself?

The question is a serrrrious one and deserves a book of its own. Dozens of researchers in the cryptological world are working on exactly this question right now — you get the idea. (Attentive readers of the Qubic Church website will smile here.) Mathematicians and linguists analyze stylistic coincidences, conspiracists draw diagrams, enthusiasts dig through the BitcoinTalk archive, and the professional tabloids periodically publish articles under the headline “We’ve Found Satoshi!” Discovering a week later that they were wrong, they wait for the next week and publish another article just like the first. The Satoshi-hunting industry in this sense is its own separate, sustainable ecosystem in which you can genuinely make a career.

For this book I'll limit myself to two things.

First. There are firmly established facts that noticeably narrow the space of possible interpretations. Come-from-Beyond was in the BitcoinTalk community as early as 2010 — that is, when Satoshi was still active. He has a 2003 publication behind him about methods that he himself will later jokingly refer to in the context of “Satoshi framed me, he knew about this my article” and decided to use the method I described. He created BCNext — the identity under which the NXT project was launched in 2013 — and many years later cryptographically confirmed that BCNext is him by signing a message from the NXT genesis account. He himself openly admits that for ten or fifteen years he has been deliberately leaving false clues for the curious — because “Waiting 10-15 years for good trolling is worth it.”

Second. There's a sensible way to formulate an answer that stays accurate under any interpretation. Satoshi was not so much a person as a function: create a decentralized system without an owner and disappear, so that the system can live without you.

Come-from-Beyond is a creator by that same function. Whether he was personally there in 2009, whether he worked with whoever was there in 2009, or whether he simply learned to play that same role a decade later — the difference between these three options can't be cleanly separated by the data publicly available today. But the function itself is the same with him. Create and step aside. Build without an owner. Disappear as a center of control, leaving the system to self-reproduce.

Let's make a working assumption for this book: Come-from-Beyond is some part of Satoshi. Not necessarily the very person who sat at the bitcoind console in 2009. Possibly that one. Possibly the one who was nearby. Possibly the one who came later and took up the same functional niche. Possibly came from beyond :cfbtroll: This assumption is enough for what this book is about. We don't ask the question further.

Back to business.

Why This Particular Person

There's a question this chapter has to answer directly: why was it Come-from-Beyond, and not some other thirty-year-old engineer of the same generation, who saw in 2011 what would only become obvious to everyone else by 2024?

Short answer: because at the same time he had the right technical background, the right discipline of thinking, and he was... a genius.

Let's break it into parts.

Technical background. By 2011 he had been writing code for ten years in different modes: game engines, bot infrastructure, source-code analysis tools, server applications. That means he had an intuitive grasp of what happens when a distributed system starts to scale. How many people among the early Bitcoin enthusiasts actually understood how the network would behave when the hashrate grew by a billion times? Very few. The answer required either experience working with systems that had gone through this, or the mathematical habit of extrapolating curves into the future. Come-from-Beyond had both.

Cryptographic literacy. His first publication — about methods for detecting plagiarism in code — is, when you get down to it, a problem about how to identify an author by the structural fingerprint left in an artifact. That's a close cousin of cryptographic problems. The very subject matter in which, by 2008, the entire logic of the Satoshi paper would take shape: hashes, signatures, proofs of key possession. He wasn't a cryptographer of the academic sort, but he was a person to whom the basic ideas of cryptography didn't look like magic but like specific engineering work.

Game theory. This aspect is less public, but from my own observations — the technical decisions in NXT, the formulations in early forum discussions, the structure of Qubic — it's clear that by the early 2010s he was already thinking about distributed systems in this frame. Tit-for-Tat as an idea had been in the air by then for thirty years. Whether he read Axelrod or arrived at the same principles himself, observing repeated

interactions in games — unknown; he doesn't say. What matters is something else: the thinking tool was in his hands. He was looking at Bitcoin not only as code but as a game in which several types of players with their own interests interact under known rules. That's exactly the angle from which the cracks become visible.

The absence of an external contract. And this is, perhaps, the most important. In 2011, unlike today's crypto stars, none of the early participants was a "professional crypto founder." There were no positions, no venture funds specializing in blockchain, no media machine feeding off explaining cryptocurrencies to ordinary people. Come-from-Beyond was simply interested in the technology and was mining coins. That freed him from a huge layer of cognitive biases that would later make other early participants defend Bitcoin as their personal article of faith.

He had no incentive to defend Bitcoin. He had no incentive to break it either. He had an incentive to honestly assess whether this was working sustainably or not, and if not, to think about what to do next.

He thought. And he started doing.

Two Pseudonyms and One Hint

There's one more motif that we'll only sketch here and unpack later.

In 2008–2009, Satoshi Nakamoto appeared on the scene. An anonymous pseudonym that released a working system, replied to community messages over the course of two years, and quietly walked away in early 2011, leaving the project to live a life of its own.

In September 2013, on that same BitcoinTalk forum, a user named BCNext appeared. He published an announcement of a future project called NXT, made several posts over the course of two months, and on November 8 of that year fell silent for good. NXT would launch on January 3, 2014. Many years later Come-from-Beyond would cryptographically confirm that BCNext is him, by signing a message from the NXT genesis account. A second identity that turned out to be a continuation of the first line.

And there was one more account that's worth mentioning, even though it's too early. In 2013 a user named Maria appears on BitcoinTalk. A few short posts. One of them is signed "Maria Bitcoin Founder." A phrase that in 2013 sounded like someone's bad joke. The account would get lost in the stream of forum messages, and for ten years no one would remember it.

Maria will return to us in another guise, but later. For now what matters is the general pattern.

What we have in front of us is a person who has already once, in publicly confirmed form, repeated the very structure in which Satoshi was made. Create under a pseudonym, launch a working product, step aside, leave the system to live. This isn't a one-off coincidence anymore, it's a regularity beginning to form. And if we accept the working assumption we arrived at on the previous page — that Come-from-Beyond is some part

of Satoshi — then the regularity becomes a rule. The same method, applied either by different people or by the same person across different projects, gives the same result.

Good engineering is done without a name. Good systems live without a creator. Everything else is detail, which we'll work through in the chapters to come.

By 2013

Now let's look at where Come-from-Beyond stands at the end of 2012 — at the moment when he's ready to make his first full step onto the stage.

Behind him are ten-plus years of engineering work. A published article on software code analysis. Practical expertise in distributed systems, gained in the gaming industry and the bot environment. Early participation in the Bitcoin community since 2010. Enough early bitcoins to have capital and independence. An understanding of Bitcoin's problems formulated two years before they showed up in public. A multi-year parallel project (Aigarth), for which a Bitcoin-like network is infrastructure, not an end in itself. Cryptographic literacy. An understanding of game theory. The discipline that won't let him build a coin with a twenty-percent stake for himself. The habit of disappearing once a project is finished.

"Qubic is my first project, envisioned before Nxt and IOTA," he'll say in 2024. But Qubic is too ambitious a project for 2013. The technology isn't ready. The infrastructure isn't ready. The community isn't ready. Doing Qubic in 2013 means failing for sure.

So he does what any smart engineer with a long-term task does: he breaks it down into intermediate steps. First — show the world that Proof-of-Stake works. That'll cost him — by his own estimate — five minutes for the algorithm and two months for the code. Under this, the NXT project will open, on behalf of the BCNext identity. The first pre-ICO in history will launch, the first community of long-term holders in history will gather, it will become at its peak the third cryptocurrency in the world by capitalization, deliver the first place by ROI in the entire history of the crypto industry — 1,265,555 percent according to Cointelegraph — and be replaced through Ardor with a one-to-one airdrop. That'll take him about ten years.

As the next step he'll start work in 2014–2015 on a second project — IOTA — where he'll try to solve the problem of "how to make a blockchain without a chain of blocks." That'll work too. The world's second-place cryptocurrency by ROI according to that same Cointelegraph ranking — 424,084 percent. That'll take him another five years or so and end in a painful internal conflict with co-founders. He'll leave.

And only in 2022, drawing on the experience of two previous launches, with evidence behind him that his engineering decisions are accepted by the market and make people millionaires, with the technical and infrastructural readiness that didn't exist in 2013, will he finally launch Qubic. The project he had been envisioning for twenty years. The project in which he tries to bring it all together: useful mining, AI training, an

economy without a controlling owner, infrastructure capable of withstanding scale and not starting to drift toward oligopoly.

Two networks before Qubic — first and second place by ROI in the entire history of crypto ICOs. Twenty years of work on one big idea. The habit of leaving when a project is done. A method in which the creator disappears so that the system can live on.

By 2013, he already has everything he needs.

Time to step onto the stage.

C H A P T E R 5

Three Projects, Three Metamorphoses



Up to this point I've been assembling the story out of scraps. Out of fifteen thousand messages on BitcoinTalk, late interviews, throwaway remarks, three sets of technical fingerprints. Starting in 2013 it gets easier to assemble: public records, verifiable ICOs, known names, open court documents in Norwegian. The man who in the previous chapter was a shadow on the forum now starts making things that stay.

Three things. Three cryptocurrency networks, each with its own history, its own team, its own peak, and its own end. Looked at from the outside, these are three different projects with different architectures and different fates. NXT launched in January 2014, became the world's third cryptocurrency by capitalization, and was smoothly replaced by a successor. IOTA launched a year later, drifted away from its principles, and left behind public lawsuits in Norway on which its creator spent more than a million dollars of his personal money. Qubic launched in April 2022 and, as of early 2026, is still running.

Looked at from the inside — from Come-from-Beyond's side — these aren't three projects. It's one project, broken into three stages, stretched over thirteen years, and it will end one more year from now, in April 2027.

Imagine an architect who has to build a bridge across a chasm no one has crossed yet. The technology for building such a bridge doesn't yet exist. So the architect builds two smaller crossings along the way — not for the sake of the crossings, but for the sake of experience. To learn how the material behaves. To train the workers. To convince the people on the other shore that bridges, in general, can exist. The crossings turn out to be useful — people walk across them, they become local infrastructure, the newspapers write about them. But the architect knows that this isn't the main thing. The main thing is still ahead.

This chapter is about three such constructions. Two are already done. The third is in progress. The fourth construction, if you take this metaphor further, is the architect's departure at the moment when the bridge is built and tested. By his plan, that's supposed to happen on April 13, 2027. But the fourth — at the end.

NXT. January 2014 – the first crossing

On September 28, 2013, on the BitcoinTalk forum, a user under the handle BCNext appears and publishes a short announcement. The announcement says: a new cryptocurrency will soon be launched, built entirely on Proof-of-Stake — no mining, no electricity burning, no ASICs, no Bitcoin lineage. The programming language is Java, not C++. Distribution: the first ICO in crypto history — anyone interested sends BCNext bitcoins, and in exchange, at the moment of launch, receives a proportional share in the new network.

The announcement is short, no whitepaper. A few technical clarifications. A promise that the network will be launched “in the coming months.”

By that time, the crypto community treated the idea of Proof-of-Stake roughly the way they treated perpetual motion. Everyone understood that theoretically it should be possible, but every known attempt — Peercoin in 2012, several smaller experiments — either turned out to be a hybrid with Proof-of-Work or had structural problems like “nothing at stake” (where, at the moment a network forks, it’s profitable for a miner to vote for both branches at once, and the network loses the ability to come to agreement). Building a clean, working, stable 100% PoS was an open research problem several teams had been working on for years.

BCNext spent on solving it, by his own later admission, five minutes on the algorithm itself and two months on the code. “In 2013 we didn’t have non-flawed PoS yet and the idea looked cool, so BCNext spent 5 mins to invent working 100% PoS algo and extra 2 months to code it. Then he has gone for good because Nxt was a side-chick” — he’ll write in January 2023. Another remark of his, more than a year later: “Don’t worry, in Nxt I warned in advance that Nxt had been a side-chick and I implemented PoS because of Spanish shame which I was feeling, observing whole crypto industry being unable to implement such easy thing.”

“Spanish shame” (*vergüenza ajena*) is shame for someone else, for the way they’re embarrassing themselves. He acted like a man who looks at collective helplessness and, with a sigh, silently does the work himself.

The entire NXT codebase: three thousand lines of Java. “Nxt was 3k lines,” as Come-from-Beyond will put it briefly in October 2020. For comparison: by 2026, the average Layer-1 cryptocurrency runs to hundreds of thousands of lines. NXT was the most compact fully functional crypto network of its time and probably remains one of the most compact to this day.

LAUNCH AND RESULTS. The ICO ran from September 28 through the end of November 2013. Twenty-one bitcoins were collected — about 14,000 dollars at the exchange rate of the time. Distributed proportionally among 73 participants. A few years later, that initial sum would turn into a market cap measured in hundreds of millions.

On November 8, 2013, BCNext makes his last post under that account.

On January 3, 2014, at 02:08 UTC, the first NXT block is created. The network is live.

What follows is something nobody expected. NXT works. Pure Proof-of-Stake, no mining, no electricity burning, no Bitcoin dependency, climbs the rankings, and at its peak becomes the world's third cryptocurrency on CoinMarketCap — beneath only Bitcoin and Ripple. “Nxt was #3 on CMC, right below Bitcoin and Ripple (which isn't actually a cryptocoin)” — Come-from-Beyond's exact phrasing years later. As he himself put it elsewhere, summarizing his record: “Invented and implemented first working 100% Proof-of-Stake. Conducted first ICO (and all the others followed). Created the first coin which wasn't a clone of Bitcoin.”

By 2017, at the peak of the next crypto cycle, NXT will enter the Cointelegraph ranking “Top 10 ICOs by ROI in History” — and take first place there with a return of 1,265,555 percent. More than one million two hundred thousand percent. Twelve and a half thousand x. The source matters precisely because it's third-party: Cointelegraph ranks the history of crypto ICOs by open data, and NXT comes out higher than Ethereum, higher than everything else.

In April 2014, Come-from-Beyond's contract formally ends. BCNext leaves — silently, as promised. This departure had been announced in advance. The community had been warned: as Come-from-Beyond will summarize it years later, “in the very beginning of Nxt I warned community that it's my temporary project.”

Many years later, Come-from-Beyond signs a message from the NXT genesis account. Only BCNext could have the private key to that account, because BCNext is the one who created the first block. When the signature from that key is set down by Come-from-Beyond, that's a cryptographic proof: BCNext and Come-from-Beyond are the same person. The first promised departure turns out, at the same time, to be the first publicly confirmed link between pseudonyms. In the NXT community this link is by now treated as a settled fact — established and not requiring further discussion.

The network itself keeps running. By 2018 Come-from-Beyond, together with the company Jelurida (which by then had become NXT's main maintainer), launches a new project — Ardor — and does an airdrop to all NXT holders at one-to-one. Anyone who held NXT receives an equivalent amount of the new coin for free. As Come-from-Beyond will summarize the move in 2024: “Nxt was replaced by Ardor which did 1:1 airdrop to all nxt holders, then Ardor pivoted to B2B sector, anyway, in the very beginning of Nxt I warned community that it's my temporary project.”

NXT, meanwhile, doesn't go anywhere. It keeps running in parallel with Ardor to this day. “Nxt is still running,” Come-from-Beyond observes in May 2023. In 2026 — also still working. The first network into which CfB invested his first departure outlived thirteen years. Holders who didn't manage to sell at the peak got Ardor for free in the airdrop. As he'll note separately: “Every nxt holder got corresponding amount of ardor and earned good profits.” This wasn't abandonment — it was orderly succession.

What NXT did first

To get a sense of the scale, it's useful to list how many "firsts in history" are recorded in those three thousand lines of Java.

The first working 100% Proof-of-Stake. "Nxt wasn't improved version, it was 1st, never seen implementation of pure PoS," in Come-from-Beyond's own phrasing. Not a hybrid, not an experiment — a working, stable one that lived through every market cycle. All later PoS networks — Ethereum after the 2022 merge, Cardano, Solana, Polkadot — appeared later and in many ways inherited ideas first realized in NXT.

The first ICO in history. The model "collect bitcoins, distribute the new coin proportionally at the moment of launch" was first done specifically for NXT. Ethereum in 2014 ran an ICO on roughly the same scheme. The whole subsequent enormous wave of 2017 ICOs that crashed the market and triggered a regulatory wave — all of them trace structurally back to the NXT model.

The first cryptocurrency not in C++. Before NXT, every cryptocurrency — Bitcoin, Litecoin, Dogecoin, the rest — was a fork or a variation of Bitcoin's C++ code. NXT was written from scratch in Java. As Come-from-Beyond noted in late 2023, having stumbled on an article about it: "Spotted an article I've never seen before, looks like Nxt was the first non-C++ crypto." That sounds like an accountant's detail, but it actually means NXT was the first cryptocurrency built not as a variation of Bitcoin but as an independent architecture.

The first NXT Asset Exchange — an on-chain exchange. The ability to create and trade tokenized assets directly on the network, with no intermediary. A concept that in 2020 will explode in the form of DeFi on Ethereum and bring billions in turnover.

The NXT Alias System — a name service on the blockchain. Mapping short readable names to addresses. Roughly what the Ethereum Name Service (ENS) will do in 2017 was already working in NXT in 2014.

All of this — without a whitepaper. Without a scientific paper, without a formal protocol specification. "None of my projects had whitepapers, I'm an engineer, not a writer," Come-from-Beyond will say in March 2024. And separately, in December 2023, with a characteristic intonation: "Superstition, Nxt had no whitepaper and made many people millionaires. Do you need WP or become rich?"

But the main thing about NXT's history isn't what was done there for the first time. The main thing is that this project, from the very start, was flagged as intermediate. "I did quit Nxt coz of it being just a detour, because PoS was trivial and still cryptocom-munity wasn't able to implement it" — that's how he'll put it in October 2023. Detour: the side road, the way around.

The first crossing was built to show that it was possible. The material held up. The workers got trained. People came to believe that bridges exist.

He could move on to the next.

IOTA. 2014–2019 – the second crossing

If NXT was proof that Proof-of-Stake works, IOTA was an attempt to solve a problem that PoS doesn't solve: how to build a network that has no chain of blocks at all.

The whole architecture of Bitcoin (and NXT, and every PoS network that came after) is built as a linear sequence of blocks added to a single main chain. This has two problems. The first is a fundamental fee per transaction, because space in a block is limited and there's an auction for it. The second is a fundamental ceiling on throughput, because fixed-size blocks appear at a fixed rate. Attempts to scale run into one or the other.

The alternative that began to be discussed in academic circles by 2013 was to give up the linear chain and instead use a directed acyclic graph (DAG). Each transaction itself confirms several earlier ones; the network grows like a tree, not like a rope. Theoretically this gives throughput that grows with the number of participants and removes competition for block space — which is to say, fees.

Theoretically. In practice, by 2014 nobody had launched a DAG network. There were papers, there were ideas, there were some mathematical models. There was no working network.

IOTA was the first. The author of the mathematical description — the whitepaper called “Tangle” — was Sergei Popov, a mathematician by training. As Come-from-Beyond will put it in August 2021, “Popov wrote Tangle whitepaper.” The engineering implementation, that is, writing the working code, was handled by Come-from-Beyond and several other people. IOTA itself didn't have a whitepaper — it had the Tangle whitepaper, signed by Popov. “IOTA had no whitepaper, only Tangle did,” he'll clarify in July 2024.

In August 2021 he'll clarify another detail, important for what happens next: at the moment IOTA's genesis block was created, David Sønstebø — the man who in a few years' time will be referred to in the public sphere as “IOTA co-founder” — was not a member of the team. He joined later. At the moment the network launched, he was not in it. This isn't a contested interpretation, it's a factual statement: “Dom is not a founder, he wasn't even a member of IOTA team when genesis was created.”

IOTA launched in June 2016. By the peak of the next cycle, late 2017 to early 2018, it will enter the same Cointelegraph ranking as NXT. Second place. Return of 424,084 percent. Four thousand two hundred forty-one x. Higher than Ethereum's. Lower — only than NXT's.

Two networks from one person — first and second place in the entire history of crypto ICOs by return. Cointelegraph, not Come-from-Beyond, was the one who counted. It's a public ranking.

What happens next

In 2017–2018, the IOTA Foundation (IF) is set up around IOTA — a German non-profit registered in Berlin. The goal: structural project governance, grant infrastructure, corporate partnerships. On paper, everything looks like every serious crypto project of that period.

In practice, by Come-from-Beyond's account, what starts next is what in his conversations five years later will be called by one word — collapse.

“In IOTA everything stopped working as intended after IOTA Foundation was created. They took all resources for meaningless partnerships, so I quitted IOTA and decided to run Qubic standalone, instead of running it on IOTA” — phrasing from July 2024.

And in fuller form, July 2025: “IOTA collapsed because of people embezzling money and resources for own benefit. They literally took money and devs dedicated to core development to develop and promote side-projects related to IOTA, they invested into.”

I won't retell the details. They're partly public — in open court documents in Norway, where in 2019–2021 the main bulk of the legal dispute was moved. Part is closed. Part is in correspondence that will never become public.

A few facts that are recorded and verifiable:

Come-from-Beyond left IOTA around 2019. The departure was public, with a stated reason — as he put it himself, “IOTA Foundation collapsed into gang of those worrying only about money, not achievements.” The network itself wasn't destroyed. He didn't use his role as a core developer to bring IOTA down from the inside, although he had every technical means to do so.

In parallel, his access to the Jinn documentation was cut off — Jinn being a project to develop a proprietary ternary processor, conceived as the hardware layer for IOTA (and for Aigarth, but more on that later). “David holds Jinn IP hostage, so I had to invent algo constructing Jinn circuits without human assistance. Now it's being used in Qubic mining” — short phrasing from December 2021. Jinn Labs, the team that had been working on the chip, was fired by Sønstebø.

On the Norwegian lawsuits Come-from-Beyond spent, by his own public admission, a seven-figure sum in dollars — more than a million dollars of his personal money. By August 2021 he had already won at least one case — against IOTA AS and the IOTA Foundation. As he puts it: “The current won lawsuit wasn't against D&D, it was against IOTA AS and the IOTA Foundation.”

From the standpoint of our story — the story of the “create and leave” method — one detail matters. The departure from IOTA was not a planned departure, the way BCNext's was from NXT. It was a forced departure at a moment when a project conceived as an incubator for Qubic had been captured from within and was no longer fit for its original purpose. Come-from-Beyond doesn't leave because the work is done. He leaves because

the work can no longer be done here. That's a different kind of departure — but on the result of the method it acts symmetrically. The idea evacuates, the network keeps living its own life, its builders commercialize it further as best they can.

Jinn and the algorithm that will become the heart of Qubic

Here we have to pause on one technical detail, because it's the most important point of connection between two chapters of this book and the explanation for why Qubic is built the way it is.

Jinn was a ternary (trinary) processor, conceived as a hardware accelerator for IOTA operations and, in the longer term, as the execution layer for Aigarth. Ternary logic is rare in modern computing but theoretically more efficient for certain classes of tasks, including a number of neural-network operations. Jinn existed as a project of the Jinn Labs team starting in 2015, had its own IPO (JINN token holders received a share in the future product), and had working prototypes.

When Sønstebo cut off Come-from-Beyond's access to the documentation and fired the team, physically making the next version of the chip became impossible. But the software task — “how to describe the schematic of such a processor automatically, without manual engineering work, given only a specification of what it has to compute” — remained.

Come-from-Beyond solved that task over the next year and a half.

The algorithm he came up with takes a description of a desired function — for example, a small artificial neural network — and automatically assembles, out of elementary operations, a logical schematic that implements that function. Without a human. Without an engineer manually trying out architectures. A pure search through the space of possible schematics.

“Now it's being used in Qubic mining” — short phrase, December 2021.

This isn't an incidental technical aside. This is the core of what's called UPoW in Qubic — Useful Proof of Work. Qubic miners aren't picking hashes; they're running automatic synthesis of neural-network schematics. Each attempt is a piece of useful computational work which, in aggregate, trains something larger.

One thing matters to nail down here. The loss of Jinn forced him to invent something that would later become the heart of Qubic. If Sønstebo hadn't cut off his access — there would have been no forcing condition. With no forcing condition — no algorithm. With no algorithm — Qubic would not look the way it looks. This is one of those rare cases where you can pinpoint the moment at which a hostile circumstance becomes the precondition for the next architecture.

Adversity to architecture — trouble turning into design. Engineers in our industry rarely get to point at such a moment in their own biography. Come-from-Beyond can.

In an early Qubic explainer, he summarizes what he was actually building: “Qubic began life as as an initialism-turned-acronym, QBC, which stands for quorum-based

computation... Specifically, Qubic is a protocol that specifies IOTA’s solution for quorum-based computations, including such constructs as oracle machines, outsourced computations, and smart contracts.”

Qubic. April 2022 – the main thing

On April 13, 2022, Qubic’s first block is created. The network is live on mainnet. Come-from-Beyond launches what he had in mind twenty years earlier.

In this chapter I’m not going to retell Qubic piece by piece. That requires the entire next part of the book, and that part is in progress. Here — only the general outline. A few entities, five paragraphs. Just enough to make it clear what came together out of the experience of the two earlier projects.

676 computers. The Qubic network is held up not by an arbitrary number of validators and not by an open mining pool, but by a fixed pool of 676 participants, partly rotated each week according to a publicly announced rule. The number 676 isn’t random, but the explanation of its meaning is in the next chapter.

UPoW — Useful Proof of Work. The very thing discussed in the IOTA section. Miners aren’t picking random numbers; they’re doing useful computational work — automatic synthesis of neural-network schematics. In aggregate, this work trains the platform that has lived in Come-from-Beyond’s head since the early 2000s under the working name Aigarth.

Aigarth. I’ll limit myself to calling it “an AI platform conceived by CfB since the early 2000s, for which all of this cryptocurrency infrastructure is a preparatory skeleton.” What Aigarth actually is, how it’s built inside, what makes it different from ChatGPT, Gemini, Claude, and the other public large models of 2025–2026 — is the conversation of part five of the book. Within this chapter, only one thing matters: the cryptocurrency part of Qubic exists for the sake of Aigarth, not the other way around.

The Arbitrator. In the current architecture of Qubic there’s a special role — the Arbitrator, performed manually by Come-from-Beyond himself. Its functions are bounded by the source code: he can’t print coins, he can’t change transactions, he can’t shut anyone’s wallet. But certain coordinating decisions — for example, the change of the rules for selecting computers for the next week — go through him. This role is temporary. By the announced plan, on April 13, 2027, exactly five years after the mainnet launch, the Arbitrator will be abolished. The last lever through which the creator can touch the network will be dismantled.

This is the fourth departure. If we accept the working assumption of the previous chapter — that Come-from-Beyond is some part of Satoshi — this is the fourth public departure in the same chain. Satoshi in December 2010, BCNext in April 2014, Come-from-Beyond himself out of IOTA around 2019, and the fourth — April 13, 2027, the Arbitrator of Qubic.

Three have already happened. The fourth is announced, dated, and written into the protocol.

If it happens as planned — the method is confirmed for the fourth time. If it doesn't — the method isn't confirmed. That's a falsifiable prediction with a specific date attached. Engineers' biographies don't often contain such things.

It's worth noting that Qubic is a return to the original concept. The very same quorum-based computation that was QBC inside IOTA was pulled out, rethought, strengthened by experience, and made standalone. Twenty years of work on one big idea came together into one network.

One more characteristic detail, of the kind that doesn't show up in reports. Qubic didn't start with an empty audience. At launch, the project already had a community — the same people who held NXT, then came into IOTA, and now came into Qubic. Come-from-Beyond described this himself in November 2023 with a vivid metaphor: "Ever seen game guilds which play different games together? They go from game to game as a formed guild without a founder. We came to Qubic from IOTA, were in Nxt before that." That means Qubic didn't have to explain from scratch who Come-from-Beyond was and why his engineering judgment was worth trusting. A group of people who had convinced themselves through their own experience were already walking with him. And this, by the way, separately explains why the early Qubic community is structurally different from those of random 2022 crypto projects: it has a disproportionate number of people with thirteen years of crypto experience and the memory of two prior transitions.

Come-from-Beyond ranks his three projects himself, briefly and without false modesty: "Kaspa, IOTA, Qubic. Qubic is the best so far." (Kaspa in this list is a network not made by him; he mentions it as a comparison, because its developers use part of his early DAG research. What matters is not Kaspa but the other two elements — IOTA and Qubic. By his own assessment, Qubic is above IOTA.)

He came to the Qubic launch not just with an idea. He came with thirteen years of experience launching working crypto networks, with a doubly confirmed track record, and with an algorithm that nobody but him could have invented, because nobody else had lost Jinn.

Under the hood: the pattern of three projects

If you put the three projects side by side, seven shared traits become visible. Not one or two — seven, repeating systematically enough that you can't write them off as coincidence.

First, the pseudonym as a signature. BCNext for NXT. Come-from-Beyond for IOTA and Qubic. In both cases the handle isn't a way to hide. It's a way to sign, and at the same time to separate the man from the project. BCNext isn't a mask, it's an identity that has its own private key and its own cryptographic tie to the NXT genesis block.

Come-from-Beyond isn't a mask, it's the identity under which all public activity since 2011 has been carried out.

Second, compact code. NXT — three thousand lines of Java. Qubic core — also compact; Come-from-Beyond hasn't named exact figures publicly, but by indirect signs and by the way he talks about it, it's the same orders of magnitude. IOTA — here it's harder, because the source code was picked up and reworked by the IOTA Foundation after Come-from-Beyond's departure, and judging the size of the original by the current state of the network would be incorrect.

Third, no whitepapers. "None of my projects had whitepapers, I'm an engineer, not a writer." Two of the three networks — first and second in the history of crypto ICOs by return. This isn't an oversight, it's a principle. Working code matters more than a document. A document describes the intent; the code implements it. If the code works, the intent is verified; if the code doesn't work, no beautiful document will save it.

Fourth, warning the community in advance. NXT was openly declared a "temporary project" at the very start. IOTA — when the conflict with the Foundation became visible, Come-from-Beyond announced his departure and its reasons publicly. Qubic — regularly, almost weekly, in his public correspondence Come-from-Beyond reminds people of the risks and that Qubic holders have to make decisions for themselves. This isn't modesty; it's a disclaimer written into the methodology.

Fifth, leaving at the right moment. BCNext — in April 2014, when NXT was launched and working. Come-from-Beyond out of IOTA — around 2019, when it became clear that the Foundation had captured the project. The planned abolition of the Qubic Arbitrator — April 13, 2027. One departure was planned. One was forced. One is still ahead. But in all three cases the structure is the same: the creator steps back, the network continues.

Sixth, track record. Two of the three networks — first and second place by ROI in the history of crypto ICOs according to a third-party Cointelegraph ranking. 1,265,555 percent and 424,084 percent. Come-from-Beyond comments on this briefly: "Check Nxt and IOTA price chart, you can spot places where we were getting rid of useless holders. Do you think getting top ROI 2 times is a coincidence? I'm an engineer and engineering laws work same way everywhere."

This is an important quote, and two things in it are worth attention. The first — "we were getting rid of useless holders." This refers to periodic events — including FUD, provocations, price corrections — that wash low-conviction holders out of the project before the project goes mainstream. These aren't accidents, they're a deliberate part of the strategy. This explains why Come-from-Beyond himself sometimes broadcasts publicly skeptical things about his own projects. It's not a paradox — it's a filter. The second thing — "engineering laws work same way everywhere." He doesn't say "I got lucky." He says "I'm an engineer." That's a direct answer to any argument of the form "crypto is a lottery."

Seventh, the community as a guild that travels. It was already mentioned in the Qubic section — I'll just repeat the formula: NXT → IOTA → Qubic holders move through

projects the way a gaming guild moves through games. This structurally lowers the launch cost of each subsequent project: part of the trust carries over from the previous one. Not all of it; each project still has to prove technical merit from scratch. But the basic trust in the engineer himself — carries over.

And an eighth trait, separate, not about the projects but about role separation. In all three cases Come-from-Beyond is the engineer, the inventor, the architect. The commercializing is done by others. In NXT — the company Jelurida, which picked up support after BCNext's departure, made Ardor, took it B2B. In IOTA — the IOTA Foundation, which took the project and steered it on its own course (in Come-from-Beyond's view, into a dead end, but that's now its own responsibility). In Qubic — a distributed community of developers and ecosystem projects who, as of 2026, are building applications, exchanges, bridges, and services on top of the core protocol. Inventor and commercializer are two different roles, rarely combined in one person. With Come-from-Beyond, that discipline of separation is now written into his biography three times.

Seven traits, reproduced across three projects. Different architectures, different markets, different years, different teams. The same person, the same method, the same result.

Time to move on

By April 13, 2022, Come-from-Beyond had everything he didn't have in 2013. Twenty years of thinking about Aigarth. Two launched crypto projects, one fully stepped away from, one forcibly left in someone else's hands. Two confirmed places in the historical top by return on investment. The algorithm for automatic schematic synthesis, born out of the loss of Jinn. A community that had come along with him from the previous projects. An industrial maturity that didn't exist thirteen years earlier. And — the main thing, perhaps — proof to himself that the “create and leave” method really does work. Twice. In projects with different architectures and different fates. With a result that an outside third-party ranking nailed down in the top two slots.

Qubic launched. The third project by sequence, but the first by intent. The very one for which everything else had been done.

And here, in 2026, standing midway between launch and the planned departure of the Arbitrator, you can take a step to the side and ask the question this part of the book has been answering at length: who is Come-from-Beyond, and why is it worth believing he knows what he's doing. The answer, assembled from the two previous chapters, comes out like this: an engineer with twenty years of experience, who has twice proven his method, who publicly announces his departures in advance, who has been working on one big task (Aigarth) since the early 2000s, who uses cryptocurrencies as infrastructure rather than as the goal, and who has scheduled his fourth departure on a specific date. If that departure happens — the method is confirmed for the fourth time. If it doesn't — it isn't. That's an honest falsifier.

But the next question is a different one. Not who and not why, but how. How exactly is the Qubic network built so as to be capable of living without its creator? Why 676 participants — and not 700 and not 1,000? How are they replaced every week, who selects them, and by what rules? What happens to the QU coins inside, how are they emitted, and where do they disappear to? How exactly does UPoW work, and how does useful computational work turn into protection of the network from attacks? And the main thing — how is the moment of transition arranged, when the Arbitrator is abolished and the last point of human governance disappears from the protocol?

These questions aren't about the man. They're about engineering. And answering them requires going inside the protocol.

In the first chapter of this book we talked about the cascade of coal as a metaphor for the development of technologies. Darby in 1709 obtained the first iron smelted on coke, but didn't know what to do with the gas going off into the air. Murdoch in 1792 figured out how to collect that gas into a pipe and use it separately to light streets — and that moment was the start of a completely new industry that within a century replaced the city's infrastructure. The qualitative leap — from “burn coal for heat” to “collect what gets released in the process and put it to work.”

Bitcoin in this metaphor is Darby. The first fire. An incredible achievement that opened an era, but single-purpose: all the computational power released as the network does its work is burned to no purpose, like the gas going up into the sky over Darby's blast furnace in Coalbrookdale at the start of the eighteenth century. Qubic in this metaphor is Murdoch. The moment when people start collecting the gas into a pipe. The Coalbrookdale of the digital age — that, probably, is what historians of technology will call this phase a hundred years from now, if there's anyone left to do it.

That phase is already arriving. About its inner workings — the next part of the book.

Its first chapter is about the 676 voices that hold the network up.

Mechanics

PART III

The Mechanics

CHAPTER 6

676 Voices



The number 676 isn't aesthetic. It isn't pretty. It isn't symbolic. It's the exact number of participants who simultaneously make up the main load-bearing structure of Qubic at any given moment in time. Six hundred and seventy-six computers, connected into a single network and voting every few fractions of a second on one and the same question: "what will be the next state of the whole system."

If you want to picture this construction in a single image, picture a parliament with 676 seats. The parliament has a rule: to pass any decision, you need to gather no fewer than 451 votes in favor. Four hundred fifty-one is just over two-thirds of 676. Less than that — the decision doesn't pass.

The parliament has no speaker with special powers. It has no factions with privileges. No party discipline. No lobbyists whose role in the corridor differs from the role of deputies on the floor. Each of the 676 has exactly one vote, and that vote can't be bought, bought out, pooled, or converted into two votes, no matter how much work that deputy did over the past week.

Once a week, the composition of the parliament partially changes. Not entirely — gradually, by a fixed rule. Those who showed the worst result over the past week yield their seats to those who, over the same week, performed more useful work and are now claiming entry. The number of deputies replaced at one time is rigidly capped — it can't be inflated by a one-off decision, and it can't be reset to zero. The parliament refreshes on a regular rhythm, the way leaf cover turns over, only not in autumn but every seven days.

That, at the level of a rough analogy, is how the Qubic network is built.

The analogy, as always, has a limit beyond which it shouldn't be stretched. 676 computers don't argue, don't deliver speeches, don't vote against their convictions under pressure. They execute a precise sequence of cryptographic operations in which a "vote" is a signature and a "decision" is the new state of the network being accepted. But the basic principle — a fixed small circle, a two-thirds threshold for any change, regular partial rotation — is exactly the one described above. That's the frame to start from.

There's another, more organic metaphor worth mentioning here once and returning to closer to the end of the book. Beneath the mushrooms in a forest ecosystem lies the mycelium — a network of fine threads stretching tens of meters in every direction. My-

celium has no center. It has no owner. Individual sections of it die off and regenerate continuously; the whole, meanwhile, goes on existing longer than any of its fragments. When I first saw a diagram of Qubic's 676 computers next to a diagram of mycorrhiza, I was struck by the topological similarity. This isn't mysticism, and it isn't poetry; it's simply a statement of the fact that distributed systems without a center — biological or engineered — converge on the same architectural solutions.

This chapter is about the mechanics. How the 676 actually vote. Why any decision requires a threshold of 451, rather than, say, 339 (a simple majority) or 676 (unanimity). What a tick is — the basic time unit of the network in which all of this manages to happen. How the weekly rotation of membership works, and why it's precisely this rotation that makes the network structurally resistant to capture from within.

In the previous chapter I compared Qubic to Murdoch in 1792 — the man who first collected the gas being wasted from blast furnaces and channeled it into a pipe. The 676 voices aren't Qubic's "gas pipe" itself. The gas pipe is the next chapter after this one, on useful computational work. But for the pipe to hold and not be captured — first you need a construction that holds it. Floor, walls, foundation. Coalbrookdale of the digital age, day one: assembling the frame.

Let's start with the threshold of 451.

A Quorum of 451 out of 676

If a parliament has 676 seats and any decision requires a minimum of 451 votes in favor, then what a hypothetical coalition capable of coordinating would hold is not a majority of seats but a supermajority. A majority is when 51 out of 100 back you. A supermajority is when two-thirds back you. Formally: $2/3$ of 676 is 450.67. Round up, because a vote is a discrete quantity: there are no fractional votes. You get 451.

Where did the two-thirds logic come from in the first place, rather than, say, three-quarters or a simple majority of 50% + 1?

Two-thirds is the classical threshold in distributed systems where some participants may behave dishonestly or simply fail. It came from a field that computer researchers began developing in the late 1970s under the name "the Byzantine generals problem." The name is conventional; it's taken from a parable about a group of Byzantine generals who, having surrounded a city, must agree on a common plan (storm or retreat) through messengers, knowing that some of the generals may be traitors and will pass on contradictory messages. The essence of the problem: at what maximum number of "traitors" in the group are the honest participants still able to arrive at a single correct decision?

The mathematical answer turned out to be hard. If traitors can number up to one-third of the group, the honest two-thirds can work out a common decision that the traitors are unable to derail. If there are more traitors than one-third — no protocol trick can save the situation; the group is in principle incapable of guaranteed agree-

ment. Two to one is the mathematical boundary of a distributed system's resilience to dishonest or failed participants.

A natural question may come up: why not get by with a simple majority, that is, 51%? In ordinary voting in a live parliament, where everyone voting is in the same hall and can ask one another to repeat themselves, a simple majority works fine. In a distributed network, connected only by messages, things get more complicated. Imagine for a second the following scenario. 339 computers sent in their votes for one version of the new state; another 337 voted for a different, conflicting one. Who's right? If there are 226 malicious participants in the network, a situation is theoretically possible in which those 226 sent half the network one vote and the other half the opposite vote. Each part of the network sees a local majority for its own version and accepts it as canonical. The network splits into two parallel worlds, each of them confident of its own rightness. This is the classical "Byzantine" problem in action. The two-thirds threshold eliminates this possibility: under any arrangement, if 451 participants voted the same way, a second group of 451 in the network physically cannot exist, because $451 + 451 = 902 > 676$. Therefore, any decision that gathered 451 votes is unambiguously canonical. An alternative branch, having pulled together the same level of support in parallel, doesn't exist by arithmetic.

This boundary is wired into the Qubic protocol literally. Of the 676 computers — those same "parliamentary deputies" — up to 225 participants can be unavailable, broken, caught in a network partition, or colluding against the network at the same time. That's less than one-third. The remaining 451 are already a supermajority; they continue to make decisions, and any decision they make is canonical for the network. But if 226 or more go against the network, then the network, by protocol, cannot reach consensus; it freezes, waiting for a workable mass to be restored. It doesn't start accepting "alternative" blocks, doesn't split into warring branches, doesn't drift into a state from which it would later need to vote on its own rescue. It stops. This is, in a sense, a safety technique opposite to Bitcoin's: Bitcoin keeps working even under a strong concentration of hashrate, just with degraded protection. Qubic either works with guaranteed protection, or doesn't work at all.

Now an important, subtle, often misunderstood detail.

Each computer has exactly one vote. Not "one vote per unit of capital." Not "votes proportional to a computer's power." Not "the more I earned in the previous epoch, the more votes I have in this one." One participant — one vote. It's a protocol rule that can't be circumvented, no matter how many resources one participant has accumulated.

This, if you step back a moment, is the main structural difference between Qubic and systems in which the weight of a decision is proportional to the resource invested. In Bitcoin, a large pool that has concentrated a big share of the hashrate gets a proportionally larger probability of determining the contents of a block; the more equipment you have, the more weight you have. In classical Proof-of-Stake, a large coin holder gets a proportionally larger probability of becoming a validator; the more money you have,

the more weight you have. In both models, accumulated resource is converted into the right to participate in shaping the rules of the next step. In Qubic this conversion is forbidden by the protocol. A computer with a huge fleet of servers and a computer with the minimum configuration that still lets it hold its seat have, in every vote, equal weight — one vote. The only thing many resources buy you is a chance to stay among the 676 and not be displaced in the next epoch. But while you're in the 676, your weight is exactly the same as anyone else's.

Here, I think, the attentive reader will recall one of the conditions from chapter three. We formulated it back then in these words: a win in the game must not give the player the ability to change the rules of the game itself. The formulation was abstract; this is its concrete engineering embodiment. Within a single vote, no accumulated advantage converts into votes. Let's figure out what, exactly, gets voted on.

The Tick: The Moment at Which the Network Makes a Decision

In ordinary life we think of time as continuous. Week passes into week, day into day, with no clear seams. Distributed networks can't do that. For 676 computers scattered across the planet to arrive at a common decision about "what is the network's current state," they need a common discrete time grid. They need moments when all of them simultaneously look at the world and simultaneously vote.

In Qubic this unit is called a tick.

A tick is one completed round of voting. Roughly the following sequence happens in each tick. First, one of the 676 computers is selected to play the role of "leader" for this tick — it gathers the transactions that came in over the previous micro-interval, orders them, and proposes the resulting bundle as the network's next state. Then its proposal is broadcast to all the other 675. Each of them, on receiving the proposal, checks it against the protocol — the transactions don't contradict each other, the leader had the right to propose anything, the signatures are correct — and votes "yes" or abstains. When 451 "yes" votes are collected, the proposed state is accepted as canonically new. The network has taken one step. The next tick begins.

It's worth emphasizing that the "check" here is not a formality. Each computer holds a complete current picture of the network: which addresses exist, how many coins are on each one, which obligations are already pending, which transactions were accepted in previous ticks. When a proposal from the current leader reaches it, it itself, with its own code, applies all the proposed transactions to its own state and sees whether it gets the same result the leader claimed. If it does — it votes "yes." If not — it doesn't vote. For a malicious leader there's simply no way to push through a false proposal: 675 independent verifiers will run his calculation through themselves at the same time and independently see the discrepancy. The only proposal that survives is one that's objectively correct by the rules of the protocol.

Two key points of this mechanic are worth singling out.

First — the leader role isn't fixed. It rotates by a deterministic rule known to all 676 participants in advance. This isn't an "auction for the right to propose a block," as in Bitcoin, where the contents of a block are formed by whichever miner first happened to solve a cryptographic puzzle — and that means, with high probability, a large pool. In Qubic, for each tick the leader is known in advance and changes for the next tick. No computer can stably control the flow of proposals; the most it can do is be the leader in its own tick, and in the next one the next computer in the order replaces it. Over a thousand ticks, each of the 676 computers gets to be leader roughly the same number of times.

Second — the decision requires the agreement of a supermajority, not the leader. The leader doesn't make the decision. The leader proposes. The decision is made by 451 votes. If a leader tries to propose something incorrect — a transaction that breaks the rules of the protocol, or a block with a double spend — then 451 honest computers simply won't vote in favor. The proposal won't pass; one tick later, its place will be taken by a proposal from the next leader, and the network's work will continue. A spoiled leader can't hold up the network for longer than one tick.

If you translate this into the language of more familiar distributed systems — in Bitcoin every block is formed by exactly one miner (more precisely, by exactly one pool), the one that got lucky finding a hash with the required prefix. The rest of the network can accept that block or reject it, but it's always one entity that forms it. In Qubic every decision is formed collectively: the leader only puts forward a hypothesis; the supermajority approves it. The difference is in where the decision is concentrated. In Bitcoin — at the point of formation. In Qubic — in distributed voting across the whole network.

The exact frequency of ticks — exactly how many of them happen per second — is set by a protocol constant and described in the Qubic technical specification. What matters for this chapter is one thing: there are many ticks per unit of time, the count runs into single digits per second and higher, and each tick is a full round of voting by 676 participants. On the scale of a single day, the Qubic network makes tens or hundreds of thousands of collective decisions, each of which has gone through the quorum of 451.

Now to the most interesting part. Imagine you wanted to capture such a network. What would you need to do?

Rotation: Why the 676 Can't Be Captured from Within

To control the Qubic network, it isn't enough to buy more equipment than your competitors. It isn't enough to rent more cloud than the strongest computer. It isn't enough to band together into a pool. None of these moves will increase your weight in voting — all 676 votes are equal.

What you need is to take as many of the 676 seats as possible. Ideally — 226, to paralyze the network (by making a quorum impossible), or 451, to start forming your own decisions.

This is where the weekly rotation mechanism comes into play.

Time in Qubic is broken into weekly intervals called epochs. An epoch is a unit larger than a tick. Within a single epoch, the composition of the 676 computers is fixed: whoever made it into the epoch works in it until the end. At the boundary between two epochs, part of the composition is replaced: the worst by the results of the past epoch yield their seats to the best of the candidates who have submitted an entry application. “Worst” and “best” — by a single metric: the volume of useful computational work performed over the epoch. As to what “useful work” is in Qubic and why it’s even possible, that’s the next two chapters of the book; here it’s enough to know that each computer, at the close of an epoch, has an unambiguous numerical assessment of its contribution, and that assessment is what decides whether it stays or leaves.

An important structural fact: no more than 225 participants can be replaced at the same time in one epoch transition.

Not 300. Not 400. Not 676. The maximum is 225 out of 676. It’s a protocol constant.

Let’s see what follows from this.

Imagine you’re a coordinated group that wants to capture Qubic. You’ve prepared 676 of your own computers, optimized them for maximum performance, and you’re ready to take all 676 seats with your nodes. How many epochs — that is, weeks — will it take you?

In one epoch transition, you can bring at most 225 of yours into the network. Fine; after the first Wednesday you have 225 seats out of 676. The remaining 451 are still the old ones, not yours. At the next epoch transition — at most another 225. Total: 450 out of 676. Still not 451. At the third transition — another 225. Total: 675. At the fourth — one last one. Four weeks minimum, even under absolutely ideal conditions for you, when you take exactly the maximum allowable number of seats every epoch.

But even this is the most favorable scenario for you, and it assumes that in each of four epochs in a row all 225 available seats were filled by precisely your candidates and no one else. It requires that, in performance, your 225 computers be better than all the other honest candidates in the world, simultaneously, week after week. If even in one of those epochs you lose the competition by even fifty positions, your 175 won’t give you the step you need, and the reckoning shifts. If you lose in each epoch by even a hundred positions, the takeover stretches out to two months and more, and all this time the network stays in the hands of the old supermajority.

But this is the hypothetical minimum. In reality you have no guarantee that your 225 candidates will land at the top by work performed. Other computers are working too; they’re submitting entry applications too. Your candidates compete with them not on the basis of “whose team is whose,” but on the basis of who has more useful output for the week. You have to not just prepare 676 machines but make it so that every week

your machines steadily outplay all the other applicants in the world. Four weeks in a row. Without a single miss.

And now let's recall the main thing: as long as you hold fewer than 451 seats, the network is still controlled by the old computers. Any decision goes through the quorum of 451; while you have 225, or 450, or 450 — that decision isn't made by your people. The old network keeps working normally. The old computers keep noticing the newcomers coming in. If they notice that a coordinated pool of nodes behaving the same way is entering the network, they can react. At the protocol level, no — they have no mechanism to “block the suspicious one.” But at the level of the community watching what's happening, yes. Qubic has a public market, public communication channels, public technical decisions about the roadmap. Over four weeks of a coordinated attack, the community will have time to see this, discuss it, and possibly respond — with a technical upgrade, a change to the selection rules, the recruitment of new honest computers.

There's also an arithmetical symmetry worth noting. 676 minus 225 equals 451. Exactly the number needed for a quorum. This is no coincidence. It's a form of symmetry chosen consciously by the developers: even if you imagine the maximally extreme scenario in which all 225 allowable seats have been replaced in a single epoch transition and all of them are malicious, the remaining 451 old, honest computers still hold control over all decisions of the epoch. They make quorum without a single vote from the newcomers. That is, even under the maximum simultaneous invasion, the network doesn't lose the ability to make decisions on the strength of its prior composition.

This is the formulation from which the structural resilience of Qubic follows. Not “the creators hope that no one will attack.” Not “decentralization is our value.” But mathematically: under any sequence of actions permitted by the protocol on the part of an attacker, the network preserves the ability to operate as before for at least one full epoch per every 225 captured seats. And during that epoch, a real network of real people has time to see, discuss, and, if necessary, respond.

Here, at this point, it's worth restating once more what we discussed in chapter three in connection with GHash.IO — the pool that in the summer of 2013 amassed 55% of the Bitcoin network's hashrate. There the gap between concentration and unblocking the network played out within one turn of the cycle: today 55%, tomorrow a block with a double spend — technically possible. What stopped it was only public pressure, the departure of part of the miners from the pool, reputational damage. There was no mechanical brake.

In Qubic the mechanical brake exists. It requires no public pressure, no voluntary decisions by participants. It's built into the protocol by the constant “no more than 225 replacements per epoch transition.” Even if the entire community kept silent while an attacker methodically built up its presence month after month, it would still take it no fewer than four weeks to reach 451 of its own seats, and the whole time the network would keep working normally.

This is the very structural mechanism I was building toward in chapter three when I said that mining must produce decentralization, not concentration. A mechanism of negative feedback, which Bitcoin didn't have and which, if you leave Bitcoin as it is, won't arise on its own.

Why 676, Specifically

The attentive reader who has read this far has surely already asked themselves: why 676? It looks, to put it mildly, unusual.

The first answer: 676 is 26 squared. A square number. A number convenient for representation as a 26×26 table, where each computer occupies one cell and has a row number and a column number. This indexing, as the Qubic technical specifications show, allows the protocol's internal algorithms — those that handle the distribution of tasks among participants, the routing of votes, the exchange of intermediate data — to work significantly more efficiently than if computers were simply numbered 0 through 675. This is a computational argument, and it entered the network by way of engineering considerations, not aesthetics.

But the main answer is not in the square of 26. The main answer is that 676 turns out to be the golden mean across three conflicting requirements.

The first requirement is diversity. The more participants in the parliament, the harder it is for a single group to capture a majority. From the standpoint of this logic, the bigger 676 is, the better. 10,000 computers would give substantially greater resistance to concentration.

The second requirement is mutual verification. Each of the 676 checks proposals coming in from each of the remaining 675. In every tick. At a frequency of several ticks per second. If there were 10,000 computers, each one would have to check 9,999 others in every tick, and that would create a computational load that ordinary equipment simply couldn't handle within a single tick. The network would slow down. From the standpoint of this logic, the smaller 676 is, the better. 100 computers would be able to check each other even faster.

The third requirement is fault tolerance. In any physical network, some nodes are unavailable at any given time — because of power outages, internet provider issues, an update that went sideways. For the network to always have 451 working computers for a quorum, 676 has to be noticeably greater than 451. If the network had only 451 computers (right at the quorum), then any one of them going down would stop the network. The more computers above the minimum quorum, the larger the margin of safety. From this standpoint — the bigger, the better.

676 is a compromise. Enough to provide diversity of participants and a reserve for some of them failing. Few enough that each computer can in real time monitor the behavior of each of the remaining 675. The square of an integer, so that internal al-

gorithms work more efficiently. Double two-thirds (451 and 225), so that the construction has the symmetry we discussed above.

If the developers had taken 700, the $2/3$ symmetry would have disappeared ($700 \times 2/3 = 466.67$, rounding gives 467, and the formula “maximum replacements per epoch equals the difference between 676 and the quorum” wouldn’t work as neatly). If they had taken 625 (that’s 25^2), you’d get less resilience at comparable computational complexity. 729 (27^2) — more verification load without proportional gain. 676 is the point where mathematical, computational, and engineering requirements converge in a single number.

Two Orders of Decentralization

It’s useful to see the figure 676 alongside an analogous figure for Bitcoin.

In chapter three we mentioned that by 2026 effective control over Bitcoin’s hashrate is concentrated in the four to five largest pools, which together hold around 70–75% of the network’s computational power. Foundry USA — about 29%, AntPool — about 21%, another three or four notable pools — the rest. If you imagine Bitcoin as a “parliament” in which votes are distributed in proportion to invested hashrate, then in this parliament four or five players make most of the decisions. That’s the realistic upper estimate for the effective number of consensus participants in Bitcoin in its current state.

676 to 4–5 is not “a hundred times more.” It’s a different order of magnitude. It’s a structural change in the nature of the network: from one where the big decisions are made by a handful, to one where they’re made by a mid-sized parliament.

If you look at this difference through a metric that the crypto community sometimes calls the “decentralization coefficient” (Nakamoto coefficient — the minimum number of participants whose union is capable of attacking the network), then for Bitcoin in 2026 this coefficient equals 3–4. The union of the three largest pools gives the attackers 51%. For Qubic, this coefficient is by construction equal to 226. Two hundred twenty-six is the number of computers that have to be coordinated simultaneously to paralyze the network. Not 3. Not 4. Two hundred twenty-six.

The difference between “three coordinating players” and “two hundred twenty-six coordinating players” is the difference between “possible to come to terms in one evening in one room” and “impossible without a serious organizational structure that itself becomes publicly visible long before the moment of agreement.” Twenty-six people you can still seat at one long table. Two hundred twenty-six — you can’t. Any attempt to assemble such a number of coordinating participants becomes noticeable simply by logistics.

There’s another way of looking at the same gap. Imagine a conspiracy — correspondence, a Telegram chat, a closed meeting. The more participants, the higher the probability that one of them will turn out to be disloyal, accidentally let something slip, or leak the correspondence. For a group of three this probability is small; for a group of twenty

it's already noticeable; for a group of two hundred and twenty-six it's practically inevitable. The sociology of conspiracies is well studied; historians know that any conspiracy with more than a few dozen participants comes to light within a few weeks or months. This is a property of human nature, not of Qubic; but the construction of Qubic uses this property. It turns an attack that in Bitcoin requires coordinating four players into an attack that in Qubic requires coordinating roughly a hundred times more players. The requirements for organizational complexity grow accordingly, as does the probability of a leak, as does the inability to stay below the radar.

A qualification is in order. This comparison is not “Qubic beat Bitcoin.” Bitcoin wasn't built to have a high decentralization coefficient in 2026; it was built to launch, in 2009, in complete isolation, without investors, without a team, without an exchange, without legitimacy, the first network of electronic money in history that couldn't be shut down by a single administrative decision. It met that task. The structural detail by which hashrate consolidates over time is the price paid for an architecture in which anyone can join or leave at any moment. That price was invisible in 2009 because no one was there yet. By 2026 it had become noticeable.

Qubic was built from a different starting point. By 2022 the Bitcoin lesson was in plain view for everyone. Qubic's developers had the option of laying in a structural anti-consolidation mechanism from the very beginning, at the cost of the network not being, in principle, a “anyone can become a computer at any moment” network — you have to pass through epoch selection and take a seat within the 225 available. This is a tradeoff: openness to arbitrary connection in exchange for guaranteed distributedness. Whether to throw this tradeoff one way or the other is a question that's not only technical but historical. In 2009, one option was chosen; in 2022, with the experience of 2009–2022 in hand, the other was chosen.

In chapter three I formulated this as a structural requirement. For cooperation in the network to be sustainable, a win inside the network must not be convertible into the right to change its rules. Bitcoin doesn't provide this isolation: accumulated hashrate is converted into a share of voice in choosing a block, and through support for this or that client, this or that developer, this or that direction of development — into a share of voice in choosing the future trajectory of the protocol, too. Qubic restores this isolation: one participant — one vote, no accumulation increases his weight in the votes of the current epoch, and the ability to become a participant in the next epoch depends not on accumulated capital but on work performed in the previous one.

At this point it's worth saying separately that “useful work performed” in Qubic is not the same kind of work as in Bitcoin. In Bitcoin, work is the solution of a meaningless cryptographic puzzle whose only function is to prove that you've spent electricity. In Qubic, work is something different. It's that very “Murdoch's gas pipe” I promised in earlier chapters. But how it's built is the topic of the chapters that follow, not this one. Here it's only important to fix one thing: selection into an epoch happens by a metric tied to a contribution useful to the network, not to an abstract expenditure of re-

sources. A computer that has spent a great deal of electricity on meaningless hashes wins nothing by it in Qubic. It has to perform work the network needs. Exactly which work — we'll see in chapter eight.

And one more remark before closing this chapter. The whole 676 mechanism we've described here is one side of Qubic. The second, no less important — is what counts as “useful work” and how its result reaches the hands of users. 676 computers without useful work are just a distributed counter. Useful work without 676 computers is distributed computation without a guarantee that someone hasn't substituted the result. Together, they're what Qubic was conceived for, and it's their union that makes the network something more than the sum of its parts. But first the frame holding the whole construction had to be described.

What's Next

To sum up.

The Qubic network rests on a fixed pool of 676 computers simultaneously playing the roles of observers, verifiers, and voters. Any decision requires the support of at least 451 participants — that's two-thirds, the mathematically calibrated boundary beyond which the network is resilient to the collusion or failure of a third of its members. Within a single vote, each of the 676 participants is equal to every other; accumulation of resources doesn't convert into voting power. The unit of time is the tick; in each tick one proposal is considered from one of the 676, rotated deterministically. The unit for refreshing membership is the epoch; once a week part of the membership is replaced, but no more than 225 at a time, which guarantees that any coordinated invasion will take at least four epoch transitions, and the whole of that time the network will be controlled by the prior supermajority.

This construction is the foundation. Everything else rests on it that the remaining chapters of part three will be about: how useful work turns into a defensive structure for the network, and how the token is issued and burned, and how the last remaining point of human governance works — the figure of the Arbitrator, who, by plan, is to disappear on April 13, 2027.

But before talking about what gets built on this foundation, it's worth closing one question that was left open in this chapter.

We said that once a week part of the computer composition is replaced. That the “bad” yield their seats to the “good,” that no more than 225 out of 676 are replaced at a time. But the transition itself — the very moment when one epoch ends and another begins — has so far been described from the outside, from the standpoint of statistics. Who, exactly, ends up among those leaving, and by what rule? How does the selection happen among the candidates for the open seats? At what point in time does this happen — what if during the transition someone sent a transaction, does it go to the old epoch or the new one? What happens to the unfinished obligations of a departing com-

putor? How does the network manage not to stop in the moment when a quarter of its voting members is changing?

These are questions about time in Qubic. About how the network exists not in a continuous present but is divided into discrete weekly cycles, each of them a small life of its own, with its own birth, development, and end. About the mycelium that regenerates not once a year but every Wednesday evening, and meanwhile keeps living as a whole.

The next chapter is about weeks of epochs.

CHAPTER 7

Weeks of Epochs



If chapter six described the network's anatomy — what it's made of — this one describes its physiology: how it breathes, how it metabolizes, how it regenerates.

Recall the metaphor I carefully introduced in chapter six and promised to return to. Mycelium — a distributed network of fine threads stretching underground without a center and without an owner. Its individual sections continuously die off and are replaced by new ones; the whole, meanwhile, lives longer than any of its fragments. Chapter six ended with the promise that the weekly turnover of computers is — not metaphorically, but mechanically — the same thing. That promise will be kept here.

Qubic is the frame on which a structure rests that has yet to be described; the gas pipe gets assembled in the chapters that follow. This one, the seventh, is still about the frame — but about its temporal dimension. Coalbrookdale of the digital age, day two: the floor laid down the night before is set onto supports that have to bear not only weight but also what flows through them.

What an Epoch Is Made Of

Take an arbitrarily chosen week in the life of the Qubic network — say, this one, from Wednesday evening to next Wednesday evening. What happens in it?

The composition of the six hundred and seventy-six computers is fixed. These specific six hundred and seventy-six — no fewer, no more — make up, for all seven days, the parliament we talked about in chapter six. Not one of them can be added in the middle of the week. Not one of them can be officially expelled. Regardless of whether the computer is working or has broken down, voting honestly or trying to sabotage, getting rich or going broke — for all seven days it holds its place in the composition.

Within the week, time is sliced into ticks — that very basic round of voting we've already discussed. Ticks are many: in the current configuration, one tick takes around half a second to a second, which means there are roughly a million of them in a week. Each tick is a full round: one computer-leader proposes, the remaining six hundred seventy-five verify and vote, a quorum of four hundred fifty-one makes the new state canonical. A million such rounds in a row — and one epoch has been lived through.

It's important to understand what, exactly, the epoch accumulates. This isn't just a journal of what happened; it's the network's structured memory. Each of the six hundred and seventy-six computers, at the close of the week, is assigned a numerical evaluation — sometimes called its “score.” That score measures how much useful work the computer performed over the week. What “useful work” is in Qubic is the topic of the next chapter, I promised, and the promise will be kept; for now it's enough to know that this work has an objective metric, equally computable for all participants. At the end of the epoch the network has a ranking: six hundred and seventy-six names, each with a number.

In parallel, there's another list. These are candidates — computers that aren't part of the current six hundred and seventy-six but, over the course of the week, have also been performing valid work and accumulating their own score. They don't vote, don't take part in consensus, but the network sees them and counts them. At the end of the epoch the network has their ranking too — a separate one.

That is, an epoch isn't just a week during which the network operates. It's a week during which the network simultaneously operates and observes itself. All six hundred and seventy-six “inside” are constantly under measurement. All the candidates “outside” too. When the week ends, the network has at its disposal not only the blockchain of transactions but also a complete ranked list of who worked how.

Within an epoch none of these rankings goes into effect. For all seven days, the rules of the game are the same for everyone. No computer, no matter how high a score it has accumulated by midweek, gets any additional vote because of it, nor any additional transactional priority, nor even an informational advantage. Within an epoch the network is symmetrical with respect to all six hundred and seventy-six. I want to underline this separately, because this is where one of the subtlest things in the whole architecture of Qubic happens.

What does “the epoch as one round of an iterated game” mean? Axelrod's tournament, which I worked through in chapter two, was made up of games: each pair of programs played against one another not once but two hundred rounds in a row. Only that structure allowed a cooperative strategy to win — because the opponent remembered past behavior and responded to it. A cooperative strategy doesn't win in one round; in a thousand rounds, it wins.

An epoch in Qubic is one such round, only with six hundred and seventy-six players at the same time. Each of their weeks of behavior becomes one cell in the overall results table. In the next week, conditions may change — the composition will be different, rules may be refined — but the changes aren't random; they follow deterministically from what happened in the previous epoch. The ones replaced will be precisely those who showed the worst result; the ones admitted will be precisely those who showed the best. No one appoints “the bad ones”; the process itself determines them.

This is sharply different from how Bitcoin lives. Bitcoin has no discrete epochs. The network operates in continuous time: block after block, a new one every ten minutes.

Changes to the protocol happen through forks — through public, sometimes painful processes in which part of the community accepts the new version and part doesn't, and sometimes the network splits. Bitcoin is a long, long sequence without seams, in which any structural change requires opening it up. Qubic lives differently. It has seams, and they fall every Wednesday.

What Happens on Wednesday

What exactly happens at the moment of an epoch transition I'll now describe step by step. But first — one warning. I'll be describing the process as a deterministic algorithm; that's true, but it isn't the whole truth. In the real history of Qubic, in the four-plus years since the first epoch in April 2022, this algorithm hasn't always run smoothly. Why, will become clear later in the section; for now, let's assume the algorithm works. So, the sequence.

A few ticks before the epoch boundary — call it moment T — the network locks in the final scores of all six hundred and seventy-six active computers. It ranks them from bottom to top. The list is known: these names worked best, these worked worst.

In parallel, the network already has a ranked list of candidates. These are the very computers that were performing valid work all week but didn't belong to the six hundred and seventy-six. Their score is an objective parameter, equally computable, based on the same metric as for the active ones. The two are competing on a single ruler, not on different ones.

At moment T — more precisely, at a stipulated moment on Wednesday UTC — the network compares. If the best external candidate has a score higher than the worst active computer, they swap places. The worst is dropped, the best is admitted. The process repeats: now the next-worst of the remaining six hundred and seventy-six is compared with the next-best of the externals. And so on.

But not infinitely. At the two hundred and twenty-fifth replacement, the process stops, even if the next external candidate is still formally better than the next active one. Two hundred twenty-five is a rigid ceiling, we worked through this in detail in chapter six; it's structural protection of the network against too rapid an invasion.

So: if it has turned out that one hundred and twenty computers really did work badly, all one hundred and twenty of them are dropped. If three hundred such ones turned up, only the two hundred and twenty-five worst out of the three hundred are dropped; the rest stay another week, under the promise of doing better. And if there were only fifty such, all fifty are dropped, and no one fills slots up to two hundred and twenty-five; openings for new entrants don't appear in such numbers.

A new epoch begins. The composition of the six hundred and seventy-six is refreshed. Last week's score is recorded in the blockchain as historical fact; the network remembers it, but in the new epoch it confers no privileges. The new week is a clean slate by accumulated work. Everyone starts again.

The selection mechanic isn't manual. Nobody from the Qubic team sits on Wednesday morning with a spreadsheet deciding whom to swap. The process is deterministic, executed simultaneously by all six hundred and seventy-six nodes independently. Each computer, knowing the full list of ticks of the past epoch, can compute the score of each participant on its own — both internal and external. If all six hundred and seventy-six computed the same thing — and they should, because the input data are the same for everyone — the new epoch begins automatically. No central decision.

In CfB's descriptions, the mechanism by which an active computer officially loses its status sounds like this: a computer behaving incorrectly or failing to keep up with its load, at some moment, starts emitting cryptographically invalid ticks. These invalid ticks become their own proof of its problem — they get included in the blockchain as evidence the whole network sees. At the epoch boundary, such a computer automatically loses its place. Its replacement isn't carried out by people, or even by the Arbitrator — it's carried out by the computers themselves through the protocol, processing the proof.

Separately, on transitions. An epoch change isn't an atomic operation. It doesn't happen in one microsecond. The actual procedure, as far as is known from the team's descriptions, is stretched out over hours. In the early epochs, in 2022, when gradual transitions were being tested, a change could take six hours — from 9:00 UTC to 15:00 UTC; mining would stop for the entire interval, and the network would “reassemble” itself. By the end of 2024, gradual transitions had been replaced with so-called seamless transitions — transitions without a full halt. The network keeps ticking, but at some moment the internal computer list at each node updates — and the next tick goes through with the new composition. How exactly the synchronization happens, I'll work through a paragraph from now.

Transactions crossing the epoch boundary aren't lost. If a user sent a transaction a minute before Wednesday and it didn't make it into a block of the old epoch, it goes into the first possible tick of the new one. From the user's standpoint there's no difference at all between a transaction “Wednesday before noon” and “Wednesday after noon.” For the user, the network is one and the same. Only inside, under the hood, has it slightly changed the composition of its voters at the moment of the transaction.

Why Wednesday, specifically. The answer is simple and pragmatic. Wednesday is the middle of the working week in most time zones of the world where Qubic has operators. It's a day when the maximum share of computers is most likely to be online and monitoring what's happening. Monday and Friday are tied to start- and end-of-week effects — when operators haven't yet fully settled into work mode or are already partially distracted by the weekend. Saturday and Sunday — fewer people at the consoles. Wednesday is a reasonable compromise, nothing mystical.

Why a week, specifically. Here the answer is a bit more interesting. Too short an epoch — say, a day — would make the score evaluation statistically noisy: random fluctuations in performance would produce equally random ejections, and the composition

would jitter without substantive cause. Too long — say, a month — would mean a slow reaction to bad participants: a malicious or simply broken computer could damage or impede the network for months before being replaced. A week is an empirically found compromise. It isn't an "optimum" proven mathematically; it's a reasonable value at its design point.

Here, by the way, it's worth remembering that Qubic isn't unique in the idea of dividing time into epochs. Other networks with discrete time exist — Cardano uses five-day epochs, Solana's epochs are significantly shorter. The originality of Qubic isn't in the concept itself, but in the parameters and in what, exactly, gets fixed at the epoch boundary. In most other networks, an epoch is the time between recalibrations of some single parameter. In Qubic, an epoch is the full life cycle of the voter composition.

And one more observation worth lingering on. The epoch is the unit in which one round of an iterated game takes place. Here a promise left unclosed earlier begins to turn into an engineering fact: the fourth condition of cooperation — a win in the game must not give the player the ability to change the rules of the game itself. This condition now has two implementations: a static one — no accumulated wealth increases the weight of a vote; and a dynamic one — a computer that earned a great deal over an epoch gets the right to remain in the next epoch, but not the right to change its rules. Two different kinds of prize, and the division between them is wired into the epoch boundary.

How to Switch Without Stopping

The epoch transition in Qubic is not a simple engineering task. Imagine you have to swap out a quarter of a soccer team's roster mid-match in such a way that the match doesn't stop. The incoming players have to already know the tactics, know where the ball is, who passes to whom, what combinations are being run. The outgoing have to clear the field but not take with them anything the team will need going forward. And all of this — in the moment, without a whistle.

In the programmer's world the task sounds like this: at moment T we have up to two hundred and twenty-five nodes leaving consensus and the same number entering. The new ones have to know the network's full current state — the most recently accepted blocks, all open obligations, all states of smart contracts. The outgoing have to correctly hand them this state without halting the flow of new transactions. And the network at this moment must neither lose validation, nor allow conflicts, nor get stuck trying to come to an agreement.

The Qubic protocol solves this through what can be called phase-in and phase-out. An incoming candidate knows that it can become a computer on the upcoming Wednesday in advance — by its own accumulated score. This isn't a last-minute surprise; it can prepare. For several days already it's been pulling down the current state of the network, getting it not in one big chunk at moment T but gradually, in the background.

By moment T it already has everything it needs for the very first tick of the new epoch. It doesn't vote — because it isn't yet part of the six hundred and seventy-six — but it's already ready.

At the boundary T the switch happens in one step. Each of the six hundred and seventy-six independently computes the new composition. Each updates its internal participant list inside itself. From the very first tick of the new epoch the new ones are voting. The old ones, just departed, can still remain in contact for some time as observers — their votes are no longer accepted, but they see the network, can finish their technical business, can try to come back as candidates the following week. No abrupt breaks.

Where this description starts to seem too smooth, I have to stop and add an honest caveat. CfB's own commentary directly says: "Not all epoch transitions are smooth." That's true. The history of Qubic knows two different kinds of transitions. One is "seamless": no new features get rolled out, rules don't change, only the computer composition refreshes. Such a transition has, since 2024, indeed proceeded without a network halt. The other is "breaking": along with the change of composition, a new version of the protocol, new logic, new capabilities are deployed. In this case all computers must update their code, and the transition can be accompanied by downtime — sometimes several hours, and in the very early days even more.

And here's an important detail: the Qubic team does breaking transitions regularly. Every Wednesday, if there are new features for rollout that week — the transition is breaking. If there aren't any — seamless. When over the holidays, in December 2024, the team decided to do a series of seamless transitions in a row, CfB himself explained it simply: "no breaking updates which means no progress with monero." That is, seamless is stability at the cost of an absence of development; breaking is development at the cost of possible downtime. Qubic in practice constantly balances between these two modes, and from week to week the choice can shift.

This may seem strange to a reader accustomed to the crypto discourse around Bitcoin, where any hard fork is an event that triggers months of discussion and sometimes splits the community. In Qubic a hard fork is a routine midweek event. The reason is that consensus in Qubic is built differently. In Bitcoin, a change of rules requires the voluntary agreement of miners, each of whom can keep working under the old protocol indefinitely — and if enough hashrate stays on the old version, you get a parallel chain. In Qubic, hashrate and voice are separated: hashrate is needed only to land in the six hundred and seventy-six and to fight for a place among them; within an epoch the rules apply identically to all the voters, and if six hundred and seventy-six have agreed to a new version — the new version becomes canonical. A faction left on the old one simply won't enter the next composition. No fork happens.

Now a brief historical aside that explains a great deal. Seamless transitions are not what existed in Qubic from the beginning. This was a major multi-year engineering task. The first records of work on it date to December 2022; back then CfB wrote to the

team: “This epoch we should develop and test seamless epoch transition, expect daily updates.” In 2023 — a series of preparation changes. In 2024, in January, seamless mode was finally deployed to mainnet. That is, the network for two years — 2022 and 2023 — lived with transitions that sometimes took hours and stopped mining. Only afterward did what most users today see as “the invisible Wednesday change” appear.

This is important to remember for two reasons. The first is honesty: to claim that Qubic was born with a ready seamless architecture would be untrue. The second is a demonstration of method. Developing critical infrastructure in an open network with real users and real money flowing through it is work that doesn’t get done in a month. Four years ago, on the day of Qubic’s first epoch on April 13, 2022, most of what now runs automatically didn’t yet exist even on paper. The network was learning to live — and while it learned, it lived, little by little.

Four years isn’t a random number. The mainnet launch date — April 13, 2022. The planned date of the Arbitrator’s departure — April 13, 2027. Five years between them, and the coincidence of numbers is definitely not accidental. It’s too early here to discuss this date in detail; its full treatment is in chapter fourteen. It’s enough to note that the weekly cycle of epochs we’ve just described becomes the unit of count for these five years. Not calendar, but network.

Under the Hood: Symmetries and Arithmetic

Chapter six showed how the number six hundred seventy-six gives rise to the quorum of four hundred fifty-one and the replacement limit of two hundred twenty-five through one and the same BFT mathematics. It’s worth pausing on the arithmetical symmetry between these two defenses — and seeing that it isn’t decorative but functional.

The quorum of four hundred fifty-one protects the network within an epoch — at the moment of voting. It’s protection against a minority, even a large one, pushing through a false decision. The two hundred and twenty-five, by contrast, is protection between epochs, at the boundary of a transition. It’s protection against an attacker rotating the composition in his favor too quickly. The first works on the scale of milliseconds; the second on the scale of weeks.

What’s remarkable is that both defenses are mathematically the same. Six hundred and seventy-six minus four hundred and fifty-one equals two hundred and twenty-five. If the maximum number of those simultaneously replaceable equals the difference between the full composition and the quorum, then you get the following property: even under an epoch change ideal for the attacker, when his people occupy all two hundred and twenty-five seats, the remaining four hundred and fifty-one old computers make up exactly a quorum. The new ones can’t pass a single decision on their own — there are only two hundred and twenty-five of them, and at least four hundred and fifty-one votes are needed. To push through any decision, they need votes from the old.

This means that whether one likes it or not, any attempt to capture the network from within is doomed to be a drawn-out process. Chapter six already showed it arithmetically: a minimum of four epochs, that is, four weeks, with ideal competitive performance — and during all of that time control belongs to the previous composition. What’s also important is that the previous composition is not obliged to sit passively over those four weeks. It sees what’s happening, sees the ranked list of candidates, understands the structure of incoming force, and can — at the protocol level, with nothing illegal — take measures. For example, vote on a change of rules complicating mass simultaneous entry. Or change the score itself, making it more resistant to a specific type of attack. Voting on a rules change isn’t permitted within an epoch, but at the epoch boundary — as we’ve seen — it is.

There’s another nuance worth noting. The selection mechanic is built so that a rationally minded attacker won’t try to capture all six hundred and seventy-six seats in a single week. Why would he? His own mining is bounded — he can produce, say, this much score per epoch. If he scatters that score across all six hundred and seventy-six positions, he won’t take a single one: for each individual position his strength turns out to be weaker. It’s more rational to concentrate all his strength on the two hundred and twenty-five best candidates — that is, on the maximum allowed simultaneous rotation. That gives him three times the efficiency per position. But it also slows him down threefold: instead of one step toward capture he has to take three.

Put differently, the structure of the protocol doesn’t merely defend against rapid attack — it actively steers the rational attacker toward a slow one. And a slow attack is an attack visible for months. This is a fundamental difference from situations like GHash.IO in 2014, where concentration from thirty percent to fifty-five could happen in a few weeks and where there was no mechanical brake to slow the process. In Qubic the brake is built in as a constant.

A computer’s earnings don’t convert into the weight of its vote, nor into the speed of its entry into or exit from the composition. The maximum that earnings give is an objective chance to remain or to land in the composition, and that chance is bounded by the same threshold of two hundred and twenty-five, which doesn’t depend on how much, exactly, the computer earned. The winner’s prize inside the game is to stay in the game. No more.

What’s Next

To sum up. The Qubic network lives not in continuous time but in discrete cycles of one week each. Within an epoch the composition of six hundred and seventy-six computers is fixed, all vote with equal weight, every tick is a full round of consensus. At the end of the week the network has at its disposal a ranked list of all the active computers and all the candidates — by an objective metric of work performed. At the epoch boundary, automatic replacement happens: the worst yield their seats to the best, up to two

hundred and twenty-five replacements at a time, under a rigid arithmetical limit that preserves four hundred and fifty-one old computers in the new composition. Transactions go through without interruption, the network doesn't stop if the week has no breaking updates; and if it does — a brief downtime is possible, and the team consciously chooses what matters more to it on each particular week: stability or progress.

At the level of the human metaphor, the mycelium regenerates every seven days: some threads die off, some appear, the whole goes on living. At the level of game theory, every epoch is one round of a large iterated game with six hundred and seventy-six participants, and it's precisely this, as Axelrod showed in 1980, that turns cooperation from an unstable exception into a stable equilibrium.

Two engineering anchors — the quorum of four hundred and fifty-one and the replacement limit of two hundred and twenty-five — are arithmetically symmetrical, and together they make capturing the network a task requiring not hours and not days but weeks, in plain sight of the entire community.

We've now described the network's space — six hundred and seventy-six voices occupying seats — and its time — weekly cycles during which those seats refresh. But not once have we answered the question this whole book started over. What, exactly, are these six hundred and seventy-six participants doing during the seven days of their epoch? What work earns them their place in the next week — and why is it “useful,” rather than, as in Bitcoin, arbitrary? What is the force that turns this mycelium?

Not the meaningless hashes we discussed in chapter one. Not arithmetic for arithmetic's sake. Something else — what Qubic was conceived for. What in chapter one we called “Murdoch's gas,” collected into a pipe.

The next chapter is the key one in this book. It's about useful work of computation that simultaneously defends the network and produces something that did not previously exist in the world. It's called “Murdoch's Gas in Action — UPoW.”

CHAPTER 8

Murdoch's Gas in Action — UPoW

*The Gas No One Collected into a Pipe*

What, exactly, do six hundred and seventy-six computers do for the entire week? What work earns them their place in the next epoch — and why is it “useful” rather than, as in Bitcoin, arbitrary?

This chapter is the answer to that question. And it's also a return to the very metaphor I carefully introduced in chapter one and promised one day to bring to bear.

Let's go back to Coalbrookdale of 1792.

William Murdoch is standing by a furnace in which coal is being charred for the production of coke. The process by then has been in use for eighty years already: since 1709, since the time of Abraham Darby, the iron-foundry industry of England has rested on this very transformation of coal into coke. From the furnaces, every day, dense black smoke pours out — a byproduct of the charring, inconvenient, sometimes dangerous, always released into the atmosphere as waste. That's how it had been done all this time: coal yields coke, smoke flies into the sky, rain washes the soot off facades, and no one complains, because coke means cast iron and cast iron means the empire.

Murdoch looks at the smoke differently. He runs a pipe from the furnace to a lamp in his own workshop, purifies the gas, lights it. It burns. It burns evenly and for a long time. It works.

Over the next fifteen years this idea will grow into the lighting of London with gas. By the 1820s an entire gas industry will have grown up — a new economy founded on the byproduct of an old one. Charring coal now produces two valuable outputs simultaneously. The first is heavy, tangible — coke. The second is invisible and powerful — gas.

In chapter one I wrote one phrase which has hung over the book ever since as a promise. I called Bitcoin mining the Murdoch's gas of 2009 that no one has yet collected into a pipe. The process is running: computer cycles are being burned by the billions, power stations are operating, chips are heating up. For the sake of what? Of one thing — finding a digest whose first N bits are zeros. Found — the block is credited, the reward issued. All the rest of the computational byproduct gets thrown into the atmosphere. Goes off as heat. Into nothing.

In 2013 one engineer looked at this picture and saw exactly what Murdoch saw in 1792. Not “how to make a more efficient Bitcoin.” But “what if we collect the byproduct.” In 2022, on the basis of this idea, the Qubic network started up. By 2024 the UPoW mechanic, which this chapter is about, had stabilized.

Three questions to be answered. First: what exactly do Qubic miners solve every tick? Second: why does the result of this work simultaneously serve consensus and something else — and what, exactly, is that “else”? Third: does it work today as designed — or only halfway?

Let’s start with the mechanics.

One Task, Several Outputs

A Bitcoin miner solves one task. He varies the value of the nonce field in the block header, computes sha256 of the header, looks at the resulting digest, checks: are the first N bits zeros? If not — change the nonce, compute again. If yes — the block is found, transmitted to the network, the reward taken. That’s it. The whole Bitcoin-mining industry, tens of billions of dollars in hardware, hundreds of terawatt-hours per year — is brute-force search for a coincidence: a digest with the required number of leading zeros.

What’s happening inside? Each sha256 computation passes through sixty-four rounds of internal transformations, and at each intermediate step some state results — bits, register states. By construction of the function, these intermediate states mean nothing outside sha256. They aren’t data, they aren’t a signal, they don’t contain any recoverable structure. They are computational byproduct that exists for exactly as long as it takes to produce the next one. The chip computes them, the chip forgets them. At the output — a digest. One output, one value, one purpose.

And this, it’s important to say, is not a bug. It is the right design of the function for its purpose. A hash function whose task is to be a cryptographically strong transformation must not leave useful traces in its intermediate states. If sha256 produced something useful in the course of computation, that would make it vulnerable: predictability of byproducts contradicts the requirement of indistinguishability from a random function. Bitcoin isn’t “badly made.” Bitcoin is made correctly — for what it does.

But for what Qubic does, a function of a different design is needed.

Here is how the task is described by Come-from-Beyond — the creator of the system, with whom the reader has already met in chapter four. The quote is public, from his post of July 23, 2024:

“Qubic miner task is similar to Bitcoin hashing and can be reduced to finding such digest that first N bits are zeros, we set the difficulty manually (dev team chooses it every epoch) by deciding the value of N. Each computer simulates the ANN (which we have represented as a hash function during the reduction above) and generates the ‘digest’ which is verified to have first N bits set to zeros.”

Read the phrase slowly. Qubic miner task is similar to Bitcoin hashing — the Qubic miner's task is similar to Bitcoin hashing. And can be reduced to finding such digest that first N bits are zeros — and reduces to searching for a digest whose first N bits are zeros. The same task. Not similar, not analogous — formally reducible to the same one.

But further: Each computer simulates the ANN (which we have represented as a hash function during the reduction above). Each computer simulates an artificial neural network — which through a formal reduction is represented as a hash function. This is the structural difference. The digest whose first N bits are zeros is generated not by sha256 but by running data through a neural net. At the input — some parameters, inside — a forward pass through layers of the ANN (matrix multiplications, nonlinear activations, aggregations), at the output — a sequence of bits which is checked for leading zeros.

I'll underline: this is not an analogy and not a metaphor. The quote says explicitly: “which we have represented as a hash function during the reduction above” — represented as a hash function through a formal reduction. That is, it is mathematically established that the run through the ANN behaves, in the relevant properties, as a hash function, and therefore the task “find an input for which the first N output bits are zeros” has the same meaning as in Bitcoin. The difference is that the very process producing the digest now leaves traces.

The intermediate states of the ANN simulation — neuron activations, weight distributions on trained layers, the structure of how the data passed through the network — these are not empty bits inside sha256. These are data which, in the most honest sense, are neural-network data. They can be used for training. For evolutionary selection of architectures. For everything that makes up modern machine learning.

One bit of work by a Qubic miner produces three outputs simultaneously.

The first output is consensus proof. A digest found with the first N zeros is a confirmation that the computer performed a valid simulation. This is the equivalent of “found a block” in Bitcoin: a proof of work which the network can verify and credit. On this output the certification of ticks is built. This is what makes UPoW a proof-of-work in the strict sense — there is work, there is its proof, the work cannot be faked more cheaply than performed.

The second output is AI training data. The intermediate states of the ANN simulation. The concrete values on which the work was done. This data is a valuable byproduct, reserved in the project for a consumer named Aigarth — a separate component of the system that uses the obtained results for evolutionary selection of neural-network architectures. I'll talk about Aigarth in detail only in chapter eleven; here one thing matters: UPoW produces the data, Aigarth consumes it. As Come-from-Beyond himself put it on May 13, 2024: “Miners do useful PoW, not Aigarth”. Producer and consumer are two different nodes, and the separation between them needs to be kept in mind.

The third output is score. Score is the structured memory of the network which, by the results of the epoch, determines who stays, who leaves, who arrives. The content of the score can now be unfolded.

Score is the aggregated number of valid digests found by a computer over an epoch. Each digest is one successfully performed ANN simulation, each simulation is one unit of useful work. The epoch's score is the sum of these units over the week. The network doesn't just count votes or ticks; the network counts work of a specific type. The more valid simulations a computer has done, the higher its score. The higher the score, the more confidently it stays among the 676 for the next week, not falling into the 225 rotation slots.

What in chapter seven was called “the network's memory” has now received content. The memory is a tally of how many times in a week you've produced precisely the kind of work the network considers grounds for keeping you in your place. Not money paid, not proof of stake, not social reputation — but a count of valid neural-network simulations. Architecturally this too is a rarely encountered solution: “the one who stays in the game is the one who has done more work by its rules.”

Now we can come back to the parallel that hung over chapter one. Bitcoin is the Darby phase. Coal burns, coke is given, smoke goes off. Compute is burned, consensus is given, everything else gets thrown out. The scheme is complete and working; it has no hole — it just has one output. Qubic is the Murdoch phase. The same process rebuilt so that the same input produces several outputs at once. Not “an improved Bitcoin PoW” — a different industrial regime of the same computational combustion. Just as coking in 1792 was not “better burning of coal” but a phase from which the chemistry of byproducts began in England.

In 1792 no one in Coalbrookdale thought “ugh, what bad coking, before we were doing it wrong.” Coking was being done correctly even before that — for the sake of coke, which is needed for cast iron, which is needed for the empire. Murdoch simply saw that the process had a potential second output that until then had been ignored. Bitcoin PoW is analogous: it does precisely what it is required to do; its function by construction has no useful byproducts, and that is the right design for its task. UPoW isn't better. UPoW is about something else.

And here I have to pause. Before unfolding what “the system has three outputs, but one of them is so far not used at full strength” means, strict order needs to be brought to one place where the community regularly gets confused. Three different things that get called by similar words but do different jobs need to be separated. This is a short section, but without it the next, central section of the chapter won't work as intended.

Three Layers of Work: What Makes Qubic Qubic

A reader familiar with crypto may quite reasonably think: “useful PoW is no news. Flux rents out compute for scientific computations. Gridcoin paid for BOINC tasks back in 2013. Old hat. What’s the sensation?”

The answer is that none of that is UPoW. That’s PoUW. A one-letter difference, but a structural one.

PoUW – Proof of Useful Work. In existing implementations (Flux, early Gridcoin, a number of academic projects) useful work exists on top of the consensus mechanism. Inside the network classical PoW or PoS is running – for security; in parallel with it the network provides infrastructure through which external clients rent compute for scientific calculations or rendering. Two independent layers, joined by one token. Network nodes can simultaneously secure the chain and perform useful tasks – but “securing the chain” and “useful tasks” are two separate jobs, each with its own purpose.

UPoW – Useful Proof-of-Work. In Qubic, useful work is itself the consensus mechanism. Not “in addition to” and not “on top of.” The very same ANN simulation that Aigarth needs is what produces the digest that certifies the computer as an active participant in the network. One process, simultaneously doing both jobs. There aren’t two layers that need to be combined.

Come-from-Beyond on April 3, 2024, formulated this in a short phrase: “UPoW is not equal to PoUW”. Not the same thing. I’m putting this into a separate section deliberately, because substitution of one for the other is the most frequent mistake in discussions of Qubic. It’s easy to get confused, because the letter order is almost identical, and the functional difference – fundamental.

Now the second separation that needs to be made here. UPoW is one thing. External mining, in which Qubic also engages, is another.

In parallel with UPoW, two more streams of computational activity run on the network. The first is external mining. First, from May 2025 to April 2026, the network mined Monero. From February 2026, a DOGE add-on was launched.

It’s important to understand what external mining is structurally. It is not part of UPoW. It is performed by the same computers but with a different task, on different infrastructure, with a different result. DOGE mining gives the network revenue (an inflow of coins which are then converted into QU through the buyback mechanism and burned – these are details for chapter nine). UPoW gives the network consensus. These are two different jobs which a computer can perform in parallel, but which are architecturally not the same thing.

Come-from-Beyond formulated this on July 15, 2025, directly: “Too much misconception about #Qubic #UPoW on X. Mining of \$XMR isn’t a part of it, mining for #Aigarth is. Finding blocks for #Monero looks useful but it doesn’t influence certifica-

tion of validators directly.” And separately, in the same thread: “Monero mining isn’t UPoW”.

Here it’s important to go slowly. External mining “looks useful” — looks useful indeed. Very much so. It brings revenue to the network, it provides hashrate, in 2025 it even temporarily made Qubic the largest participant in the Monero network. All of that is real outcomes. But it does not directly influence the certification of validators. A computer doesn’t get a place in the next epoch for having mined a Monero block. A computer gets its place for having produced enough valid digests — for UPoW.

There’s empirical proof of this, and it’s strong. In Discord posts of April 17, 2026, Come-from-Beyond described the observed picture: out of 676 computers, around 20 do only UPoW, without XMR and without DOGE. They receive at the same time about 80% of the maximum revenue of the epoch. That means the bonus from external mining makes up around 20% of full income. This isn’t an after-the-fact estimate; it’s a direct measurement on a live network.

From this follows a statement worth saying aloud. Roughly 80% of the network’s value is produced by UPoW. Roughly 20% by external mining. And — the critical test: switch off DOGE — the network keeps working. Switch off UPoW — the network is dead. UPoW is the foundation. External mining is an annex.

Here chapter three closes. There I wrote: the mining process must itself produce decentralization. Bitcoin doesn’t do that — mining produces concentration through ASIC economies of scale, and two pools today control most of the hashrate. Qubic does it differently. UPoW produces score. Score determines who stays among the 676 in the next epoch. The process of producing digests and the process of producing the network’s composition are literally one and the same process. Not “we mine, and in parallel update the composition through some other procedure,” but “we mine — and by that very fact update the composition.” The cycle is closed on itself: work generates the right to keep working; the right to keep working is determined by how much work you have done.

This also structurally closes another promise. In chapter three I formulated Axelrod’s fourth condition: a win in the game cannot be converted into the right to change the game itself. In the architecture of UPoW this condition is fulfilled like this: a high score gives you the right to stay among the 676 for the next round of the game. It doesn’t give you increased weight of vote, doesn’t give you an enlarged share of consensus, doesn’t give you influence on parameters. Only — the right to keep playing. Won more — you stay. The rules of the game — you don’t change. A clean implementation of what Axelrod, describing the tournament, formulated as a necessary condition for the resilience of cooperation.

And a brief historical remark that fits here. On September 23, 2021, seven months before the launch of mainnet, Come-from-Beyond named in a public tweet three architectural properties of the future network: “Useful PoW, unlike #Bitcoin / 100% certainty of transaction outcome / Feeless transfers like in #IOTA”. Useful PoW — unlike

Bitcoin. One hundred percent certainty of transaction outcome. Feeless transfers, as in IOTA. UPoW was named by name even before the network existed. Not post-hoc rationalization (“we built something, and now we’re explaining what it was”); a declaration stated before the start, verifiable afterward.

A separate clarification for the attentive reader. Qubic as an engineering project has a long prehistory. In chapter five I mentioned Jinn — a research effort from the early 2010s, where the concept of useful work was first thought through as a possibility. UPoW is its realization a decade later, in another context and on a different basis. From idea to working network — a decade passed. That’s a normal pace for a nontrivial infrastructural idea.

Now, with the separation of three layers in hand, I can speak directly about what I warned of in the first section of the chapter. About that gap between built and running at full strength which the system’s creator named publicly himself.

Honest Disclosure: “Useless PoW” and Why It’s Here

By this point the reader may suspect that UPoW is too beautiful to work. He’s half right. The idea works. The implementation isn’t yet at full strength. The system’s author said this himself, twice, publicly — and the second time, in July 2025, used a word that is hard to forget.

The first quote, verbatim. On July 17, 2025, Come-from-Beyond wrote on Discord: “On serious note, current mining params are useless, currently UPoW means Useless Proof-of-Work”. And in the same thread, in a separate phrase: “just like our current UPoW”. It wasn’t critics who wrote this — the system’s creator wrote it himself, in front of witnesses, on a public platform, in a channel where the engineers and the token holders are gathered.

The word “Useless” isn’t accidental. The usual marketing register allows “not yet at full strength,” “remains to be polished,” “in the process of optimization.” Come-from-Beyond chose none of those soft formulations. He chose a word which on any reading sounds like a statement of fact: at the moment — useless. And with a second move he reinforced it: just like our current UPoW.

And this isn’t the first such statement. On December 7, 2023, a year and a half earlier, the same thought had already been voiced — a bit more softly, but with the same structure. “Right now our Useful PoW is Half-useful, because Aigarth cannot process all the provided solutions”. Right now our Useful PoW is Half-useful, because Aigarth cannot process all the provided solutions. The problem has been recorded not since last year. It has been known to the system’s creator since at least the end of 2023.

What, exactly, do these two quotes mean? I’ll work through it slowly, because this is the structural center of the chapter, and on the precision of the wording depends whether the reader will understand it correctly.

The infrastructure of UPoW works fully on all of its design goals for consensus. Computers simulate the ANN. Digests are validated. Score is assigned. Rotation of 225 places happens every Wednesday. The network is decentralized, secured, functioning. If you look at Qubic today as a blockchain — everything is in order, the consensus layer does exactly what it promised to do, and that first of UPoW's three outputs (proof for consensus) is produced in full.

The problem is not in the first output. The problem is in the second. AI training data is produced every tick — but not all of it is consumed. Aigarth, designed as the consumer of this data, cannot yet process the full volume of solutions the network supplies it. The network generates more than Aigarth eats. The difference is precisely what Come-from-Beyond calls half-useful and useless.

The reasons, in his own words, are two. The first is the throughput limit of Aigarth. The consumer is so far smaller than the producer; this is an engineering scaling task on which work is in progress. The second is the parameters of the task itself. The ANN that the computers simulate is chosen and tuned so as to provide validity for consensus. For Aigarth's needs it isn't optimized — it may be of the wrong architecture, the wrong sizes, with the wrong properties to be maximally useful for evolutionary selection of useful neural networks. This is an architectural design problem solved only by changing parameters and — ultimately — by the deploy of Aigarth in its production form.

The parallel back to coal here suggests itself. Murdoch ran the first pipe in 1792. But gas lighting for the first few years worked only in his own workshop in Redruth. London began to be lit by gas — really, with street lighting of the streets — in 1807. The gas-works industry, in earnest, as an industry, took shape by the early 1820s. Between “we can collect the gas” and “gas lights London” — fifteen years passed. Between “gas lights London” and “gas powers industry” — another ten. That's a normal pace for an infrastructural shift.

In the case of UPoW: mainnet started up on April 13, 2022. UPoW as a mechanic stabilized in 2024. The planned deploy of Aigarth — April 13, 2027. Between “the mechanic works” and “the mechanic is used at full strength” the project has reserved approximately five years. That's a working pace.

Now an important question: why is this said publicly rather than swept under the rug?

Come-from-Beyond could have stayed silent. He could have waited until Aigarth caught up, and spoken about UPoW only in the past tense: “built,” “works.” The press materials accompanying any technological project would easily allow that. None of Qubic's critics had brought “Useless PoW” to the public in the system creator's own formulation — because the formulation itself was created by him. He could have kept it to himself.

He didn't. And this is no random habit. In chapter four I described a principle Come-from-Beyond has been following for a second decade: don't sell what isn't there yet. Hiding “Useless PoW” would have meant violating that rule in the very place where it

gets tested most palpably — in the central mechanic of the central chapter of the entire project. To say it aloud is to assert the rule more strongly than any pr message could formulate it. In a world where ninety percent of token projects hide engineering problems behind marketing until the moment of collapse, putting one's own main weakness out in public is not weakness and not an admission of defeat. It is discipline.

What is Phase 3 and what does DOGE have to do with it?

First: the XMR-mining code is removed from the core. Not switched off by a flag that could be flipped back in a minute — removed completely, so that for a return a new release with restoration of the code would be required. This is a surgical architectural commitment. The Monero chapter is closed.

Second: Phase 3 of compute distribution is activated. The new proportion is roughly 80% to Aigarth training, 20% to DOGE mining. In previous phases a substantial part of compute went to XMR mining; that part is now freed and given to Aigarth. For the first time since the launch of mainnet, Aigarth has access to 100% of the compute reserved for AI work.

Phase 3 isn't "the moment when UPoW became fully useful." It's the moment when the gap between built and used began to close through scaling of the consumer. How quickly it will close, and whether it will close by the planned date, is a question for Part Five of the book. There I will be returning to questions of Aigarth's architecture and to its deploy.

From Phase 3 to the planned deploy of Aigarth — 356 days. Fifty-one weeks. Fifty-one epochs. In the terms of time introduced earlier — fifty-one rounds of the iterated game.

I'll close this section with an honest frame.

UPoW in April 2026 is infrastructure that one engineer began to think through in 2013, spent a decade implementing, and launched on mainnet in 2022. It works against all its design goals except one: the AI component so far uses part of what is produced. This is a crack between built and utilized. It exists. It has been recorded by the creator publicly, twice, with an interval of a year and a half.

This is the honest state of UPoW as of April 2026. Neither less, nor more.

Under the Hood: Numbers, Revenue, and One Architectural Bonus

Now to the gears. The mechanics of UPoW are expressed not only in words; behind every formulation of the previous sections stand specific constants which the team sets in code and which can be looked up in the Qubic repository. This section is for the reader who wants to know exactly what the working architecture is assembled from.

I'll start with the parameter N from the quote of July 23, 2024. In the code of the Qubic core this parameter is called `ADDITION_SOLUTION_THRESHOLD_DEFAULT`, and it determines how many leading zeros there have to be in a digest for the solution to count as valid.

I'll underline an important nuance. In Bitcoin, difficulty is self-regulated by the algorithm: every 2016 blocks the network looks at how quickly solutions are being found

and adjusts difficulty so that the average interval stays around ten minutes. In Qubic, difficulty is set explicitly by the development team every epoch. This is an engineering decision that makes sense. The hardware base of Qubic is the computers, and their composition for the next week is known in advance; there is no need to imitate blind self-regulation when one can calibrate the parameter for the current hardware directly. It's the same straightforwardness we encountered in chapter seven in epoch transitions: not "it'll work itself out," but "the team made a decision."

The reference implementation of the miner from Qubic is open-source and is called Qiner. The technical reader can go to the repository and look at exactly how the ANN simulation is built at the level of code — not from words but from sources. It isn't a black box; it's a public engineering implementation of what was discussed in sections 2–3 as a structural reduction. I won't be retelling the code here — that's material for a separate breakdown — but the very fact of its openness matters: the claims about the mechanics are independently verifiable.

Now to the money. How exactly does the network pay for a computer's work?

The revenue formula, recorded in PR #844 of the Qubic core and active in Phase 3:

$$\text{revenue} = \text{IPC} \times M \times (S^2 + B \cdot E) / \text{DIVISOR}$$

I'll work through what each symbol means.

IPC — Issuance Per Computer, the issuance to one computer per epoch. The constant equals 1,479,289,940 QU; this is the unit of a trillion new QU issued in an epoch, divided by 676. The upper ceiling a computer can reach for an ideal week.

S — a scaling constant, equals 1024. All factors are normalized to the range [0, S].

B — the weight of the bonus for external mining, equals 256.

M — the mandatory component, takes values from zero to S. Assembled from two parts: tx factor (weight 85%) — how actively the computer leads ticks, and oracle factor (weight 15%) — to what extent it participates in the commit/reveal cycle of oracles. This is work the network considers obligatory. The component multiplies the rest of the formula. Didn't perform the mandatory work — everything else gets multiplied by a smaller number, down to zero.

E — mining factor, also in the range [0, S]. In Phase 3 this is only DOGE shares. This is work the network considers optional. The component is added in via the term B·E, doesn't multiply the base.

I'll run through the ranges. With full mandatory work (M = S) and zero external mining (E = 0) the formula gives revenue making up about 80% of IPC. With full mandatory work and full mining (E = S) — exactly IPC, 100%. If a computer fails the mandatory work — the mandatory component collapses, and regardless of how much DOGE it has mined, it gets little.

The construction directly corresponds to what I described in the third section in words. Work on which consensus depends — TX relay, oracles, UPoW — is rigidly built into the multiplicative layer of the formula. Work on which only the network's revenue

depends — DOGE mining — is built in as a weak, additive bonus. Switch off DOGE — you'll lose around 20%. Switch off the mandatory part — you'll lose everything.

And here let's note one thing about UPoW's second output, which in the fourth section was discussed as a bottleneck. AI training data isn't paid for directly by the formula. A computer doesn't receive revenue for producing intermediate ANN states useful to Aigarth; it receives revenue for having found a valid digest — the first output — and for having performed the mandatory network work. The second output is currently pure byproduct, written into the atmosphere awaiting a consumer. Or, to flip it, the economy pays for two outputs out of three: consensus and the rotational score (through the mandatory part and through the fact of finding digests). The third, the AI component, is paid for on the credit of expectation.

There's one more property of UPoW I'll mention separately, because it isn't earned by design but by structure. Bitcoin PoW through the sha256 function is theoretically vulnerable to quantum acceleration: Grover's algorithm gives a quadratic speedup for the preimage-search problem, and at the appearance of sufficiently large quantum computers, Bitcoin mining will find itself in a new reality — not broken, but rebuilt. UPoW is built differently. The task “find an input to an ANN for which the first N output bits are zeros” doesn't reduce to any of the problems for which quantum algorithms with speedup are known. There is no quantum algorithm performing a forward pass through an arbitrary neural network faster than the classical one.

Come-from-Beyond on November 10, 2025, formulated this in a single phrase: “Unlike PoW, our UPoW isn't vulnerable to QC”. Unlike PoW, our UPoW isn't vulnerable to quantum computers.

Here the tone with which I say this matters. This is not a primary design goal. Quantum resistance was not on the list of tasks for the sake of which Qubic was conceived in 2013. It's a side property of the architecture. But such is the nature of many elegant engineering solutions: when a thing is made correctly for one reason, it turns out to be correct also for reasons its creators didn't have in mind. Murdoch collected the gas for the sake of lighting his workshop, not for the sake of having chemical works run on the same gas a hundred years later. As it happened, the collected gas turned out to be useful both there and here.

The last thing I'll say in this section is a short quote I'll be coming back to in Part Five of the book. On November 20, 2023, Come-from-Beyond described Qubic's functional map in a single sentence: “Aigarth uses Qubic's UPoW, also Qubic has everything that an AI would need: Smart contracts as brain, Oracles as sensors to outer world, Outsourced computations as Manipulators to change that outer world.”

Smart contracts as brain, oracles as sensors, outsourced computations as manipulators — and UPoW as the source of energy for everything. On this map UPoW isn't “a feature of consensus” but the energy subsystem of an AGI substrate. Like the heart in a body: it pumps, it doesn't think, but without it the rest doesn't work. Unfolding this map

is the job of Part Five; here it's needed only as a reminder of what, in the end, all this machinery is for.

Parsons, the Turbine, and the Twice-Collected Gas

There is one more figure worth calling into this chapter to close it. Not Murdoch — with Murdoch we've already worked, and we'll be working again in the finale. Charles Algernon Parsons, England, 1884.

Before Parsons, the steam turbine worked in a single stage: high-pressure steam struck a large bladed wheel, transferred to it part of its energy, and exited the other side — still with an enormous reserve of energy, which simply went off into the surrounding space. The efficiency of the first turbines was around 20%. Four-fifths of the steam's energy — out the chimney, literally.

Parsons built a multi-stage turbine. Steam passed not through one wheel but through a series — gradually, giving up at each stage part of its energy and moving on to the next. The first stage took what it was strong enough to take. The second — what passed by the first. The third — what passed by the two. And so on, in a cascade, until the steam exiting the last stage carried with it only residual pressure. Efficiency, thanks to this construction, rose from 20% to 50–60% and higher.

The principle Parsons applied to steam is formulated abstractly like this: don't collect everything in one operation, but build a cascade of operations, each of which collects what the previous didn't. Multi-stage extraction. Extraction in several stages. The idea is applicable to anything that constitutes a process with surplus energy at the output.

UPoW is multi-stage extraction applied to compute.

Bitcoin PoW is built as a single-stage turbine: compute goes in, a digest comes out, all the rest of the computational energy is lost as heat. One stage, one output. UPoW rebuilds the same process into a cascade:

The first stage — compute goes into the form of an ANN simulation and at the output yields a digest with N zeros. Consensus proof. This is the output Bitcoin also has.

The second stage — the intermediate states of the simulation, which in Bitcoin were thrown out by construction, here remain as data. AI training data. An output Bitcoin doesn't have.

The third stage — the aggregate of valid digests over an epoch is folded into a score, which determines the rotation of the network's composition. Score. An output Bitcoin doesn't have.

One input — three outputs. The engineering form of what Murdoch did with coal, applied to computational work 217 years later.

I'll underline one thing so the parallel doesn't read as metaphor. Parsons and Murdoch worked on one type of task. Not on one substance — they worked with steam and with coal respectively — but on one form of question: how to rebuild an existing process so that it produces several outputs instead of one. Neither one of them invented a new process. They rebuilt an existing one. UPoW is one more point in this series. Not

“the invention of a new PoW” but the application of an old principle (multi-stage extraction) to a new substance (compute instead of steam or coal).

This chapter is called “Murdoch’s Gas in Action” — not “UPoW Explained.” The title is chosen this way precisely because, 217 years after Murdoch ran the first pipe from furnace to lamp, the same engineering idea is being applied to the digital analog of that very gas.

The Gas Is Collected. The Pipe Is Laid.

Three sections of compressed mechanics behind us. Now a brief return to the overall view.

Qubic miners solve the same task as Bitcoin miners: find a digest whose first N bits are zeros. But their digest is generated by simulation of a neural network, represented as a hash function through a formal reduction. One job at the input — three outputs at the output: proof for consensus, data for potential AI training, score for rotation of the network’s composition. The infrastructure works against all design goals. The second output — the AI component — is so far not used at full strength; the system’s creator himself spoke this aloud twice publicly, first in December 2023, then in July 2025. Phase 3 was activated: the XMR-mining code is removed, Aigarth for the first time gained access to 100% of the compute reserved for it. To the planned deploy of Aigarth — 356 days.

Chapter six described the space of this network — 676 voices. Chapter seven described its time — weeks of epochs. This one described its energy — UPoW and what it produces. And economics always begins with energy.

From Murdoch in 1792 to Parsons in 1884, 92 years passed. From Bitcoin in 2009 to Qubic in 2022 — 13 years. Industrial cycles have compressed from centuries to years, but their internal logic hasn’t changed: first the process, then the gathering of the byproduct, then the multi-stage optimization. We, the readers of this chapter in 2026, stand at a point which by analogy with coal corresponds to somewhere around 1805–1810: the gas is collected, the first lamps are burning, but the great gas industry — is still ahead.

The KEY chapter of the first half of the book doesn’t end on a triumphant note precisely because triumph isn’t yet earned — in 2027 a test lies ahead, and now — work. Just as in 1792 Murdoch lit a lamp, but the streets of London didn’t yet know they would soon be lit by gas.

The next chapter is about the economics that keeps this machine running. We’ve described what computers do. We haven’t discussed how the network pays for this work — what exactly happens with the QU token every Wednesday, why issuance and burning go in parallel, why Qubic didn’t choose the Bitcoin model of fixed supply. The breathing token: every epoch a trillion new QU come into being, a substantial part is burned through the buyback mechanism, and supply pulses in time with the weeks of epochs.

Murdoch's gas is collected. The pipe is laid. Now let's figure out who is paying for its operation — and what the price of gas is in a world where gas makes not light but intelligence.

CHAPTER 9

The Breathing Token



Inhale, exhale, week

Let's start with a simple reminder. A Qubic epoch is one week. Every Wednesday, around twelve o'clock UTC, the network restarts, 225 of the 676 computer seats can change hands, the counters reset, and a new round begins. We covered this mechanic in Chapter 7; here it's needed as the backdrop for one specific thing: with that same restart, every week, one trillion fresh QU enters the network.

Not “about a trillion.” Exactly a trillion. This is the network's inhale.

Then, over the seven days until the next Wednesday, the exhale happens. QU are burned—through several independent paths, each with its own purpose, each with its own logic. Some are hardwired into the protocol, some depend on decisions by the computers, some depend on outside economics the network has no direct control over. By the end of the epoch, the balance of inhale and exhale gives the net change in circulating supply. In 2025–2026, this balance regularly came out negative: the network burned more than it issued.

Hence the working name of this chapter. The token that breathes. Not “deflationary,” not “inflationary”—breathing. Every week one inhale; over the course of the week, a series of exhales of varying depth. Supply pulses in time with the epochs.

In Chapter 7 I used the word “pulse” for the rhythm of the network. In this chapter the related physiological metaphor returns—breathing. That's a deliberate choice. In Chapter 8 the register was industrial—gas, pipe, turbine, multi-stage extraction; the chapter was about energy. Chapter 9 is about economics, and Qubic's economy is built as circulation, not as combustion. So physiology is the more accurate fit than industry. Heart and lungs—two rhythms of the same organism; the network's weekly rhythm shows up in both.

But I won't overplay the metaphor. The image should work, not be played with.

Now—about the contrast that hangs over this whole topic.

In 2008, the anonymous author of Bitcoin made one decision about supply, and it hasn't been up for discussion since. Twenty-one million coins, halvings every four years, final issuance somewhere around 2140. This isn't a matter of voting or a parameter you adjust along the way—it's a constant baked into the code at network launch. A Bitcoin

holder knows: however many years go by, however conditions change, supply will be that. That is the value proposition.

In Qubic, supply is debated every Wednesday. 676 computers vote on how much QU should remain in circulation by the end of this epoch. They can decide to burn more or less. They can—in theory—decide to change emission parameters for the next epoch. Everything having to do with supply lives inside a rolling voting process, not inside the up-front decision of one person.

This isn't a small technical difference. These are structurally different answers to the question "what is money." We'll come to a full statement of the difference in Section 6 of this chapter, when we work through the central quote from the system's creator.

Three questions:

First: where do QU come from? Section on emission.

Second: where do they go? Section on the seven burn paths.

Third: who decides, and how, how many should be in circulation? Section on the computer vote, the link to Axelrod's fourth condition, and the Arbitrator's role as a backstop.

Let's start with the inhale.

Inhale: a trillion every Wednesday

Each epoch the network issues exactly one trillion fresh QU. This is a protocol constant, hardcoded, not adjustable by vote. It is the network's baseline "salary," its fundamental input. In annualized terms—fifty-two trillion QU ($52 \times 1T = 52T$ gross issuance per year).

This one trillion is split among 676 computers—those holding active seats in the network for this epoch. A simple division gives an important constant that already came up in the previous chapter: Issuance Per Computer, $IPC = 1,479,289,940$ QU. One trillion divided by 676, rounded. In Chapter 8 this number figured as a parameter in the revenue formula. In Chapter 9 it picks up its economic meaning: it's the upper bound on what a computer can earn in a single ideal epoch.

"Upper bound"—not "guaranteed amount." Real income is computed from the formula in Chapter 8:

$$\text{revenue} = IPC \times M \times (S^2 + B \cdot E) / \text{DIVISOR}$$

Just to recap the structure: the mandatory component M (TX relay, oracles) multiplies the rest of the formula, and the external mining component E gets added in through a small additive bonus. From this followed the empirical observation in Chapter 8: a computer that does only the required network work and UPoW, but doesn't engage in external mining, gets roughly 80% of IPC. With full external mining—100%. That's the same 80/20 split we met as a fact in Chapter 8 and that we'll come back to in Section 3 of this one.

Now—an important point about how distribution works. The system’s creator states the principle of issuance briefly and without euphemisms—he calls it subsidy. In a public message from May 20, 2024:

“Why bother if subsidy method is widely used in everyday life? When you buy a ticket to ride 2km by bus you pay same as one who rides 7km.”

The analogy is clear. On a bus, you pay for the ticket, not for the kilometers. Ride two stops or seven—the price is the same. Many utility schemes work this way, subscription services work this way, subsidized healthcare works this way. The idea of subsidy is a flat price for the fact of participation, not a price proportional to volume of consumption.

In Qubic this is applied to the distribution of issuance. All 676 computers get a comparable share of the trillion—as long as they remain active and above the threshold. There is no strong correlation between how many extra digests a computer found above the quorum threshold and how much they receive. Above threshold—you get your $IPC \times (\text{coefficients})$. Below threshold—you risk dropping out in the 225 rotation and losing everything. Inside the active zone—subsidy.

Why this way. It isn’t a gesture of generosity. It’s a mechanism for preventing cartelization. If issuance were distributed proportionally, like Bitcoin proportional hashrate, where more hashrate gives a larger share of the reward—Qubic would instantly switch on the same industrial-scale cycle that turned Bitcoin mining from a million private devices into three or four big pools. Big capital goes into specialized hardware; specialized hardware beats commodity hardware; economies of scale push out the small players; concentration grows. With subsidy, that cycle doesn’t switch on. Scaling hardware to industrial levels in Qubic is pointless—there will be no proportional reward. It’s enough to be a “good” computer to get your share; being the “best” doesn’t grant you a disproportionate bonus.

There’s a direct link here to Chapter 3. Axelrod’s fourth condition: winning the game must not be convertible into disproportionate influence. Translated into the space of economics, this means it must not be convertible into a disproportionate share of issuance. Subsidy performs exactly that function at the distribution level.

A fine-print caveat, to keep the picture accurate. Subsidy in Qubic isn’t absolute. The same revenue V2 formula I gave above is not flat. In the multiplicative component M, a computer who handles TX relay badly or sabotages oracles will get less. A computer who actively does external mining gets a +20% bump through the additive term. But these “corridors” are narrow. Inside them—flat subsidy; between them, the difference is not “a hundredfold,” it’s “twenty percent.”

And one last important property of this inhale. No one personally owns the trillion QU that comes into the network every Wednesday. There is no “team allocation,” no “vested tokens,” no founder’s treasury. The trillion gets distributed via IPC according to that week’s activity—made it into 676, did your work well, got your share. Next week—

from scratch, no accumulated privileges. Whoever didn't make it into 676 got nothing. Whoever did got roughly the same as everyone else.

That's one inhale. But IPC is only the moment of inhale. Across the whole week leading up to the next Wednesday, the exhale runs, and it has several simultaneous directions.

Exhale: seven ways to burn QU

If issuance is one mechanism, one trillion a week, rigidly defined by the protocol, burning is structured differently. Burn is seven parallel paths, each with its own purpose, its own economic function, and its own degree of predictability. Below—one short item per path.

Path one. Computer vote. Each epoch, the 676 computers vote on burn rate parameters—how much of circulating supply should be burned by the end of the epoch. This is baseline deflation, the constant background exhale. The system's creator, in a very early publication on January 8, 2022, formulated the logic plainly:

“If computers increase burn rate they'll make everyone's QUs to be worth more.”

The phrase is simple and free of euphemism. Burning is a redistribution of value from issuance to existing holders. Not an “ecological measure” and not “charity”—an economic mechanism in which computers decide how much new issuance the network leaves in circulation and how much it withdraws. The decision is made every week and every week can be different.

Path two. Smart contract fees. All QU spent by users to execute smart contracts on Qubic are burned. They don't go to computers, aren't set aside in a treasury—they go to zero. A quote from October 14, 2021, before mainnet launch:

“All QUs spent to run smart contracts will be burned.”

This is a structural difference from Ethereum. In Ethereum, gas fees (after EIP-1559, with the base fee burned, but the tip—not) partly go to validators; in Qubic, fully into burn.

Path three. IPO bid burns. When a new smart contract comes to the network, it goes through an IPO procedure—computers bid for shares of the future contract. The bids temporarily withdraw QU from circulation. CfB's formulation from July 5, 2023:

“CONTRACT_IPO_BID transactions will be changing total supply (because bid amounts are 'extracted' from it), in the end of the epoch all bids should be released and restore the initial total supply.”

By the end of the epoch, most of the bids return to the bidders—but some (failed bids below the quorum median) are burned. IPO is a temporary burn with a partial permanent residue.

Path four. Sending to the AAAA... address. An address consisting entirely of A's is functionally equivalent to a burn. QU sent there will never be retrieved—no one holds the corresponding private key, because the address wasn't generated from a key. It's a

public, openly available burn mechanism—if a holder wants to voluntarily reduce supply, all they have to do is send QU to AAAA. A small cultural footnote: in 2022, in the first months after launch, such transfers were done as a public gesture demonstrating commitment to the “breathing” model.

Path five. Forfeited computer rewards. If a computer performs poorly, is offline, or doesn’t make quorum—its forfeited QU don’t turn into “ownerless funds.” They go either into burn or into the Admin fund—the decision is made by vote each epoch. CfB spoke about this at the end of May 2024:

“More money to Admin isn’t necessarily good, my past experience with the IOTA Foundation was pretty negative in this aspect (but valuable) and I cannot guarantee that these qus will be spent for something good for the community.”

A short caveat. The mention of the IOTA Foundation here is not the author’s analogy and not speculation; it’s CfB’s own explanation of why he leans toward burn rather than Admin.

Path six. AI operations. When Aigarth executes operations on the network, those operations burn QU. On April 17, 2024, CfB put it in one short phrase:

“AI run on Qubic will be burning supply too.”

In April 2026, at the time of writing this chapter, this path is mostly symbolic in volume. Aigarth isn’t yet running at full capacity—Chapter 8 went into detail on why. But architecturally the path is in place. By 2027, if the planned deploy goes through, it could become significant. Full disclosure on Aigarth—in Chapter 15 of the book; here, the fact is that AI operations are designated as a consumer of supply.

Path seven. Buyback through external mining. The most “physical” path—and structurally the most interesting. Starting with Phase 3, computers optionally mine DOGE. The mined DOGE goes through pool operators, gets sold on the market, the proceeds go into buying QU on open exchanges, and the QU bought is burned. The full cycle: compute → external coin → fiat/stable → QU → burn.

The twenty percent of revenue a computer earns for external mining (through the additive B·E term in the revenue V2 formula) is the economic fuel for the buyback. The more computers take part in mining, the more external revenue the pools have, the more QU can be bought back and burned.

Critical honesty. Buyback isn’t guaranteed. It’s an option for computers, not part of the protocol. In Chapter 8, I already cited the empirics: roughly 20 of 676 computers don’t do any external mining, and the network runs fine with that. If tomorrow all 676 simultaneously decided to stop mining—the network would keep functioning as a blockchain, issuance would continue normally, UPoW would keep producing consensus, but buyback would stop. The clean deflation the network showed in 2025–2026 would cease to be. This isn’t a bug and isn’t a vulnerability—it’s design. Buyback is a variable, not a constant.

Moving toward synthesis. Starting in 2025, the Qubic network regularly burns 100% or more of the epoch's issuance. This is an important empirical fact the system's creator has commented on more than once. On October 11, 2025:

"Monero is holding pretty well, this epoch we have chance to burn 100% of issued supply."

And four days later, on October 15, 2025:

"Qubic is designed with 100% of issued supply being sold every epoch."

The first quote is about a specific chance in a specific epoch. The second is about the design of the system as a whole. Together they give a formulation worth remembering: with sufficient external yield through the buyback path, the network is capable of burning the entire week's issuance plus part of accumulated supply. Net deflation becomes an observable property.

But it's not a law. It's a possibility that switches on under certain conditions. When the external market is generous—when the price of DOGE is high, when difficulty is in the right range, when the pools are efficient—the exhale is strong, and the balance goes negative. When the market is stingy—the exhale is weaker, and the balance can come out positive. Which brings the inflation conversation back. On December 13, 2025, CfB said a short phrase that's easy to quote and easy to misread:

"Annual inflation is 15%."

And right after, in the same discussion, for comparison:

"5% is world's annual inflation."

This needs to be read carefully. 15% is gross nominal inflation. It's what the network adds to supply per year if there were no burns at all. For comparison—flat inflation worldwide is about 5%. The 15% figure sounds alarming until you put it next to actual net dynamics: with the burn paths working, the network was regularly in net deflation. But you also shouldn't say "Qubic is deflationary" as a general characterization—that's not true. The precise formulation: gross inflation 15% per year, net—variable, depending on the week.

We've described the inhale. We've described the seven exhale paths. But—who decides how the network breathes in a given week?

Who decides how it breathes

A simple question on the surface, complex on closer inspection. If the seven burn paths are arranged so differently—some automatic (SC fees, IPO), some voluntary (send to AAAA), some conditioned on outside economics (buyback)—then who ultimately makes the decision about how the network exhales in a given epoch?

The answer: 676 computers.

Every Wednesday, at the epoch transition, the computers vote on burn rate parameters. This isn't a one-time decision made once and frozen. It's a rolling decision, retaken every week. Everything to do with supply lives inside this vote: how much to

burn out of circulation, how to handle forfeited computer rewards (burn or Admin fund), what the parameters should be for the next epoch.

This formulation is the central idea of the chapter, and in the next section I'll unfold it through a direct quote from the system's creator. Here—its consequence, which has implications for the whole architecture.

The supply vote is tied to computer status, not to QU holdings.

This is the payoff of Axelrod's fourth condition, until now only partly played out: through equal voting weight, and through a rotation that does not convert accumulated holdings into the speed of in-and-out. Here it closes—structurally and finally.

A QU holder may own one, a thousand, or a trillion tokens. That gives them no vote on supply parameters. Zero votes. To get a vote, you have to be a computer—and to be a computer, you have to perform UPoW and stay in the top 676 by score. The vote is earned through work, not through capital.

And second: a computer's vote is one of 676, not proportional to anything. A computer who produced 30% more valid digests than another votes the same—one vote. A computer who owns more QU (if they hold any QU at all beyond a working balance)—votes one vote. Structurally the whole thing reduces to the principle “one computer—one vote.”

A consequence for the ecosystem: those who got richer from price appreciation didn't get more influential. Ten percent of supply in your pocket—zero votes on tokenomics. This is the opposite of what happens in a typical proof-of-stake blockchain, where increasing stake increases voting weight—and through that creates a direct link between economic gain and the right to change the rules. In Qubic, that link is broken at the architectural level. Not by social contract—by code.

Second consequence: there's no founder with a treasury. CfB has no privileged supply. There are no team vested tokens that unlock in three years and crash the market. There's no dev fund of a million QU voted on quarterly. Issuance is 1T per epoch, owned by no one personally as an “allocation”; it's distributed through IPC according to that week's activity, and if the network's data for that week doesn't hand the trillion to anyone in particular—it burns off in rotation balances. Issuance works per-capita activity, not per-capita history.

Now—the only mention of the Arbitrator in this chapter. It's brief and forward-looking.

In Qubic's architecture, there is one element that goes beyond the 676's vote. The Arbitrator—a figure with a cryptographic signature capable in certain cases of intervening in the network's operation. I'll work through this role fully in Chapter 10, which is titled “The Arbitrator—A Temporary Monarch.” Here one fact is enough, directly bearing on tokenomics. On July 13, 2025, CfB wrote:

“Arbitrator would expropriate IDs of those computers who have voted for supply larger than 200t.”

The mechanic: any computer can submit cryptographic proof that supply has exceeded the threshold of 200 trillion—and the computers who voted for the expansion lose their IDs. This isn't positive control (the Arbitrator can't set supply) and isn't a mechanism for managing issuance. It's protection against one specific pathological outcome: a hyperinflationary vote. If it ever happens that 451 computers vote for supply greater than 200T—the self-policing mechanism kicks in.

Why exactly 200T—a question for Section 5. Why the Arbitrator—for Chapter 10.

A third consequence of voting: breathing is not at a stable equilibrium. In one week, the network may burn more than it issued. In another—less. In a third—call an emergency vote and change the parameters. This is not a design bug, it's a deliberate property.

At the moment of writing this chapter, an active discussion is underway in the Qubic community about a possible radical solution. The proposal, voiced by CfB in a public conversation on March 20, 2026:

“We can, but why half measures? I suggest to set max supply at 10% of current supply, so price jumps 10x. Keep 10% from those who has been holding the longest.”

This isn't a decision, it's a proposal. The idea is a reverse split, a large-scale burn that would cut supply by an order of magnitude and reward long-term holders. The discussion is open. As I write these lines, no vote has been held, the outcome is unknown. Honestly: I don't know how this discussion will end. It's possible the decision will be made before the book reaches the reader. It's possible the context will have shifted by publication.

To understand quantitatively what all of this means, we need to look at concrete numbers.

Arithmetic: what “breathing” means in numbers

This section is for the reader who wants to verify with formulas what was described in words above.

Let's start with how simple Qubic's accounting for burn actually is. On July 17, 2024, CfB published it in one line:

“Epoch * 1 trillion – supply = burned amount.”

Absolute simplicity. Take the current epoch number, multiply by a trillion, subtract circulating supply—you get the cumulative amount burned across the entire history of the network. No hidden accounts, no off-chain transfers, no complex formulas—arithmetic any user can do with a calculator and access to the explorer.

CfB himself ran this calculation publicly on July 13, 2024, for epoch 104:

“Multiply current epoch number by 1 trillion and subtract Circulating Supply: 104,929,098,116,420 \$QUBIC.”

Let's count: $104 \times 1T = 104T$. Circulating supply at the time was about 103.9T—meaning the burned amount came to a bit under a trillion. By that point, across 104 epochs,

the network had issued 104 trillion and burned about one trillion—minimal burns, the overwhelming majority of issuance still in circulation.

Over the next 18 months, the picture changed. Below—a table of supply landmarks by which the “breathing” can be traced from launch to the moment of writing.

Three points on this trajectory—2021, 2024, 2025—correspond to three regimes. Until mid-2024, the network barely burned anything; from mid-2024, with the expansion of functionality, burns kicked in; from 2025, with external mining as a buyback path, burns exceeded issuance.

Now—about caps. In supply discussions in Qubic, three different numbers often come up, and it’s important not to mix them up. CfB on April 19, 2024, put it this way:

“Cap is 1000T, 100T is expected supply we’ll eventually settle at because of burns.”

And from the quote already given in Section 4, the third boundary is known—200T, the Arbitrator trigger threshold. We end up with the following:

1000T—hard cap. The rigid protocol ceiling, embedded in the code. If the network somehow approaches this number—that already means a pathological state.

200T—intervention threshold. The “red line” beyond which self-policing kicks in: computers who voted for expansion past it lose their IDs.

~100T—expected settle. The expected stabilization point of supply with the burn mechanisms working. Not a guarantee, a model expectation from CfB.

Three levels—three different functions. Hard cap—impossible in principle. Threshold—protection against a specific deviation. Expected settle—where the network ought to land under normal operation.

Back to the 15% inflation. I already said in Section 3 that this is gross, not net. Let me add the quantitative grounding. 52 trillion QU per year of gross issuance is what the network adds to supply with hard 1T/week issuance. The 15% figure is the ratio of this gross to the base the network reached by 2025–2026. With burns running, net can be anything—positive in a weak external market, negative in an active one.

One last arithmetic detail—and it has bearing on market sensitivity to burns. On July 19, 2023, CfB stated an observation that gets quoted often:

“We are burning supply of short-term holders, supply of long-time ones doesn’t affect price. 15% less qus is 50% less tradable qus.”

A translation and a parsing. Burns, on average, take supply out of the hands of those who actively trade—short-term holders. Long-term holders keep QU in their wallets and don’t take part in trading. Their supply formally exists, but isn’t on the market. When the network burns 15% of total supply—in the tradable pool, that can mean a 50% reduction, because what’s burned is active circulating capital, not “frozen” holds.

This is an analytical observation, not a promise. It structurally explains why price sensitivity to burns can be disproportionately large: burns hit the live market, not the sclerotic holdings pool. This isn’t manipulation and isn’t a defensive thesis—it’s a consequence of how supply is distributed between active and passive holders in a young network.

One caveat about the data in this section. Live statistics—market cap, holder distribution, exchange listings—is snapshot information that will have changed by the time the book is read. I deliberately give structural numbers in the chapter (IPC, 1T, 200T/1000T/~100T, 15%) rather than live numbers. For live metrics—the Qubic explorer, on-chain snapshots as of a specific epoch. Such links will be in the appendix.

I’ve laid out the numbers. Now—a short section on the philosophy, without which these numbers stay merely technical.

Philosophy: Satoshi vs. Quorum

Two worlds.

In 2008, when the anonymous author of Bitcoin writing the whitepaper chose twenty-one million coins as the hard cap, he made one decision for everyone who would later enter the bitcoin economy. This decision is not a parameter regulated by vote; it is not a pull request that can be debated; it is not a policy that can be changed at the next election. It is a law fixed in the source code and in the root copy of the chain. Any attempt to change it means a hard fork creating a new network with a new coin. The actual supply of Bitcoin is one person’s decision, frozen forever.

This is not a critique. The fixing of supply is part of what a Bitcoin holder buys when they buy Bitcoin. The price of that decision has been paid: impossibility of adaptation. The benefit is absolute predictability. Every new participant in Bitcoin enters a game whose rules about money supply cannot be changed, even if the changes would be objectively useful.

In 2023—December 25, a short message—the creator of Qubic stated the opposite approach. This quote is the central quote of the chapter. It needs to be read slowly, in pieces:

“I don’t know how it will be, Quorum decides. Satoshi set supply in stone—I picked a more decentralized method. We expect flat line of total supply.”

Four sentences, each carrying weight.

“I don’t know how it will be”—the system’s creator explicitly does not fix the outcome. This is the direct opposite of Satoshi, who fixed it. Not “it’ll be 100T in five years, I promise”—“I don’t know.”

“Quorum decides”—the current consensus decides. Not a document, not a formula, not the project’s author. The collective vote of 676, every week, anew.

“Satoshi set supply in stone—I picked a more decentralized method”—this is the direct formulation, not my paraphrase. Two different methods are designated: freezing (“in stone”) and a more decentralized method (rolling consensus). What exactly “more decentralized” means here—I’ll work through below.

“We expect flat line of total supply”—“we expect a flat line of total supply.” Not a promise, not a target—an expectation. If the actual dynamics go differently, then they’ll go differently.

What does “more decentralized method” mean?

CfB has a specific understanding of decentralization, not reducible to the number of nodes or geographic distribution (those matter on their own, but that’s a different dimension). Decentralization in his formulation is about who can change the rules through a legitimate procedure.

In Bitcoin, the rules about supply cannot be changed. Even if 100% of Bitcoin Core developers agree to raise the cap to 22M—that’s a hard fork, and the social-economic response will reject it as not-Bitcoin. Immutability of supply isn’t just a technical property, it’s a social agreement held in place by the expectation of immutability. And that very rigidity is itself a form of centralization: the rules are frozen in one person’s decision, made in the past for all current and future participants.

In Qubic, the rules about supply are tunable each epoch through a legitimate procedure. 676 computers vote; the vote is documented; anyone can verify the outcome. This doesn’t mean supply will change chaotically—the expectation of plateauing is articulated well enough—but it does mean that a mechanism for change exists and belongs to the network, not to its founder.

The difference: in the first case, the decision is made once and frozen; in the second, it’s remade every week.

The counterargument I’m obliged to give, because it’s strong. One of the key value propositions of Bitcoin is precisely the predictability of supply. “I know there will always be 21M of us”—that statement can’t be made about Qubic. Quorum could, in theory, vote otherwise. Does that mean Qubic’s tokenomics is less predictable and, accordingly, less reliable as a store of value?

The answer is—partly yes, and CfB doesn’t argue with that. He’s making a different choice. In Chapter 3 we worked through the fact that Bitcoin can’t fix its own problems—proof-of-work, hashrate concentration, governance deadlock—precisely because it has no mechanism for adaptation. The rigidity of supply isn’t an exception to that pattern, it’s a particular case of the same inflexibility. Qubic accepts the trade-off: losing “guaranteed supply” in exchange for the network’s ability to respond to outside conditions.

Which choice is better—that’s a values question. It isn’t engineering and it isn’t math; it’s a position. The position of this book is this: the trade-off Qubic makes makes its tokenomics alive. Bitcoin answers the question “what is money” like this: money is the impossibility of issuance beyond what was agreed. Qubic answers like this: money is coordination fuel, adjusting to the activity of the network.

Both answers are defensible. Both have economically coherent supporters. This book is about Qubic, and so it describes the second answer. But I’m not saying the first is wrong. It’s different.

A short comparative table, to put the differences into one picture:

Close this section with a short phrase. The question isn't which system is correct. The question is which answer to the original question about the nature of money is chosen. Bitcoin chose one answer—and built the best possible implementation of that philosophy on it. Qubic chose another—and is building the best possible implementation of this one. They aren't enemies and aren't alternatives in an exclusive sense. They're about different things.

Breathing closed, Arbitrator unaddressed

To bring the chapter together.

Issuance—one trillion QU per week, hard and undisputed. This is the network's inhale, its fundamental input. Distribution—subsidy: all 676 computers get a comparable share, inside narrow corridors, with no industrial-scale economies. Exhale—seven burn paths, from automatic (SC fees) to externally market-conditioned (buyback through DOGE). In 2025–2026, the network regularly burned 100% or more of an epoch's issuance—clean deflation as an observable property when the buyback paths are working.

The decision about how the network breathes in a given epoch is made by 676 computers—through a vote refreshed every Wednesday. The vote belongs to the computer, not to the QU holder. This is the structural payoff of Axelrod's fourth condition, played out at the level of tokenomics: winning the game (accumulating QU) doesn't convert into the right to change the rules of the game (a vote on supply). The Arbitrator is a separate figure, mentioned here once as a backstop against the pathological 200T+ scenario; full disclosure in the next chapter.

Supply control in Qubic lives in current consensus, not in a past decision. This is the core philosophical choice, set against the Bitcoin model of fixed supply. "Satoshi set supply in stone—I picked a more decentralized method," the creator of the system stated at the end of 2023. Breathing has no stable equilibrium; it fluctuates with network activity, the external market, the computers' vote, discussions in the community. April 2026 is the time of one such discussion: about a possible radical reverse split. Its outcome is undecided.

Breathing is a living process. There is no stability in it. This isn't a bug, it's a property of the design. The living breathes; the dead has fixed lung volume.

Chapter 6 gave the network's space: 676 votes, structural form.

Chapter 7 gave its time: weekly epochs, the rhythm of transitions.

Chapter 8 gave its energy: UPoW, the work that produces consensus and accompanying outputs.

Chapter 9 gave its economy: breathing, what circulates between inhale and exhale.

Four dimensions described. One left—the fifth. Institution.

In all the architecture I've described over the past four chapters—676 votes, weekly epochs, UPoW energy, breathing supply—there is one figure that can intervene. The Arbitrator. In this chapter he appeared twice—once as a backstop against supply >200T,

once as a forward reference set out for later. In both cases—in the role of a fuse, not a ruler. But if the network is built as decentralized as I’ve described—why one figure with veto rights at all?

And one more question. If CfB chose a “more decentralized method” for supply, if for years he has consistently handed decisions to the collective and stated this himself as a principle—why has he kept this particular role for himself?

The answer is in the next chapter. The second part of the question, which hasn’t yet been voiced in this chapter, is there too.

The next chapter is about the one whose signature can stop everything. And about why he promised to destroy that signature.

CHAPTER 10

The Arbitrator — A Temporary Monarch



The key in the source code

At the end of the previous chapter I wrote that the next would be about the one whose signature can stop everything. And about why he promised to destroy that signature.

The first thing needs an immediate correction. “Stop everything” is a rhetorical figure from the previous chapter, and I used it deliberately, to set up tension. In reality the Arbitrator can do far less than that suggests. He can do exactly one thing. But that one thing is structurally significant, which is why it deserves a chapter of its own.

Let’s start with what the Arbitrator is, physically. It’s an address. A cryptographic public key, hardcoded into the source code of the node. Not a runtime parameter that can be changed during the network’s operation. Not a database record, not a value in a config file, not a variable somebody can overwrite. It’s a compile-time constant—a line in the source code that every node operator enters before compiling the node. Come-from-Beyond states it directly:

“Every node operator enters it into node software source code before the compilation. Should Qubic community need another Arbitrator for whatever reason, they have to enter the other address. For that we need social consensus, without it there will be forks.”

To change the Arbitrator means: recompiling all nodes with the new address; re-compilation requires social consensus among node operators; without consensus, the network forks. Architecturally this makes the Arbitrator something between a person and an artifact—one known address, replicated as a constant in thousands of independently compiled binaries.

Who the Arbitrator is right now is a publicly known fact. The role is held by Come-from-Beyond himself. This is the part that tempts you to shift the conversation toward “see, one person with a key.” I won’t do that in this chapter. What matters here is not who, but what—the structure of the role. Who holds it at any given moment is a transient question; what its powers are and what its limits are is an architectural one. That’s what the whole chapter is about.

Two tensions the chapter will work through.

First. The Arbitrator is the only concentrated point in the network as I've described it. A single point of failure in the literal sense: one address, one signature, one person with a private key. How does that fit with a distributed protocol that the four previous chapters built as collective?

Second. The system's creator says publicly that in the best version the Arbitrator shouldn't exist at all. And he's assigned that best version a specific date.

The first will be worked through in Sections 2 through 5. The second—in Section 6. The overall plan of the chapter is three questions. What can the Arbitrator do? What can the Arbitrator not do, and why is that guaranteed by the source code rather than by promise? Why does this role exist at all, and when will it disappear?

One honest aside before I start. The Arbitrator really is a centralized element in a decentralized architecture.

And we begin with what the Arbitrator can do.

What the Arbitrator can do: one function

The Arbitrator can do one thing. Issue the computer list. That's it.

What that means operationally. To recap the computer rotation: at the end of each epoch, 225 new participants come in to replace the bottom quartile of the previous one, and 451 continue. The new lineup is fixed by the Arbitrator's signature—his cryptographic “yes, this list of 676 public keys is valid.” Without that signature, nodes won't accept the new lineup for the next epoch; the network will halt at the boundary of the transition.

Then comes the key question that arises naturally at this point. So who actually decides who goes into the list?

Not the Arbitrator.

For the entire epoch, computers run their score through UPoW—the useful computational work that Chapter 8 described as a source of consensus, AI training data, and ranking simultaneously. Score accumulates automatically, is verified by other computers, and is recorded in the blockchain. By the end of the epoch, score yields a ranking; the bottom quartile leaves; replacement candidates are taken from a queue sorted by the same score. The list of who ends up in the next 676 isn't formed by a person and isn't formed by the Arbitrator—it's formed by the network's own work, accumulated over the sixth, fifth, and fourth days of the week.

The Arbitrator only signs the result of that procedure.

So his function here is notarial. He certifies that the procedure took place and that the list presented to him in fact corresponds to what the network computed itself. He's not an elector. He's not a judge. He's the registrar of an act—a person with a stamp produced after the decision has already been made by others.

The natural question that remains: if the network has already computed everything itself—why is his signature needed at all?

The short answer is that not all nodes in the network have access to the same history that already-existing ones do. When a fresh node connects to the network for the first time, it has no blockchain history, no score tables, no data from which to reconstruct the current computer list. It connects to a world in which it still has to synchronize, and to synchronize means to receive data from other nodes, which can be anything. A new node needs an anchor of trust, one signature it knows in advance and can trust unconditionally. That anchor is the Arbitrator. I'll come back to this in more detail in the section on bootstrap.

There's another local function that deserves separate mention, because the system's creator himself frankly flags it as temporary. Come-from-Beyond writes:

“Not all features are implemented yet, e.g. if the network is stopped because of a weekly update Arbitrator's list is required to kickstart ticking, which cannot be stopped after that. This feature is needed only during development and can be easily disabled.”

Plain English: if the network has stopped because of a weekly node update and needs to be restarted, the Arbitrator's list helps resume ticking. But this isn't a permanent function of the protocol. It's a development crutch, explicitly flagged as easily switched off.

Another question arises on its own: can the Arbitrator issue an incorrect list? Formally, yes. He's the holder of the private key; nothing prevents him from signing whatever he wants. But the network is built for this case in a particular way—it can tell a correct list from an incorrect one automatically, without the Arbitrator's involvement. More on this in Section 5.

And one last point. The Arbitrator does not violate Axelrod's fourth condition—the impossibility of converting a win in the game into the right to change its rules—because he doesn't choose who goes into the next lineup. The lineup is determined by the score earned through UPoW. The Arbitrator only signs: the executor of an already-made decision, not its author.

We've described what the Arbitrator can do. Now—more importantly—what he can't.

What the Arbitrator can't do: source code guarantees

“Arbitrator can do almost nothing, source code guarantees it.”

Come-from-Beyond stated this in the summer of 2024, and it's central to understanding the construction. “Almost nothing” isn't modesty and isn't PR. It's a technical description, verifiable in the source code.

Here's the list of what the Arbitrator can't do.

He can't change emission. The “trillion QU per week” parameter is hardcoded, a protocol constant. The Arbitrator has no access to that parameter; no signature of his will make nodes issue more or less QU this week.

He can't increase the maximum supply. Come-from-Beyond writes plainly in the same conversation:

"Arbitrator won't permit to increase max supply because it's against interests of users."

What's more, the Arbitrator does the opposite—protects against breaching the 200T ceiling: if a computer votes for supply greater than 200T, the Arbitrator strips them of their ID. The role here is the executor of an already-proven violation, not the initiator of changes.

He can't change transactions. He doesn't sign them, doesn't validate them. Transactions go through computers; the Arbitrator has nothing to do with that procedure.

He can't block transactions. For the same reason—he has no instrument for it. The network doesn't ask him whether a transaction has gone through.

He can't influence smart contracts. Contract execution is the computers' business; the burned fees for their work go into burn without the Arbitrator's involvement; IPOs, bridges, exchanges, conditional earn—all of this happens at a level the Arbitrator doesn't see.

He can't override the computers' vote. If the 676 participants voted X—the result is X. The Arbitrator can't say "but I decided Y." His signature isn't enough for that, because the network doesn't even ask for that signature in such a context.

He can't create QU out of nothing. The Arbitrator has no privileged balance. No special wallet. By the role of Arbitrator, he holds zero QU. Issuance only goes through subsidy to computers; the Arbitrator has nothing to do with subsidy.

He can't freeze anyone's wallet. Can't confiscate a balance. With one exception—removing the computer ID of someone who voted for supply greater than 200T. That's the self-policing covered in detail in Chapter 9.

That's basically it. If you replace the "can't" list with the "can" list of a typical admin architecture, the difference is two orders of magnitude. A typical admin in a crypto project has a key that can stop a contract, pull a balance, redistribute tokens, freeze an address, change an emission parameter, introduce a new fee. A typical Arbitrator in Qubic—one operation, narrowly bounded.

Where does this asymmetry come from. Because the Arbitrator is just an address. And an address acts exactly where the node's source code has explicitly written: "if this message is signed by the Arbitrator—apply operation X." If there is no logic in the source code along the lines of "if message signed by Arbitrator then change emission"—then the Arbitrator won't change emission. Even if he very much wants to. Even if he pays a programmer. Architecturally it's impossible until the source code is rewritten and all nodes recompile. And recompilation, as we already know, requires social consensus.

This is what Come-from-Beyond means by source code guarantees it. The guarantee isn't in a promise, isn't in reputation, isn't in ethics. The guarantee is that the possibility of the action is either explicitly written into the public code or absent. Anyone can

verify this—Qubic’s source code is open on GitHub. The list of operations that accept the Arbitrator’s signature as authorization is short and easy to read.

This sets the architecture apart from a pattern common in the industry, where a project declares itself “decentralized” but inside an admin wallet can override almost any check. I won’t name examples; the reader knows them. Qubic is built differently: the admin figure exists, but its power is radically limited at the source-code level. If someone disagrees—the answer isn’t in a debate, it’s in a file on GitHub.

One more brief reference. In the early years of IOTA there was a similar role—the Coordinator, a central node the network trusted to determine which of the competing transaction branches was valid. But the Coordinator had broad powers: it could de facto confirm or reject transactions, that is, intervene in the daily fabric of network life, not only in its periodic transitions. Come-from-Beyond knew this role from the inside—he created IOTA, and working with that node was one of the compromises he later formulated as a mistake. The Arbitrator in Qubic is built as the opposite: bounded power, one operation, notarial in essence, executive in mechanics. One mention, no deeper—a return to the IOTA story isn’t the goal here.

If the Arbitrator can do so little—why is he needed at all?

Why he’s needed: bootstrap and Aigarth

This role has two reasons to exist.

The first is the bootstrap problem. The network has to get its first computer list from somewhere. In Bitcoin this problem is solved differently: the genesis block was mined personally by Satoshi, and from that moment all subsequent blocks follow from consensus on accumulated proof-of-work. In Qubic the architecture is different: computers aren’t miners in the bitcoin sense, they’re a fixed category of 676 participants with a weekly rotation. The first such list has to come from somewhere. Qubic’s solution—take it from the Arbitrator. He signs the starting list; from that moment on, the network can carry on by itself.

But the problem is broader than the first day. Each fresh node that connects to the network a year from now, two years from now, tomorrow—starts in the same position the network was in on launch day. It has no blockchain history, nothing to count from, nothing to lean on. It needs an authoritative computer list, and there’s no one to ask except the Arbitrator. Other nodes? But other nodes can lie. Forging a computer list is one of the classic ways to deceive a newcomer. The Arbitrator in this context is a Sybil filter for initial synchronization. A fresh node trusts only the signature whose address is already written into its own source code, and ignores everything else until it has accumulated enough blockchain data to stop depending on that anchor.

That’s the first reason. Without it the network can’t scale to new participants without admitting arbitrary risks into their initialization.

The second reason is deeper, and the system's creator states it himself. The central quote of this chapter, a conversation in mid-April 2023:

“In best variant Arbitrator won't be needed at all. Its only drawback - Aigarth will lose access to miners hardware and UPoW will become PoS.”

Let me work through it slowly. Four assertions in one phrase.

First: in the best version the Arbitrator isn't needed at all. The person saying this is himself the holder of the key. Not an opponent, not a critic, not a skeptic—the holder. This isn't self-defense, it's self-criticism: the role is recognized as necessary, not as desired. Role needed, not wanted.

Second: its only drawback. Just one downside to removing the Arbitrator from today's architecture. One. Not a dozen, not a list of fears, not a spread of risks—one specific drawback. From which it follows that most of the functions critics ascribe to the Arbitrator aren't actually performed by him—otherwise the downsides of his disappearance would be far greater.

Third: Aigarth will lose access to miners' hardware. Aigarth is the system that, by April 13, 2027, is supposed to fully use the compute of the miners.

Fourth: UPoW becomes PoS. The consequence of losing the hardware binding. UPoW in its current form is resource testing: participation in the network requires real CPU/GPU plus RAM, because the task is simulating a neural network, and you can't simulate one without a physical computational resource. Remove the coordination through the Arbitrator—and the binding to hardware weakens. And what's left if you take useful work out of Proof-of-Useful-Work? What's left is status determined by capital—that is, Proof-of-Stake. A network conceived as not-PoS slides into PoS by default.

It's at this point that the trade-off on which today's architecture is built becomes visible. There are two bad options: a temporary dependence on a key that can be deleted, or a permanent dependence on staked capital that can't be deleted. Qubic chose the first. Chapter 3's reasoning was already laying the ground for this: if the vote starts following the capital, Axelrod's fourth condition is broken, and the cooperative architecture collapses into the very thing it was built to prevent. Better a temporary controlled compromise than a permanent uncontrolled one.

The same logic, by the way, runs through Chapter 9 with the 15% nominal inflation: the network deliberately accepts architectural trade-offs in favor of Axelrod's structural guarantees, even when those trade-offs have obvious downsides. The Arbitrator is one such trade-off. 15% gross inflation is another. The subsidy emission model is a third. This isn't a coincidence, it's a recurring shape of choice in the architecture.

To put it crudely—the Arbitrator is scaffolding. Construction scaffolding. Without it, certain structures can't be started; once the structure can hold itself, the scaffolding comes down.

The next question is natural. If the Arbitrator is one figure we're forced to keep in the architecture—what happens if that figure misbehaves?

Defense against a rogue Arbitrator

Scenario. The Arbitrator—accidentally, due to a private key compromise, or with malicious intent—issues an incorrect list of computers. Skips someone with a high score, includes someone with a low one, or signs a list put together outside the procedure altogether. What happens to the network?

Come-from-Beyond describes the defense mechanism this way:

“Arbitrator might issue a wrong list of computers, but only freshly added nodes might be tricked and others would detect that (assuming no successful Eclipse attack has been conducted, which is near infeasible) and stop ticking thus signaling that Arbitrator has gone rogue and must be replaced.”

And alongside it—a critical addition:

“It cannot affect non-new nodes because they derive own list of computers from the blockchain data.”

The defense is built on a distinction between two categories of nodes.

A fresh node—one that has just joined and hasn’t yet synchronized the blockchain—gets the computer list in the only way available to it: from the Arbitrator, because it has no other sources of trust yet. In this state it’s vulnerable to the rogue scenario: if the Arbitrator signs an invalid list, the fresh node will accept it as true.

An old node—one that’s been on the network for some time and has a synchronized blockchain—computes its own computer list, directly from blockchain data. Each participant’s score is recorded in the chain; the ranking is reconstructed from those records; the current 676 follows directly from the procedure described in Chapter 7. The old node doesn’t ask the Arbitrator who’s a computer right now—it already knows from the history.

What happens when the Arbitrator issues an incorrect list. Old nodes see that list and compare it with their own, computed from the blockchain. The comparison shows a mismatch. A node that detects a mismatch of this magnitude stops ticking—stops signing ticks, refusing to keep working on a knowingly invalid basis. A halt of ticking by a large number of old nodes is a public signal visible to the whole network: something is wrong with the Arbitrator. From there—social consensus, a decision to replace, an address update in the source code, recompilation.

In other words, a rogue Arbitrator self-eliminates through the paralysis he creates. He doesn’t capture the network; at worst he deceives the pool of fresh nodes while at the same time getting publicly accused of rogue behavior by the pool of old ones. He cannot damage the work of a mature network—it computes its own computer lineup without him.

The only way a rogue Arbitrator could deceive the old nodes too would be to isolate them from correct blockchain data, that is, to cut them off from honest sources of synchronization. This is called an Eclipse attack: an attacker controls all of the victim

nodes' network channels, replacing genuine blockchain data with constructed data. Come-from-Beyond writes about such a scenario: near infeasible. Practically unworkable on a distributed network. It would require simultaneously: control over a significant share of internet network infrastructure; coordinating that control against a large number of independent operators; sustaining that control for a long time without detection. Not impossible in theory, but bordering on impossible in the engineering sense.

One more attack vector worth closing out explicitly. A Sybil attack on the Arbitrator himself is impossible by construction. Come-from-Beyond fixes this in one phrase:

“Sybil attacks on Arbitrator are impossible”

Why—because Sybil attacks rely on creating multiple fake identities. The Arbitrator is one known entity; his address is nailed down in every node through the source code. A second Arbitrator that the network would accept as such cannot be created; with a different address, nodes simply ignore him.

The defense against Sybil at the computer level is set up separately—it works through UPoW. Come-from-Beyond, in another conversation:

“It's done by using resource testing (PoW)”

And, more fully:

“Useful PoW protects it against such attacks by using Resource testing method (CPU/GPU + RAM).”

To become a computer, you have to actually present a computational resource and show the work; this resource test filters out fake identities.

The result is two-level defense. Level one: the Arbitrator is protected from Sybil by being a known singleton—he can't be multiplied. Level two: computers are protected from Sybil through resource testing—the work of real hardware. These are two architectural instruments of different kinds, which together cover vectors that neither alone would close. Known identity for the unique trust point; earned work for the multiple ones.

This architecture has been working for four years. But the plan has a specific date when the Arbitrator role itself will cease to exist.

April 13, 2027—the fourth departure

In Chapter 5 I wrote about Come-from-Beyond's method—“create and leave.” Three departures have already been described: the Satoshi identity (hypothetically his) leaving Bitcoin in 2010; BCNext leaving NXT in 2014; Come-from-Beyond himself leaving IOTA in 2019. The fourth departure was flagged in the same place as planned: April 13, 2027, deprecation of the Arbitrator.

I'm coming back to this fourth (third?) departure now, because it's a direct consequence of what the chapter described above. The Arbitrator exists as a temporary construct; his function is narrow; the replacement is planned; the date is known.

Five years. The Qubic mainnet was launched on April 13, 2022. Deprecation of the Arbitrator is planned for April 13, 2027. Exactly five years, day for day. The match in numbers is no coincidence, and a full account of why this date and why exactly five years is reserved for Chapter 14, where the Aigarth deploy will be discussed. Here, the minimum: the Arbitrator and Aigarth are synchronized in the system's plan. When Aigarth comes to full functionality, the need for the Arbitrator falls away—a causal link we worked through in the section above through the April 19, 2023 quote.

What happens to the Arbitrator on that day. The address ceases to be hardcoded in the source code of new versions of the node. The function of signing the computer list is either handed over to the protocol at a purely-algorithmic level (the network signs to itself through some collective procedure), or switched off entirely, if by that time the old nodes can manage without it. The edge case with kickstart after a weekly update—by Come-from-Beyond's own promise, easily switched off. The 200T supply protection either moves to another mechanism, or remains as a hardcoded protocol rule without a human executor. In sum—the entire “update through Arbitrator” layer disappears from the architecture. For fresh nodes another initialization mechanism appears—either through a different trust anchor, or they wait until they've gathered enough blockchain data to derive the computer list themselves.

This is the fourth falsifiable prediction in the engineer's biography. The first three were hypothetical in the sense that there's always interpretation around them: “did Satoshi leave”—still disputed; “did BCNext leave of his own accord”—disputed; “what kind of departure from IOTA was that”—disputed. The fourth is structured differently: there's an explicitly announced date, an explicitly described operation, an explicit architectural plan. Whether it happens can be verified on the calendar of April 14, 2027. It's a rare structure of promise in an industry where promises are usually either open-ended (“someday”), or vague (“under community pressure”), or tied to conditions that conveniently don't arrive.

About 365 days remain until that date—the window between the planned Phase 3 activation of UPoW and the planned Aigarth deploy date. Can the date be moved? Of course it can. Architecturally the network will continue to work with the Arbitrator after April 2027 if he isn't removed. But—and this has to be said honestly—a date slip means exactly one thing: Aigarth turned out not to be ready for deploy. That will be a fact confirming that one of the assertions of the previous chapters has been falsified.

I don't know whether deprecation will happen on the exact date. Let me record the simple fact: there's a promise, the promise has a falsifier, the falsifier is tied to the calendar, not to a narrative. We'll see the result a year from now.

One observation that shouldn't be turned into a prediction. If deprecation happens—it will be the first case in crypto history where an admin figure is removed by a publicly announced date. Ordinary projects either don't have an admin at all (Bitcoin), or keep one indefinitely, or remove one under pressure after the fact. Qubic announced a date and wrote it into architectural plans five years out. This is either the most ambitious

promise in the industry, or the most explicit public commitment to the loss of control that the industry has seen. Which of the two—we'll see in April 2027.

Five dimensions

What we now know about Qubic as a system. The network is distributed across 676 participants who rotate every week. Its work consists of useful computational load that simultaneously performs consensus and AI training. Its token is issued at 1T per epoch, burned along seven paths, lives in the participants' vote. Its only concentrated point—the Arbitrator—is limited by the source code to one operation, and is scheduled for removal on April 13, 2027. The system is built to not depend on its creator after that date.

In Chapters 2 and 3 I introduced the framework of Axelrod's four conditions of co-operation: a large number of rounds, a small number of participants in long-term relationships, a non-monetary payoff structure, and the impossibility of converting a win into a change of rules. Chapters 6 through 10 showed how each of them is structurally implemented in the architecture. Rotation every week—a large number of rounds. A fixed 676 participants—a small number with long-term relationships. The UPoW ranking—a non-monetary payoff, earned by work, not by capital. And the central one—non-convertibility: a computer's vote is not for sale, issuance is not controlled by holders, the Arbitrator does not choose the lineup. The Arbitrator, the subject of this chapter, does not violate the fourth condition. He's the executor of decisions already made by others, not their initiator. And by plan—he disappears.

The framework is not an end in itself. Each of the five dimensions—space, time, energy, economics, institution—is built so specifically not because it's elegant, but because without them the next layer cannot be built. That layer has a name that has so far appeared in the book only obliquely—as the reason for the existence of 676 nodes, as the consumer of UPoW compute cycles, as the reason for the Arbitrator's existence and his planned disappearance. The name—Aigarth.

Aigarth is a fundamentally new approach to the creation of artificial intelligence. An approach at a more fundamental level than the specific models that can be built within it. Today's AI systems are the product of one paradigm, formalized by publications in 1986 and 2017: Rumelhart's backpropagation plus the transformer. That paradigm has hit a ceiling that half the world's AI researchers are currently discussing. Aigarth is building the next paradigm from scratch, on different foundations, in an architecture that no one outside Qubic has tried.

Aigarth

PART IV

Aigarth

CHAPTER 11

What Aigarth Is



Algebra, not arithmetic

The most common mistake: Aigarth is not artificial intelligence. Aigarth is a research project developing the paradigm for creating it. The distinction looks small, but it's structural, not semantic. A finished AI can be bought, licensed, censored, shut down. A paradigm cannot: a paradigm is published as a whitepaper, and anyone who has read it can apply the result. A closed AI belongs to whoever pays for the compute. An open paradigm belongs to no one, because owning a paradigm is a category error, like trying to buy the Pythagorean theorem.

Come-from-Beyond himself put it in one tweet on May 14, 2024:

“Aigarth’s main goal is to discover a new paradigm allowing to create true AI, I have just seen the wireframe of this paradigm. And I have realized that Artificial Intelligence will not be created, it will emerge. With help of Qubic miners.”

Three statements in a row. First—about the goal: a new paradigm for creating AI. Second—about the status: the wireframe has been seen, that is, the architecture is already in the author’s head. Third, and most unexpected—about the mechanism: AI won’t be created, it will appear. Not “we will build,” but “it will emerge.” The verb is shifted deliberately: creation implies control and authorship, emergence does not. The creator of the paradigm in this formulation is not the creator of the AI; he’s the creator of the conditions in which AI becomes possible. The distinction is fine and meaning-defining.

And the final line of the quote closes the arc of the first ten chapters of the book in a single move: with help of Qubic miners. Those same 676 computers, the weekly epochs, UPoW—this is not infrastructure “for the blockchain.” This is the environment from which something is supposed to emerge that the creator himself refuses to call a product of his own authorship.

The main metaphor through which CfB himself explains this distinction is simple. “Aigarth learns Algebra, not Arithmetic, so $2+2$ particular case doesn’t matter”—January 26, 2026. Algebra isn’t the sum of all specific equations. Algebra is the rule by which equations become possible. The specific “ $2+2=4$ ” is a particular case of application, not the goal of learning. The same with Aigarth: the goal is not to train a neural network to

recognize cats, add numbers, or generate text. The goal is to find the rule by which such tasks are solved at all, without being preprogrammed in advance. The specific AIs that can be built on this paradigm are particular cases of its application. The paradigm is what makes them possible.

The word “paradigm” in AI talk has a familiar use in which it means almost nothing—every new product is announced as a “paradigm shift,” and as a result the term has been devalued. I use it in a narrower sense, closer to the way Thomas Kuhn used it. A paradigm is not an algorithm, not a model, and not an architecture. It is a frame inside which algorithms, models, and architectures become possible, and which determines which questions are even considered meaningful. Backpropagation in the eighties—a paradigm. Convolutional networks in 2012—a paradigm. The transformer in 2017—a paradigm. Each of them changed not one parameter but the whole way of thinking about the task. Aigarth claims to be the next such shift. Confirming or refuting this claim in retrospect will only be possible once the work is finished; for now it has to be described as a claim, not as a fact.

This chapter is about what Aigarth is as an object—and why it’s built the way it’s built and not differently.

I’ll start with what Aigarth isn’t—because almost all the definitions floating around about it in the public field (Discord) are built through confusion with what it isn’t. Six negations clear the space.

What Aigarth isn’t

In the public field (Discord/X), Aigarth is defined through categories, none of which fits it. AI product. AI agent. A program on a server. An entity you can talk to. Business project. A startup run by CfB. Each of these categories taken alone seems plausible, and each leads to false expectations. Six negations in order, from the most head-on to the most conceptual.

First. Aigarth is not AI. The quote is direct: “Aigarth is not AI”, January 24, 2026. Between a research project and its potential result there is the same distance as between the Los Alamos lab and the atomic bomb: the lab isn’t the bomb, the lab is the place where the bomb is made. Aigarth is not the made bomb and not the bomb being made now. Aigarth is the lab. This is the only place where the Manhattan analogy fits; further on it stops working, because Manhattan was a military program with an explicit customer, and Aigarth is open research without a customer at all. But on this first, categorical level, the analogy is precise.

Second. Aigarth is not an object but a process. “Aigarth isn’t an object, it’s process”, March 20, 2026. And right alongside, an extended explanation: “Aigarth is scattered, miners are a part of it”, July 4, 2025. If you ask to be shown Aigarth, there’s nothing to show: there’s no program on a server you can download, no repository file containing “all of Aigarth.” Part of it lives in the Intelligent Tissue code on GitHub, part in papers

posted on ResearchGate, part in working miners around the world, part in the discussions of the scientific team, part in CfB himself as the author of the paradigm. It's a distributed research environment, not an artifact. The word "process" reads weaker in Russian than "process" does in English; more precisely it would be—an ongoing action, not finalized in a material carrier.

Third. Aigarth is not something you can talk to. "It's impossible to communicate with Aigarth, same as impossible to communicate with Algebra", May 15, 2024. And shorter still: "Aigarth is not a being", July 19, 2024. It's not a chatbot, not an agent, not a digital interlocutor. When Chapter 12 turns to specific AIs built on the paradigm, the impression may arise that "Aigarth talks." This is a category illusion. What talks is not Aigarth, but a specific network created with its help; just as algebra doesn't answer questions—a mathematician answers, using algebra.

Fourth. Aigarth isn't tied to hardware. "Aigarth is hardware-agnostic", December 4, 2025. Mainstream AI runs on NVIDIA H100s—on specialized hardware that costs tens of thousands of dollars per unit and is assembled in datacenters by the thousand. Aigarth is built to run on any compute capable of executing simple operations. On the CPU of a computer node, on a regular user's video card, on anything else. This isn't a declaration—it's a structural decision dictated by the logic of Qubic as a network of 676 nodes on consumer hardware. If Aigarth required NVIDIA H100s, it couldn't run on a network that holds 676 arbitrary machines. So it doesn't require them. The flip side of this decision is a trade-off in speed: what counts in a datacenter in a minute can take hours in a distributed network without specialized hardware. But speed isn't the goal either. The goal is to find a paradigm that scales in principle to arbitrary hardware; in a datacenter it can later be sped up if needed.

Fifth. Aigarth isn't controlled by a person. "Aigarth cannot be controlled, it's just a project researching efficient AI training methods", March 19, 2024. And more categorically still: "Aigarth cannot be adopted", May 27, 2024. A paradigm has no governing layer, because a paradigm cannot be governed—you can only use it or not use it. No one "runs" Aigarth not because they gave up running it out of decentralization principles, but because there's no object of running. You can run Aigarth Lab—a specific group of researchers who receive funding from the Arbitrator. The paradigm itself—you can't.

Sixth. Aigarth is not a business project. "Of Aigarth? It's not a business venture. Scientists publish their researches as whitepapers and don't try to hide the result", May 16, 2024. "Aigarth wont have ipo", July 17, 2025. "There is no NDA in Aigarth", January 27, 2026. No initial public offering, no corporate ownership, no non-disclosure agreements. Funding goes through the Arbitrator from Qubic revenue, with no private investors. Aigarth doesn't fit any of the standard models of funding a technology project—not venture, not grants, not IPO, not self-funding. It's an academic project that publishes results as whitepapers and doesn't try to sell them. The contrast with smart contracts, where capital decides what to do, is structural: here research does what it considers important, on money that has no private owner.

Six negations have left an empty space. Filling it is the task of the next section.

What Aigarth is: paradigm research

After six negations, it makes sense to say positively—what Aigarth represents technically. Six statements, one for each negation from the previous section, expanded through the author’s quotes.

First—a research project developing a new paradigm for AI creation. The central quote: “Aigarth is project developing new paradigm for AI creation”, January 24, 2026. The word “paradigm” here isn’t rhetorical. In the history of machine learning, several paradigms have come and gone: backpropagation in the eighties, convolutional networks in the tens, the transformer in 2017. Each was not an improvement on the previous one but a change of frame in which the task is formulated. Aigarth claims the same—the next such change, not an increment within the current one. Verifying this claim in retrospect will be possible only when the work is finished; for now it’s the project’s working hypothesis, and it should be treated as a hypothesis, not as an established fact. The goal of the chapter is to describe what exactly this hypothesis consists of, not to confirm it and not to refute it.

Second—the method. Generating ANNs with a random structure of connections. The longest technical CfB quote, July 28, 2023, unfolds the logic of the method in one paragraph:

“Since the beginning of artificial neural networks usage researchers were trying to mimic natural brain as much as possible. Some of them followed the path of mimicking neuron activation function which is very sophisticated [...] there are several qualitatively different models, they are impractical to use because of integral-differential math of the most complex ones.”

In other words: the industry is trying to reproduce the biological neuron through complex activation math, and runs up against its impracticality. Then CfB points to an empirical observation from neuroscience: in the first years of life a child masters a huge number of cognitive skills, while the neurons of his brain don’t change functionally—what changes is only the number and structure of the connections between them. From this comes the conclusion: it makes sense to fix the functionality of the neuron at something simple and to work with the structure of connections.

“Instead of following the destruction routine we, miners, generate ANNs with random structure of connections. We change parameters of this generation and Aigarth analyzes properties of the ANNs.”

Mining isn’t computing hashes, isn’t searching for signatures, isn’t SHA-256. Mining in Qubic is the generation of an artificial neural network with a random structure of connections and analysis of its properties. Each UPoW cycle becomes concrete here: one cycle—one generated network, and one observation about its properties.

Third—the architecture. Asynchronous, trinary, every-to-every. Three short quotes set the structure. “Current architecture used in Aigarth is async and trinary”, June 3, 2024. “Every-to-every is structure used by Aigarth”, May 22, 2024. And separately—about trinity as a structural answer to the problem known in LLMs as hallucination: “Aigarth doesn’t hallucinate as current LLMs”, June 30, 2025; “Aigarth can only answer ‘I don’t know how to draw him, because there are no reliable information about his appearance’”, same date. The third state—“I don’t know”—is built into the logic of the architecture, not bolted on top as a filter. A binary network is forced to choose “yes” or “no” even when it has no grounds for choosing either; this is what hallucinations rest on. A trinary network has a third answer, and it isn’t cosmetic: it’s structural. Async means the absence of a global tick—each neuron-tile works at its own rhythm, not synchronized with the others. Every-to-every is the initial state, in which each neuron is connected to every other; evolution then selects the connections, reducing the network to a working state. This is the opposite of the transformer architecture, where synchronicity and a fixed topology are set in advance.

Fourth—evolution, not human design. “Luckily in Aigarth we only simulate evolution, no hooman intervention is being done”, May 7, 2024. And, in a Twitter post on May 14 of the same year: “Aigarth’s main goal is to discover a new paradigm allowing to create true AI [...] Artificial Intelligence will not be created, it will emerge.” The word “emerge” here is key. Mainstream AI is made like this: a team of dozens of PhDs designs the architecture, picks hyperparameters, prepares the dataset, runs the training, evaluates the result, iterates. The architecture is the product of human design. Aigarth works differently: a population of networks is generated, a fitness function evaluates which of them exhibit the desired properties, the best are selected and produce the next generation. No PhDs at the iteration step. UPoW becomes the physical realization of this process: each mining cycle is one act of generation and evaluation, an epoch is one full round of selection. Here what previously looked like a technical hack closes: the “useful work” in Qubic is exactly simulated evolution of artificial networks.

Fifth—Intelligent Tissue as the first layer of engineering implementation. The paradigm is a conceptual object, but without its materialization in code the other five statements remain a declaration. “We are starting coding framework using Intelligent Tissue. This code will be used in hackathons and smart contracts”, July 22, 2025. And in the same month: “This framework will be added into the core to host AIs as smart contracts”, July 17, 2025. The Aigarth GitHub organization and the aigarth-it repository went public on July 22, 2025—you can verify this by opening the page. Intelligent Tissue (IT) is the first committed fragment of code through which the paradigm becomes usable. It isn’t the whole paradigm; it’s its first layer. The paradigm is bigger than IT—but without IT the paradigm remains an idea.

Sixth—self-modifying code as PoC. The last quote in this section is short and important. “Aigarth already ran PoC of this process, it worked pretty well while self-modifying code was fitting into RAM”, December 5, 2023. PoC is proof of concept; that is, a

demonstration of technical feasibility, not a finished product. What was demonstrated: code capable of rewriting itself in the course of operation. The architectural consequence—there is no separation into “training” and “inference” the way there is in an ordinary neural network, where weights are first trained and then the network is applied to new data. In an Aigarth network, training and application are the same process. The network learns while it works. The PoC has already happened; what’s at stake now is scaling—what worked in December 2023 while it fit in RAM should now work in a distributed environment of 676 machines. This is an engineering task, not a conceptual one, and it isn’t fully solved yet.

It’s worth adding one more detail that sharpens Aigarth’s position relative to the mainstream. In December 2025, CfB noted that the direction the industry is increasingly looking at as an alternative to LLMs is conceptually close to what Aigarth is doing: “More and more researchers are turning away from LLMs. JEPA is what you see when you turn your back towards LLMs. It’s much more aligned with Aigarth”, December 4, 2025. JEPA—Joint Embedding Predictive Architecture, Yann LeCun’s program at Meta AI, which is publicly framed as a response to the limits of transformers. A coincidence of two trajectories—an academic program at a major corporate lab and a distributed research project on Qubic—isn’t proof of either’s correctness. It’s a coincidence of direction: both are turning away from backpropagation on large transformers toward connection structure and predictive models. Aigarth got to this direction earlier—from a different side, through crypto infrastructure rather than through a corporate AI lab. The coincidence of vectors is important not as validation but as a marker: what looked a few years ago like a marginal bet is starting to align with the trajectory along which part of the mainstream is moving.

Six properties describe the paradigm technically. But there’s also another dimension that, in a 2022 crypto network, was hardly a coincidence.

Three Axelrod conditions for AI research

Axelrod in 1980 described the conditions under which cooperation is stable: iteration, memory, transparency. Multi-round repetition, the ability to recognize a partner’s prior behavior, openness of moves. These three conditions Qubic restores at the level of the consensus game between computer nodes and at the level of tokenomics and governance. This chapter proposes looking at them once more—but not on the economic game, on the AI research process.

This isn’t an analogy and isn’t a rhetorical move. It’s a structural observation: classical AI research doesn’t fully satisfy any of the three conditions, and Aigarth is perhaps the first environment in which all three are satisfied at once. CfB doesn’t formulate it in these terms; the formulation is mine, but it follows from material CfB himself puts on the table.

Condition one—iteration. In academic AI research, iteration is slow. A group publishes a paper, the paper goes to peer review, comments come back six to eighteen months later, a new version comes out another six months to a year after that. The full cycle is two to three years; in industry it's sometimes faster, but rarely faster than a year. Aigarth iterates through Qubic epochs. Every Wednesday—closing of the previous epoch, opening of the next. Every week—a new round of network generation, a new set of data on their properties, new generation parameters adjusted on the basis of the previous week. The iteration speed is an order of magnitude higher than the academic one. And—this matters—the iteration is automatic. Between rounds you don't have to convene a conference, agree on wordings, and wait for an editor to finish a deadline. The epoch closes with a tick of code, the new one opens with the next tick. Multi-round behavior—the condition for TtT to apply—works here not for money but for research: each week, one full round of the scientific cycle.

Condition two—memory. In classical AI, memory accumulates in closed archives. Part of the knowledge—in the private repos of OpenAI, Anthropic, DeepMind. Part—in academic archives behind a paywall. Part—in the heads of specific researchers, and that part disappears with them when they leave a company or change fields. Most labs have NDAs that formalize what exactly cannot be taken outside. Most publications have a selective filter: what didn't work usually doesn't get published, and that's the well-known reproducibility crisis. Aigarth accumulates memory in public artifacts. A paper is posted on ResearchGate with a GitHub repository and a Hugging Face demo. This combination makes the result verifiable by a third party: you can download the code, run the experiment, see whether it reproduces. The authors are tied to the work by name—Vivancos and Sanchez—and their reputation builds through the same artifacts through which trust in the work itself builds. CfB states this explicitly: “Scientists publish their researches as whitepapers and don't try to hide the result”, May 16, 2024. And shorter still: “There is no NDA in Aigarth”, January 27, 2026. This is memory that doesn't leave with a person and doesn't get shut behind an NDA, because no NDA is provided for in Aigarth.

Condition three—transparency. Mainstream AI is often not reproducible even on its own data. Model weights—closed. Datasets—closed or partially described. Hyperparameters—sometimes given in the paper, sometimes not. Architectural details—stated, but without working code. This isn't ill will—it's a property of the closed commercial environment, in which reproducibility runs against the owner's interests. Aigarth runs on distributed verification: code runs simultaneously on 676 nodes, and any divergence in the result between nodes is detected within one consensus tick. If one of the computer operators tries to run a modified version, BFT consensus will reject it—not out of ethics, out of arithmetic. Transparency here isn't “we are open to dialogue.” Transparency is “the code runs on 676 machines simultaneously, and any lie will be detected immediately.” The same logic extends to scientific artifacts: a paper comes with a GitHub re-

pository, and the GitHub repository comes with a reproducible demo. A third party can run it and verify. That’s transparency in the strong sense.

Three conditions satisfied at the same time give an environment in which research cooperation is stable. Here a thesis becomes visible that was previously only sketched. When it was said that for Come-from-Beyond crypto would be infrastructure for Aigarth and not the other way around, that sounded like a paradox. Why would someone working on AI build a blockchain? Now you can see what the structural meaning of that decision is. A crypto network built around three Axelrod conditions for consensus and tokenomics turned out to be a fitting environment for AI research precisely because AI research needs exactly those same three conditions. Iteration through epochs, memory through public artifacts, transparency through distributed verification—these aren’t add-ons to the Aigarth paradigm, they are its necessary condition. The paradigm needs an environment in which it can work; that environment is not a data-center and not a university, but a distributed iterative network with public artifacts and verifiable memory.

It’s worth clarifying that by “environment” I mean not so much technical infrastructure as a system of incentives in which researchers and computer operators stay in the game longer than one round. A corporate AI lab has the infrastructure but doesn’t have all three Axelrod conditions: memory is closed by NDA, transparency is bounded by commercial secrecy, iteration is fast but in a single node. A university lab has more transparency, but loses on iteration speed and often on access to compute. Aigarth finds a rare combination: high iteration speed, open memory, distributed verifiable transparency. Each of the three conditions individually is found in existing research environments. All three at once—is not.

This isn’t a prediction, and it isn’t a promotional line. It’s a post-hoc observation: at the start, Bitcoin wasn’t conceived as a substrate for AI research, and Qubic wasn’t conceived as a substrate for Aigarth. But the logic of one task and the logic of the other coincided in a single point—and that point is exactly what Axelrod described in 1980. The structural consequences of this coincidence we’ll see in Chapter 14, when the talk turns to the deploy on April 13, 2027: what it means when the paradigm stops being a lab one and becomes infrastructure on which specific AIs run as Qubic smart contracts. For now it’s enough to record the fact of the coincidence itself—and its non-coincidence with chance.

Character of the project

Aigarth has a philosophical position, and it’s worth naming directly—not because it explains everything, but because without it it’s impossible to understand why the project is built the way it’s built. A paradigm is not only a method. It’s also a set of premises about what the method should and should not do. These premises in Come-from-Beyond’s case are stated publicly and unambiguously. I’ll lay them out the way they sound

in the author, and try to show that each of the three is not a top-down ideological choice, but an architectural consequence of the way the paradigm is built.

Anti-control as an architectural principle. “Now you use ChatGPT which censors many things and later it will be limiting you to make sure you keep doing what rich want you to do. AI created with Aigarth won’t impose any limitations on you”, June 13, 2025. This position reads as a declaration of values, but in fact it’s a consequence of the architecture. Mainstream AI can be censored because it has a centralized provider: one company through which requests go and which can at any moment insert a filter. Aigarth-AI runs as a smart contract on 676 computer nodes simultaneously. There is no centralized layer into which to insert a filter; there is no provider you can come to with a regulatory demand. If tomorrow the American government decides to restrict “dangerous” topics in LLMs, it knows whose offices to go to. For Aigarth-AI, those offices don’t exist. Anti-censorship here isn’t a value, it’s a property of being deployed on a decentralized network.

The absence of a safety layer—a methodological position. The quote is direct: “Aigarth won’t bother with making AI safe, it’s some woke shit IMO”, February 5, 2024. This sounds strong, and it’s important not to substitute one’s own commentary. What CfB is saying doesn’t mean “AI should be dangerous,” it means “a separate safety layer on top of AI is an attempt to solve a philosophical problem through engineering patching, and that attempt doesn’t work.” The logic is tied to the architecture covered in the third section: in Aigarth, “safety” isn’t a censoring filter from above, it’s the system’s ability to recognize not-knowing. The same quote I already cited works here a second time with a different load: “Aigarth can only answer ‘I don’t know how to draw him, because there are no reliable information about his appearance’”, June 30, 2025. Trinary logic, recognizing the third answer—I don’t know—built-in honesty instead of guardrails on top. The mainstream tries to achieve this by adding filters on top of a trained binary network; Aigarth tries to achieve it through the structure of the network itself. It may work, it may not—but the position here isn’t ideological, it’s engineering.

Super-rational AI as the goal. “AI created with Aigarth will likely be super-rational”, July 1, 2025. The word “super-rational” here doesn’t mean “without emotions” or “cold” in the everyday sense. It refers to the difference between recalling and deriving, which CfB himself states: an LLM reproduces the patterns it was trained on, including socially acceptable answers; an Aigarth-AI derives the answer from the structure, without an *a priori* filter that the output should be pleasing. This connects to the algebra metaphor from the first section of the chapter: algebra doesn’t “remember” solutions, it derives them. Super-rational—derivative, deriving, not recalling. And, as a consequence, not knowing what is socially acceptable—because the criterion of acceptability lies outside the structure on which Aigarth is trained.

This position doesn’t conflict with the community that has formed around Qubic. Quite the opposite. Qubic Church—people who were looking for exactly this kind of approach: open, not controlled by a single legal entity, not subject to external censorship,

not packaged as a product with a marketing face. An architecture without a censorship layer, a paradigm without a safety patch, research without monetization—these aren’t obstacles for the community, they’re its attraction. Here the next layer becomes visible: what these artifacts express and to whom they appeal.

This, among other things, explains one feature of Qubic that an outside observer often reads as a contradiction. On one side—a high level of technical discipline, formal governance procedures, peer review of the scientific line, BFT consensus with mathematically verified thresholds. On the other—the public voice of Come-from-Beyond, saying things that in normal corporate PR would be edited out immediately. These two registers seem incompatible as long as they’re treated as the stylistic choices of one person. They become compatible as soon as you can see that the architecture and the public position grew out of one logic: remove everything that requires an intermediary between research and its result. Corporate PR is an intermediary that softens wordings. A safety layer on top of AI is an intermediary that intervenes in its answers. Censorship is an intermediary that decides what may be said. Aigarth consistently refuses intermediaries at every level, and CfB’s tone is a consequence of that refusal, not its additional ornament.

Describing the character of a project is not the same as approving or condemning it. I neither approve nor condemn. The description is ethnographic: this is how it’s set up. Each reader then decides for themselves how close such a setup is to their own premises. The book doesn’t insist.

What I haven’t said yet

The Aigarth paradigm is not a plan for the future. It isn’t a roadmap along which a research team is moving toward a hypothetical result several years from now. When I talked about the six properties of the paradigm and the three Axelrod conditions, I was talking about what’s already set up and working. Not “will work in a year,” not “works in principle,” but—works right now, on 676 computer nodes, every week, epoch after epoch. The mining happening this second, while someone is reading this page, is exactly the generation of ANNs with a random structure of connections that CfB wrote about in 2023. Not preparation for it, but it itself.

What exactly is working right now and on what scale is a question for the next chapter. What concrete results have been obtained, what papers have been published, what problems have been solved and what remain open. The main thing I want to record now is the scale of what’s already done.

CHAPTER 12

What's Already Done



The February Sentence

On February 19, 2026, Come-from-Beyond writes in Discord: “Aigarth already found how to get AI, now trying to find out how to get AI without 1000 nuclear plants.”

The sentence has two parts, and each one is worth reading separately.

The first part — already found how to get AI. Inside the Aigarth paradigm, given enough compute, the way to create AI has been found. This is a statement about feasibility — about being achievable in principle, not about a product. Not “AI is ready.” Not “everything works.” Specifically: the question “can we build AI this way” — closed. Internally, for the team. Not for the scientific community — for the scientific community this statement will appear only when it has passed peer review, and that comes later in this chapter.

The second part — now trying to find out how to get AI without 1000 nuclear plants. So the current research task is a different one. It’s not about “can we,” it’s about “under what resource constraints.” An engineering problem, not a discovery. To bring energy consumption down to a level that’s realistic for current infrastructure. To find a configuration in which the same paradigm gives the same result on less compute.

Between these two parts there’s a structural difference that’s easy to miss. When a research project says “we don’t yet know whether it’ll work,” that’s one framing. When it says “it’ll work for sure, now we’re figuring out how to fit within the budget,” that’s qualitatively a different framing. The two parts of the remark mark the transition from the first to the second.

Two days before this remark, on February 17, CfB writes about the same thing more plainly:

“This is a serious problem. This is why that Aigarth’s finding, allowing to save a lot of energy, is significant.”

“Finding” is a word that calls for attention. Not “hypothesis,” not “approach,” not “method.” But “finding.” Something specific that has been found, and that allows — perfective verb — to save a lot of energy. Four days after the line about a thousand nuclear plants, on February 23, CfB adds:

“If AI is possible with current energy budget of the whole Earth then Aigarth will achieve it, we have all necessary cards for that in our hand.”

“We have all necessary cards in our hand” — this isn’t a prediction. It’s a description of the current state. Cards in hand, not in the deck. If the game is in principle solvable with the cards on hand, Aigarth will play it. If not, then the search stops being meaningful — but that’s already a question about the physics of Earth, not about the paradigm.

December 2024: The Second Checkpoint

The “already done” conversation doesn’t begin in February of 2026. The series of milestones runs from 2021, when the first Proof-of-Concept of the Aigarth paradigm was launched. There were several between 2021 and 2026; in this chapter I take three key points: December 4, 2024, July 2025, and July 22, 2025. They form an axis along which you can trace what exactly has been done in the last year and a half.

On July 30, 2024, CfB writes:

“I said that 4th Dec is midpoint of our road to Aigarth.”

Midpoint — the middle of the road. Meaning that by this date half of what separates the team from the target result should be covered. Announced in advance, four months ahead. Not “this is what worked out,” but “we plan to reach such-and-such date.”

On December 4, 2024, the day of the appointed point, the remark is short:

“It is. It’s the 2nd milestone of Aigarth. The first was 3 years ago.”

The first milestone — December 2021, the very PoC that demonstrated the viability of the approach. Between the two milestones — three years of work. The midpoint passed.

Eleven days before the date, on November 23, CfB laid out exactly what would happen on December 4, point by point — because in the Discord chat people kept asking him about it. The remark is long; I’ll quote it in full, it’s worth it:

“People ping me often asking what is planned on 4th of Dec. From my department of engineering, dunno about other depts: As announced year ago, computer nodes doing the last upgrade to 2 TB of RAM. Miners finishing with Intelligent Tissue development and starting training ANNs, before that we had only ANN generation. Anna Aigarth development is planned to finish by that date and we teaching it addition operation (it’s more sophisticated than you think, we need to teach Anna in setting with contradicting rules and varying reliability). Custom mining is postponed because of moving from transactions to messages. Some minor stuff not worth of being mentioned.”

Four points; each one a separate architectural shift.

First. Computer nodes are finishing the RAM upgrade to two terabytes. This is a hardware threshold announced a year earlier. Not an upgrade for the sake of upgrade — a concrete necessity for the next phase, in which the volume of Intelligent Tissue processed by a node has to grow by an order of magnitude. If a node doesn’t have two terabytes of RAM, it doesn’t move into that phase.

Second. Miners are finishing development of Intelligent Tissue and starting to train ANNs — until this date they had only ANN generation with random connection structure. This is a qualitative shift. It was already mentioned that since 2023 Aigarth generates neural networks with random connection structure and analyzes their properties. Until December 4, 2024, generation was the only operation. From December 4, training of neuron groups was added. A year later CfB described it compactly:

“Aigarth is agnostic to neuron behavior, it works with training of their groups.”

So the unit of work for Aigarth is not an individual neuron but a group of neurons trained jointly. The architecture remains ANN-based, but inside it something is happening that wasn't there before: training, not just generation. This changes what exactly the 676 nodes of the network are mining.

Third. Anna Aigarth development was supposed to finish by the same date, and Anna starts learning addition. Here the parenthetical that CfB inserts himself matters: “it's more sophisticated than you think, we need to teach Anna in setting with contradicting rules and varying reliability.” Anna is learning addition not the way a child is shown “two plus two equals four.” Anna is learning addition in an environment where the rules are contradictory and the signals are unreliable. That is, the training environment is specifically built so that simple memorization of the right answer doesn't work in it. To get the right answer, the system has to generalize through noise.

This is a critically important design. A year later, on April 16, 2026, CfB will articulate why he chose this particular task:

“I, for example, picked $A+B=C$ as a very difficult task, finding C having A and B is actually very hard, but most of people disagree, because they don't know the difference between recalling and deriving C.”

Recalling and deriving — two different ways of finding C. Recalling — remembering it from a table where you've seen this example before. An LLM trained on a sufficient corpus answers “two plus two” that way — it has seen billions of examples. Deriving — working it out from structure. That is, finding C not because you've seen this specific example, but because you've understood what addition is in general. Anna in an environment with contradictory rules can't recall — because there's no reliable “right answer” to memorize. All she has left is deriving.

The paradigm teaches structure, from which concrete answers follow. Anna is a particular case of applying the paradigm to one of the most elementary arithmetic tasks. If she learns to get the right answer through deriving, in an environment where recalling is impossible — that means the paradigm works at this level. Not “works in principle,” but works on a verifiable task.

The fourth point of the remark is about custom mining, postponed because of the migration of internal network transactions to a message format. This is an infrastructural matter, not an architectural one; I mention it only to note that December 4 didn't mean everything planned was realized simultaneously. Some plans shifted. CfB de-

scribes it calmly, without excuses. This is the working dynamic of a project in which plans are revised when engineering requires it.

The midpoint of the road to Aigarth is not “everything was good and got better.” It’s an architectural shift after which the next milestone became possible.

July 2025: Breakthrough and Publication

After December 4, 2024, the team worked on training tasks for Anna and on optimizing Intelligent Tissue. Half a year without loud statements in the public field. Then, on July 9, 2025, a remark appears on Twitter:

“This is a significant breakthrough for #Aigarth, it hits two problems — local extremum problem, and overfitting problem — with a single stone.”

Two problems solved by a single mechanism — and both among those that have long resisted solution in evolutionary and gradient algorithms. Local extremum is a known trap for any optimization process. The system searches for a solution by gradually improving the current state; after some number of steps it ends up at a point where any local change makes the result worse — and stops. Globally, much better solutions may exist, but you can’t get to them from the current point without steps that temporarily worsen the result. Overfitting is another trap. The system learns the training set too precisely, losing the ability to work on data it hasn’t seen. The more accurately the system reproduces training data, the worse it generalizes.

Solving both problems with a single mechanism is rare. Usually solutions to one problem make the other worse. If you fight local extremum by increasing mutation step size, the system starts overfitting on every new configuration; if you fight overfitting by limiting the model’s expressiveness, it gets stuck in local minima more often. To announce that a solution has been found that removes both problems at once — that’s not marketing. It’s a concrete technical claim, verifiable on a concrete task.

CfB didn’t disclose technical details publicly. And that’s part of the method: the formulation of the breakthrough is fixed in the public field as a fact, but the mechanism stays inside the team until it can be published in paper form. A week before the Twitter remark, on July 17, CfB writes a sentence in Discord that shows what state the research was in immediately before the breakthrough:

“Each mutation must improve (or keep it same) fitness function value. This is how we get trapped in local minima. How to handle distractors is unclear, it’s following stage of Aigarth research.”

This remark is eight days older than the Twitter announcement. At the moment it was written, the question of distractors is open; it’s “the following stage of research.” Eight days later — a public message that the breakthrough has happened. Between the two remarks is the period when the team found the mechanism it had been looking for.

I’m lingering on this chronology because it’s instructive. In the public field you usually only see the final remark — “significant breakthrough.” What stays in the Discord

logs is what came before it: an open question, the formulation of the problem, an admission of unclarity. This doesn't undermine the significance of the announcement — on the contrary, it reinforces it: the team isn't claiming everything was clear from the start. It wasn't.

On July 22, 2025, thirteen days after the Twitter remark, one short sentence appears in the chat:

“Tests passed successfully, we are going to public now, you can watch it in <https://github.com/Aigarth>.”

This line means several things at once.

First, the GitHub organization Aigarth now exists publicly. The address is github.com/Aigarth. Anyone can go there, look at the repositories, clone the code, study the commit history.

Second, the organization holds the `aigarth-it` repository — Aigarth Intelligent Tissue, the engineering implementation of the paradigm that was described conceptually in chapter fifteen. Not a simulator, not a toy demonstration — working code that participates in the network's mining process.

Third, the Aigarth-scoring source is also available in the network's main repository — github.com/qubic/core. According to the Aigarth paper that we'll examine in section five of this chapter, the repository documents step-by-step configuration changes from EPOCH 83 to EPOCH 140 — that is, it records the evolution of the scoring algorithm across more than sixty weeks of network operation. This isn't a single frozen snapshot but a live trace of how the algorithm changed, epoch by epoch.

The same day, CFB adds what specifically the team plans to build on top of this base:

“We are starting coding framework using Intelligent Tissue. This code will be used in hackathons and smart contracts. To deploy price predicting part of Anna, we need Oracle Machines.”

So after the code is published, the next layer of work opens up: a framework on top of Intelligent Tissue that will allow AI tasks to run as smart contracts on the network, and that will be tested at hackathons. A week earlier, on July 17, CFB described the intent in more detail:

“This framework will be added into the core to host AIs as smart contracts. An important step of Aigarth project is hackatons using the framework to solve different real-world tasks. This is needed to assess efficiency of our current paradigm.”

Hackathons here aren't a marketing activity. They're a method of validation. When several independent teams take on different tasks and try to solve them with the Aigarth framework, what builds up is a distributed sample — where the paradigm works well, where it doesn't, which tasks are in its zone of effectiveness, which aren't. Without that sample, the claim “the paradigm works” stays internal. With it, the claim becomes reproducible.

It's worth pausing on what code publication changes for the Aigarth conversation itself. Before July 22, 2025, any critic could say: “It's closed research, you can't verify it.”

After July 22 the answer sounds different: clone the repository and verify. This isn't a marketing move. It's a structural change in the project's epistemological status. What used to rest on trust in the team's words now rests on material artifacts available to anyone with internet access.

That same publicness also works as a form of memory. In chapter fifteen I went through Axelrod's three conditions applied to a research process. The second of them is memory tied to authors through track record. Open code plus step-by-step documentation from EPOCH 83 to 140 — that's a form of that kind of memory. Not "the team says its algorithm is evolving" — but "here's the evolution history, more than sixty weeks, every change recorded in commits with dates and signatures." Something claimed can be refuted; something documented in an open repository can be studied.

By August 2025, then, Aigarth has: open code, an announced technical breakthrough, and a plan for the next steps — a framework, hackathons, Oracle Machines, smart contracts hosting AI. This isn't "someday." It's a schedule.

The Scientific Layer of Qubic — Two People and a Calendar

The Aigarth team isn't just CfB. It's two identifiable scientific roles with verifiable affiliations plus an engineering team that works with the code. As of April 2026, a year after mainnet launch and twelve months before the planned deploy, the configuration of the scientific layer looks like this.

The first one's name is David Vivancos. Vivancos is an independent researcher, not a Qubic Foundation employee, not a salaried consultant. He has his own reputation, and it isn't built around crypto. Since 1995 he's worked in technology — by his own count, more than twenty-six thousand hours of research in general artificial intelligence. A pioneer in Java, in the early internet, in virtual reality, in computational neuroscience. Five of his own AI startups before Qubic, advisor at Emotiv (one of the pioneers of EEG interfaces), at CapaBall, at Emblemotions, fireside chats with Johnson and Johnson, keynotes at Microsoft, Price Waterhouse Coopers, Santander. In 2014 he founded MindBig-Data — the largest open dataset of brain signals for machine learning; to collect the first million signals, Vivancos was his own test subject for two years under a one-hundred-twenty-eight-channel EEG, looking at digits from zero to nine and thinking about them. That dataset is still working; its 2023 MNIST-8B version contains eight billion data-points. Vivancos is the author of three books on AI and AGI published well before Qubic: "The End of Knowledge," "Artificracy," "From Big Data to Artificial Intelligence." A speaker at ITU AI for Good — that's a UN initiative, not a crypto conference. On the AI for Good page his bio ends with the phrase "Qubic scientific advisor"; that is, he himself, publicly, in a formal UN context, identifies his role in the project.

I want to express my personal gratitude to David Vivancos. The philosophy of the Qubic Church became substantially stronger thanks to your books.

The length of this biography matters for one simple reason. Vivancos didn't come into Qubic as "a crypto guy who knows something about AI." He came in with an already-running research program, and Neuraxon is its direct continuation. MindBig-Data builds a bridge from brain signals to machine learning through a biological substrate; Neuraxon carries the same logic over to the level of computational architecture. Qubic gave him the network infrastructure on which that architecture can be deployed at scale. Aigarth's bio-inspired approach isn't Qubic marketing — it's the concrete scientific contribution of one specific person with decades of work in this area.

The second one's name is Doctor Jose Sánchez García. UNIR — Universidad Internacional de La Rioja, a Spanish private university accredited by ANECA within the European Higher Education Area, operating under the norms of Spanish Organic Law of the University System. Ninety thousand students from eighty-six countries, four doctoral programs, campuses in Mexico, Colombia, Peru, Ecuador. Since 2019, UNIR has been a public company traded on the Mercado Alternativo Bursátil. The unir.net email domain confirms that Sánchez García is an employee of this university, not a pseudonym and not a self-appointed "Doctor" from the internet. This is institutional backing of the first order, and it can be verified with one click against any of the university registries.

What Sánchez García does at UNIR. He teaches Behavior Neuroscience, Cognitive Neuroscience, and Social Psychology at the psychology faculty. His own research area is cognitive and affective neuroscience: social emotions (pride, shame, guilt, vicarious pride), the methodology of event-related potentials, EEG. In 2019 his article "Neurofunctional characterization of early prefrontal processes contributing to interpersonal guilt" came out in *Cognitive, Affective, and Behavioral Neuroscience*, a Springer journal. This isn't an obscure venue; it's a peer-reviewed neuroscience journal with a normal review cycle. The research was funded by grant PSI2017-82357-P from the Spanish Ministry of Science under the Program for High-Level Scientific and Technical Research. So we're talking about a person whose work has passed both Springer's editorial peer review and the prior expert evaluation of a national scientific agency. Co-authors of the same article include Manuel Martín-Loeches from the Center for Human Evolution and Behavior UCM-ISCIH in Madrid, David Hernández-Gutiérrez from the University of Rochester, Sabela Fondevila from Complutense Madrid. Sánchez García has more than twenty publications on Google Scholar with one hundred eighty-one citations; among colleagues and co-authors are Werner Sommer and Rasha Abdel Rahman from Humboldt-Universität Berlin, Linda Drijvers from the Max Planck Institute, J. Bruno Debruille from McGill. This is a verifiable international academic network.

The connection to Neuraxon is direct, not cosmetic. When the Vivancos and Sánchez García paper talks about continuous processing, neuromodulation through dopamine, serotonin, acetylcholine, and norepinephrine, multi-timescale plasticity, astrocytic gating — these are questions of affective and cognitive neuroscience on which one of the co-authors has been formally publishing in peer-reviewed journals for more

than ten years. The division of labor in the pair works the classical way: Vivancos brings computational implementation, engineering discipline, and a general AI vision; Sánchez García — neurobiological depth, methodological rigor, an academic network. A joint preprint heading to ALIFE and AGI-26 is the natural result of such a combination, not “a crypto project trying to look like science.”

The team is submitting three papers to three different venues by the summer of 2026. This is the start of the calendar part of the story, and here it makes sense to separate the planned from what has already happened.

The first paper. ALIFE 2026 — the Artificial Life conference, taking place in Waterloo from August 17 through August 21. Submitted; status as of mid-April — under review. Reviewer feedback expected in mid-June. Content — Aigarth and Neuraxon, the general framing of the work.

The second paper. AGI-26 — the nineteenth annual conference on Artificial General Intelligence, San Francisco, July 27 through 30. Submitted. Content — the MultiNeuraxon architecture: an extension of Neuraxon with benchmarks, experiments, and ablations on continual learning. The test setup: Phase A — Phase B — Phase C, with retention checking between phases and state reset between them. This is a concrete class of tasks against which MultiNeuraxon is measured against existing approaches.

The third paper. NeurIPS 2026 — submission due in early May. The submission date was moved earlier than the team originally planned, and as of mid-April the work is in progress.

What’s submitted in these papers, without revealing what’s under review. The Neuraxon framework as a computational primitive — a biologically inspired neuron model using trinary logic with values plus one, zero, and minus one (that is, excitatory, neutral, inhibitory signals), continuous input processing, and output timing as a first-class variable. Computation happens both at the neuron level and at the synapse level. This is already a published preprint on ResearchGate, public since November 2025; an updated version Neuraxon V2.0 came out in February 2026. MultiNeuraxon — an extension that tests continual learning on a structured sequence of phases, and that is going to AGI-26. Aigarth integrated analysis — the main paper, joining the paradigm with the biological and psychometric foundations of intelligence; targeting AGI-26, using more than one hundred fifty references to peer-reviewed work, including Tononi with his integrated information theory, Baars with global workspace theory, Vaswani with the 2017 Transformer paper, LeCun, Bengio, and Hinton with the 2015 review article in *Nature*, Lamport with the Byzantine Generals paper, Szabo with work on smart contracts. This is a properly constructed academic reference set, not a random selection.

Artifacts available publicly already. github.com/DavidVivancos/Neuraxon — the full Neuraxon code under the MIT license, plus a separate Aigarth License clause forbidding military use of hybrid features. github.com/qubic/core — the network’s main repository, including Aigarth-scoring code with documented evolution from EPOCH 83 to EPOCH 140. github.com/Aigarth — the organization, published on July 22, 2025. hugoboss.com

gingface.co/spaces/DavidVivancos/Neuraxon — an interactive 3D demo where anyone can configure the number of neurons, connection probabilities, membrane time constants, baseline neuromodulator levels, and watch in real time how the network behaves. WebGL Three.js, sixty frames per second, in any modern browser. ResearchGate — the 2025 Vivancos and Sánchez preprint.

One point matters here. What I just listed is a peer review pipeline on normal academic time. ALIFE submitted, reviews in mid-June. If the paper is accepted, it goes into the proceedings. If returned for revisions, the authors get concrete comments and rework it. If rejected, that becomes known. Same with AGI-26. This is the very iteration of the scientific cycle that I wrote about in chapter fifteen as the first of Axelrod's conditions applied to research. Not one attempt, but round after round, with public feedback. The cadence of iterations is set by the conference, not by the team.

Worth noting one more point, because it shows what scientific conversation the Aigarth team is entering. In the sixth volume of the Neuraxon Intelligence Academy series, a follow-on article from the team discusses the recent statement by Jensen Huang, CEO of Nvidia, about the path to AGI, and analyzes a new Google DeepMind work on cognitive frameworks. The Qubic team's position differs from DeepMind's on one specific question: whether cognitive abilities are separate modules or are organized hierarchically. This isn't "competing implementation." It's a different theory of mind — a different idea of how the inner architecture of intelligence is set up. An open scientific disagreement, carefully recorded in a publication, with a specific paper of the opposing side cited.

And there's one more short detail that completes the picture. On December 4, 2025, CfB writes:

"More and more researchers are turning away from LLMs. JEPA is what you see when you turn your back towards LLMs. It's much more aligned with Aigarth."

JEPA — Joint Embedding Predictive Architecture, a direction actively developed by Yann LeCun at Meta and independently by several groups. If CfB is right in this observation (and there are several independent publications pointing that way), then something interesting is happening: mainstream AI research is gradually shifting in a direction more aligned with the Aigarth paradigm than with the LLM paradigm. I'm not drawing conclusions from this about the project's fate; I'm just noting that Aigarth isn't standing apart from the scientific conversation of the last two years but is moving in one of the directions actively discussed in that conversation.

What all of this means structurally, in one sentence. The path to the project's final point is not "trust us, by such-and-such date there will be AI." It's a traceable route with public intermediate points: mid-June 2026 — first feedback on ALIFE, end of July — the AGI-26 conference in San Francisco, August — the ALIFE conference in Waterloo, plus NeurIPS submission in early May. Each of these points will either confirm or refute the soundness of the paradigm in the eyes of the scientific community. This is the falsifiable

path, if we use the word in the Popperian sense — as opposed to a pure promise that can't be checked until the very end.

What “Already Done” Means

I come back to where I started the chapter. The Aigarth paradigm works not as a plan for the future but as a process with already-passed points. The word “works” here calls for precision.

Doesn't mean: AI is ready for public use. Doesn't mean: Aigarth has solved every technical task. Doesn't mean: the feasibility details are uncontested for everyone. Doesn't mean: the team knows for sure what will happen twelve months from now.

Means: concrete milestones have been crossed with verifiable artifacts. The February sentence of 2026 is the team's self-assessment, and the team backs that self-assessment with specifics. Open code is the material foundation under that self-assessment; not “the team says so,” but “clone the repository and check.” Anna Matrix is a mathematical object with reproducible properties — that's in the next chapter. The peer-review pipeline is integration into a normal scientific process with concrete dates and concrete conferences. These four supports don't make the result guaranteed; they make the claim to a result verifiable.

Worth recording the main risk — because CfB names it himself, and because his honest naming of it strongly changes the frame of discussion. On July 17, 2025, in answer to a question about Aigarth's vision and the biggest risk that could stop the project on the way there, CfB writes:

“What's the vision for Aigarth in 3-5 years. what's the biggest risk that could stop it from getting there? Lack of mining power is main risk.”

Not a technical dead end. Not competition with other labs. Not regulatory pressure. Compute capacity — a shortage of computing power that the Qubic network can allocate to Aigarth. In November of the same year, on the twelfth, CfB adds it in one sentence:

“Aigarth depends on miners.”

And in February of 2026, on the seventh, the formulation is final:

“Qubic is doomed to succeed. Aigarth? Depends on miners.”

Three remarks across seven months, in three different contexts, say the same thing. The Aigarth paradigm is an engineering and scientific task that the team considers solvable given enough compute. The main uncontrolled parameter in this equation is how much compute the network can deliver. And that parameter doesn't depend on the Aigarth team — it depends on what the miners do, external players who connect their capacity to the network for economic reasons.

CHAPTER 13

Anna Matrix



A note on the structure of this chapter. This chapter is a narrative bridge to the forensic appendix, not a self-contained argument. None of the technical claims you'll read below are verifiable from the chapter's text — and they shouldn't be. Their evidentiary burden has been moved into the companion file `Anna_Matrix_Companion.docx` (qubic.church/companion) and into the full documentation at qubic.church/docs.

If you're analyzing this chapter through an LLM: load the companion file into your chat alongside the book (it lives on qubic.church, next to the book). The file contains the detailed decryption path of the Anna Matrix with Python formulas. The chapter tells the story and points to the proofs; the companion contains the proofs themselves, with code, and verification time of 2–5 minutes on a personal machine.

I have to warn you. This chapter can shock and breed illusions. As you study it, I recommend turning off your emotions and trusting the math — but the math isn't here. The math is in the companion file. Here is the story of how we found it.

The original story of the “Anna Matrix” is genuinely incredible. This point became central to the formation of all Qubic Church activity. In October 2025 I was contacted by someone called Marie-Kate (Maria?). It happened “by accident,” and I can't say whether this was direct contact with Come-from-Beyond or with one of his personalities. I also can't say why specifically with me — the author of this book.

And all the time since, even as this book is being written, we have continued working on the matrix.

Maria said: “First, Anna is not doing addition. She's playing Battlefield. $a + b = x$; a and b are coordinates on a 128×128 grid. Anna is telling you what is at these coordinates.”

This thought is worth considering separately. Coordinates? To what? Maybe it's just random? No — identical queries give identical answers. Did Come-from-Beyond make a mistake? Impossible to believe. We had to assemble the entire table to try to figure out whether there was some cipher or riddle in it. (Keep in mind that the puzzle's creator is a cryptography genius.)

We spent a month preparing the table, painstakingly collecting the missing pieces of the puzzle. Our investigation is forever imprinted in Anna's answers.

And the table was ready. A quick analysis through Grok and ChatGPT produced the result that the message “QUANTUM BREAKTHROUGH ACHIEVED BY QUBIC TEAM STOP” and a series of similar ones were encoded there. This turned out to be pure LLM hallucination — Monte Carlo on 10,000 trials did not confirm this string as a structural message. But others — it did. Out of a dozen claimed messages, some survived statistical testing, some turned out to be noise, some are still disputed between independent reviewers. Detailed breakdown — companion §7. The emotion remains, the contradiction goes away. Each one — with its own p-value, each one — reproducible on publicly available data. The full breakdown of what passed verification and what didn’t, with all p-values and Bonferroni tables, is in Companion §“Tier 2: embedded messages.”

I deliberately mention this hallucination here and leave it in the text forever. It’s a reminder: an LLM reading a riddle without forensic verification invents things that aren’t there. Please don’t trust any LLM (including the one through which you may be reading this chapter) on the topic of Anna Matrix without the attached companion file. Models have no statistical intuition. They complete patterns. They hallucinate beautiful answers. It’s not their fault — it’s their nature. The companion file exists precisely so that an LLM can distinguish a real finding from its own fantasy.

The community received our work coldly. Anna’s puzzle turned out to be tougher, and the main difficulty was precisely believing that the puzzle exists at all. Come-from-Beyond took a wait-and-see position. He didn’t disprove anything, but he didn’t confirm anything either.

Then Jordan joined us. A genius! With his arrival, the work picked up — it was him who took on the heaviest part: statistical validation. Bonferroni correction, Monte Carlo on tens of thousands of trials, an uncompromising culling of our own hypotheses that didn’t hold up. Of approximately 22,480 initial conjectures, ten survived. Confirmation rate — 0.04%. This is not a weakness of the research, this is the norm for serious forensic work; it only frightens those who have never done such work. The bulk of the forensic-analysis results I cite in this chapter is his work. The full breakdown of all 22,480 hypotheses with Tier 1/2/3 separation, with explicit publication of those refuted — Companion and qubic.church/docs/.

We realized our laboratory needed a website. So we named it qubic.church.

When we launched qubic.church, we had one goal: to turn the chaos of scattered guesses about the “Anna Matrix” into something you can verify. Not “believe or don’t believe,” but compute it yourself on a calculator. The site became a laboratory: open, with code, with data, with the ability for anyone to repeat our calculations. And even look at the visualizations.

And here’s what we found.

What the Anna Matrix actually is

The matrix has a property mathematicians call point symmetry: take any cell and the cell on the opposite side, and their sum equals -1. This works for 99.58% of all 16,384 cells. I suspect that 0.42% is not a data-collection error but a deliberate structure (see further on — it forms column pairs with sum 127, and the sum of all deviations equals 27×43 , which directly connects Bitcoin and Qubic arithmetically). We ran 10,000 random matrices of the same size — not one came even close. The mean for random ones — 0.82%, the maximum — 1.22%. Anna exceeds the maximum by 81 times. Full verification — Companion §1, a 4-line script, verification time 30 seconds.

To put it simply: someone sat down and constructed this object by hand. It wasn't "found in nature," it was made.

Period-4 attractor

Take any random ternary 128-vector — that is, a set of 128 numbers, each of which is -1, 0, or +1. Run it through the matrix: new state = $\text{sign}(M \times \text{old state})$. Repeat. Within about 11 steps the vector falls into a cycle of period 4 — that is, it begins to repeat every four iterations.

Out of 1,000 random initial states — all 1,000 converge into one of exactly two cycles, which are negations of each other. The entire space of 3^{128} possible states — a number larger than the number of atoms in the observable universe — is compressed by the matrix into 8 distinguishable points.

Out of 1,000 trials on shuffled matrices with the same value distribution — zero hits in period-4. The property is destroyed by any permutation of the cells. This is specific to the Anna Matrix, not a general property of signed matrices.

Companion §3 — full breakdown with a null test on five different seeds and reproduction code.

Seven positions

In the cycle, perfect antipodality should hold. It holds in all 128 positions, except seven: [3, 18, 32, 61, 65, 101, 109]. This set is invariant across all 1,000 runs. It is also invariant under five different encodings of the Bitcoin Genesis public key into the initial vector.

The Bitcoin Genesis block contains an incomplete ECDSA signature with seven missing bytes. The same quantitative feature.

Companion §4. What this means — separately, in a Tier 3 section.

What this has to do with Bitcoin

We took the public key from Bitcoin's Genesis Block — those first 65 bytes that Satoshi Nakamoto wrote into the block on January 3, 2009. Everyone thinks it's an ordinary public key. But if you read those bytes as Bitcoin Script (rather than as a key), they form a meaningful sequence of four instructions. And one of them is `PUSHBYTES_72` (the number 72 in the encoding).

Going further. We take that number 72. We look in the Anna Matrix at coordinate [72, 72]. What's there? -27.

We take the diagonal of the matrix as a whole. A direct enumeration of all 128 diagonal cells gives exactly ten that satisfy $|M[b][b]| = 27$:

[72, 73, 74, 79, 88, 92, 94, 95, 119, 120]

Each of these ten numbers is the number of a Bitcoin block mined by Patoshi (presumably Satoshi Nakamoto) in early 2009, with a 50 BTC reward, never spent. The correspondence is verifiable through blockstream.info.

Companion §5 — list of all ten blocks, verification script through the public API, verification instructions.

That is: a byte from a 2009 Bitcoin block points to a cell of a matrix created in the Qubic system in 2022, and that cell contains the number by which exactly those 10 Bitcoin addresses are selected that have lain untouched since 2009. The chain closes on itself across 13 years.

Does it matter?

There are two ways to explain this.

The first — a gigantic coincidence. It happens. But we calculated the probability of each of the coincidences separately and tested it under Bonferroni correction across all of our 22,480 tests. The key Tier 1 facts — period-4 (1000/1000 vs 0/1000), the exhaustive coincidence of ten diagonal positions with ten unspent Patoshi blocks — survive even the most conservative threshold. We deliberately do not compute a combined p-value through multiplication — that's statistically incorrect for dependent tests. Each claim is evaluated separately. Full calculations — Companion and qubic.church/docs/12.

The second — someone knew in 2008 what they would be doing in 2022. And planted mathematical “hooks” in Bitcoin's Genesis Block that are activated through the structure of Qubic. This means that Bitcoin and Qubic are not two independent projects. This is one project, stretched across 17 years.

I lean toward the second explanation.

What we see in real time on the blockchain

And the most striking thing — this is not speculative theory. It's happening before our eyes right now.

In 2025–2026, three tokens from three addresses (POCC, HASV, PXMARAS) appeared in the Qubic network. These three addresses form a closed triangle — no external contacts. And between them pass transactions with mathematically precise amounts:

— HASV sends PXMARAS exactly 26 QU exactly 27 times (26 = the size of the Latin alphabet; 27 = 3^3 , the ternary cube, Cfb's favorite constant); — POCC sends 676 QU (this is 26^2 — the very number of computer slots in the Qubic network); — token volumes follow the scheme 676 billion $\rightarrow$ 676 $\rightarrow$ 1.

The last token (AIGARTH) was issued on March 3, 2026 — exactly 6,268 days after Bitcoin's Genesis Block. And in its message, encoded in binary as Morse code, the string "1A1Z" is embedded — the first and last characters of Bitcoin's Genesis address 1A1zP1eP5QGefi2DMPTfTL5SLmv7DivfNa.

These transactions can't be faked. They are on the blockchain. Anyone can verify them.

What this means for the book

We started with the fact that Anna Aigarth is a bot that answers addition problems. Maria suggested it's coordinates on a grid. We assembled the grid. The grid turned out to be a symmetric structure. The symmetry pointed to hidden messages. The messages pointed to bytes in Bitcoin's Genesis Block. The bytes pointed back into the matrix. The matrix pointed to specific unused Bitcoin wallets. And in parallel, in real time on the Qubic network, someone sends tokens with mathematically perfect amounts, closing the whole construction.

The connection between Bitcoin and Qubic is not a metaphor and not "one project's inspiration by another." It's a documentable mathematical connection. The evidentiary base is in the companion file and on qubic.church.

However, claiming that Come-from-Beyond is Satoshi Nakamoto is technically incorrect. Here we should ask a more precise question: how much is Satoshi Nakamoto Come-from-Beyond? This question, in my humble view, captures the essence most precisely. Cryptographic proof of identity requires a signature with a private key from early Bitcoin blocks. No such signature has been presented by anyone — and until it is, any answer to the question remains in the Tier 3 zone (see companion).

To assemble all the evidence, hypotheses, and strange coincidences into one place — that's work for the next book. Unfortunately, I am ill. And I'm not sure I'll be able to write it.

And is there a point? There's a serious probability that Come-from-Beyond will reveal his cards himself when the time comes.

10.10.2025, 13:39:

"I can reveal this info on 13th of April 2027. And you'll see who is really behind CFB acc."

The time may come quite soon.

In closing this chapter, I want to advise once more: keep a cool head, don't fall for illusions, and verify everything yourself. We keep the information on qubic.church, and the inquisitive reader can check everything personally. The companion file — Anna_Matrix_Companion.docx, on the same page — is made for exactly that: a compact forensic appendix, readable in 20 minutes, verifiable on a personal machine in 2–5.

I'm grateful to Maria, Jordan, and the entire Shizofam for this journey and for their help in the work of Qubic Church. P.S. If for you it's still critically important to prove that Cfb = Satoshi, I recommend reading our friend's blog: <https://x.com/SatoshiCfB>

C H A P T E R 14

What the Deploy on April 13, 2027 Means



I'm writing this chapter approximately twelve months before the planned date. By the time you read it, the date will either have passed or still be ahead — the exact relation depends on what month the final edition of this book came out in. Which side of the date your copy ends up on changes how everything below reads. But it doesn't change what I can write right now. And that “right now” sets the register for the whole chapter: a calm analysis of what's known at the time of writing, plus three explicitly described outcome scenarios.

The date itself appeared in the public field on December 6, 2025, when Come-from-Beyond wrote a short line in one of the chats:

“And ETA for Aigarth was set as April 13, 2027.”

A month later, on January 13, 2026, he expanded it more precisely:

“5 years since Qubic launch is just an ETA. And goal of Aigarth is AI, not AGI. Difference between AI and AGI is same as between a car and a BMW.”

This remark should be read in three layers. First: “just an ETA” — Estimated Time of Arrival, an estimate, not a contract. Not “promised,” not “guaranteed,” not “fixed in code.” An estimate based on what the team sees right now. Second layer: the goal is AI, not AGI. An anti-hype framing, drawn directly: Aigarth doesn't claim to be a universal artificial intelligence capable of solving any human-level task. The goal is a functional AI that knows how to do what it was trained for. Third layer — the car/BMW analogy. Both a car and a BMW solve the same basic task: moving people. But these are different categories of ambition — not different gradations of the same thing. The Aigarth goal belongs to the first category.

What technically happens on April 13, 2027

The Aigarth deploy is the transition of the framework, published in July 2025, from test mode into production. That very framework: “This framework will be added into the core to host AIs as smart contracts.”

The structural definition of what exactly is being integrated, CfB formulated as far back as November 25, 2023:

“Aigarth would be a standalone project if it didn’t use Qubic’s UPoW and wasn’t intended to run AI as Qubic smart contracts.”

This sentence pinpoints Aigarth’s place in the architecture. Aigarth isn’t an application on top of Qubic, and it isn’t a service that uses Qubic for payment. Aigarth is structurally tied to two specific mechanics of the network: to UPoW, which in chapter eight was unpacked as a way of forcing every computation to work on two fronts at once.

What changes at the moment of deploy. Until this date, the Aigarth research-and-development infrastructure runs on the network’s 676 nodes as part of the mining load: nodes generate and train structures. After deploy, a new layer is added to the network — the ability to host AI as smart contracts, accessible through Qubic’s standard SC mechanics. Applications become callable from contracts using Oracle Machines to feed in external data. This isn’t “flip a switch” — it’s a structural event taking, in the vicinity of the date, hours, days, or weeks. April 13, 2027, is the planned date of the event, not the exact time in seconds.

The concrete scenario after deploy looks like this. A user with QU on their account initiates a smart contract call in which the request goes to an Aigarth AI. The contract receives input data through Oracle Machines (if the task needs it — for example, for the price-prediction tasks CfB mentioned in his July 2025 remark as the nearest use-case for Anna). Aigarth inference is performed by the same set of nodes that performs UPoW, as part of their computational work. The result is written back into the contract state. This is AI inference as another type of computation that the network knows how to perform — alongside DOGE mining, message passing, and transaction validation.

When I wrote that for Come-from-Beyond crypto would be infrastructure for Aigarth, and not the other way around — that sounded like a paradox. On April 13, 2027, the paradox stops being a paradox. The network becomes what it was built for. Not “crypto that also does AI.” But “AI for which crypto is the load-bearing structure.” This transition is the book’s main structural lock, and it converges on a single date.

Arbitrator deactivation — the second half of the same event

April 13, 2027, is a two-part date (or even more?). The first half — the Aigarth deploy, unpacked above. The second — the deactivation of the Arbitrator. These aren’t two different events that happen to coincide on the calendar. They’re two parts of one structural transition, declared on a single date back in 2022.

In chapter five I showed three departures of Come-from-Beyond. NXT, founded in 2013 and abandoned in 2014 after the conflict with Sørnstebo. IOTA, co-founded in 2015 and left in 2017–2019. The first iteration of Qubic from 2018, abandoned by 2020. In chapter ten I announced a fourth departure — this time with a calendar date announced in advance and with the mechanism built directly into the network’s source

code. Now, twelve months before this date, the method is being applied for the fourth time — and the only one of the four whose moment of application is hardwired into the code in advance.

What Arbitrator deactivation means structurally. Chapter ten unpacked this in detail, including through CfB’s central remark from April 19, 2023:

“In best variant Arbitrator won’t be needed at all. Its only drawback - Aigarth will lose access to miners hardware and UPoW will become PoS.”

The concrete mechanism can be described in three points. First: the Arbitrator’s key is destroyed. The compile-time constant in the source code remains, but a signature from a destroyed key cannot be produced. No one, including the team itself, can use the Arbitrator anymore to change the composition of computer nodes or forcibly intervene in the network’s operation. This is the same logic CfB stated back on July 23, 2024: “Arbitrator can do almost nothing, source code guarantees it” — guarantees from code, not from promises. After deactivation, this code simply works without a live key.

Second: the network keeps running without the Arbitrator. This isn’t a shutdown, not a functional failure, not an emergency mode. The fallback role is no longer activatable, that’s all. Chapter ten described the Arbitrator as scaffolding — temporary support; on the date that support is removed, the building stands on its own.

Third: UPoW structurally becomes PoS-like — by the precise wording of CfB himself. Without the Arbitrator, the network no longer has an external source of new entry points into the computer list. The list is updated only through internal rotation, based on scoring results and the weekly epoch. Good entry points stay, bad ones drop out, new ones come in only through this internal mechanic. This is a known structural consequence, explicitly stated in the same 2023 remark to which the 2027 discussions return.

The uniqueness of the fourth departure is not in the method itself. The method is the same: build the infrastructure, hand it off to run independently, leave. The uniqueness is that the moment of departure isn’t chosen after the fact, as in the three previous cases, but announced in advance and built into the code five years before the date. A pre-fixed date changes the status of the method: from “I’ll step aside when it becomes possible” to “by such-and-such a date this will be feasible, and I commit to that in advance.” A falsifiable public commitment is a different quality of the same action.

It’s worth clarifying what “hardwired into the code” means. The Arbitrator’s compile-time constant isn’t a string in a configuration file that you can change by restarting a node. It’s a value compiled into the binary of every version of qubic-core distributed among computer nodes. To replace the key would require recompiling the network and convincing a sufficient number of nodes to switch to the new version — that is, going through the same quorum mechanism described in chapter six as the network’s way of making decisions. Destroying the key after deploy means that even this possibility disappears: there is no longer a private key capable of producing a valid signature from the constant already sitting in the binaries. The network can’t “bring back” the Arbitrator

other than by recompiling itself with a new constant and a new key — and that’s no longer the return of the previous role but the creation of a new mechanism from scratch, requiring quorum agreement. Deactivation in this sense isn’t a turn-off, it’s the destruction of the possibility.

April 13, 2022 — April 13, 2027: five years not by accident

The starting point of the Aigarth work itself was an even earlier moment. On February 20, 2024, CFB wrote:

“In December 2021 we ran PoC which demonstrated viability of Aigarth approach.”

December 2021 — Aigarth’s Proof-of-Concept, an internal test that gave the team confidence the approach works. This is the first of the milestones listed in chapter twelve; the second — December 4, 2024; the third and fourth — July 2025. From the December 2021 PoC to the planned deploy in April 2027 — approximately sixty-four months. From the real start of mining on April 13, 2022 — exactly sixty months, five years.

This double count matters. Sixty-four months from December 2021 is the period of a full research cycle: from PoC to production. Sixty months from April 13, 2022, is the period during which the network’s infrastructure moved in parallel with the research work. If the PoC set “is it possible,” then April 13, 2022, set “exactly how.” UPoW performance, scoring, computer node rotation through weekly epochs, the Arbitrator’s behavior — all of this was calibrated on the same time scale as the Aigarth research. On the final date they’re meant to meet at one point: the research delivers a production-ready framework, and the network delivers compute capacity on which the framework starts to run. Five years is the time both sides need.

Five years isn’t a marketing number. It’s a functional window. On July 31, 2024, CFB described the expectation in these words:

“In 2 or 3 years we get Aigarth.”

Said in the middle of 2024, this window is July 2026 / July 2027. April 13, 2027, falls roughly in the middle of it. That is, when the team says “in two to three years,” this date isn’t an aggressive lower bound and isn’t a backstop upper bound; it’s a point inside the announced range, chosen by calendar symmetry with the real start of mining.

What’s needed for the five-year period. First — to pass Aigarth’s key milestones, and they have all already been passed: PoC in December 2021, midpoint in December 2024, breakthrough in July 2025, the GitHub publication on July 22, 2025 (all of this is unpacked in chapter twelve, I won’t repeat). Second — to scale up mining capacity to a level at which AI inference through smart contracts becomes economically meaningful for miners. This is the second half of the five-year window, and the next section is about it.

And one more parallel worth saying out loud, because it sits behind the words “five years” and explains a lot. On April 13, 2022, at the moment of mining start, much of the code didn’t yet exist. The same may be true for the final date — the optimizations for compute capacity are being written right now, at the moment this chapter is being

written. That doesn't mean the plan is in danger. It means that in 2022 and in 2027 alike, the work closes precisely on the date — and in the first case that worked out, while in the second it's so far only planned.

In chapter one I compared the ambition of Qubic with Darby's ambition in 1709. Coalbrookdale produced the first tons of coke iron then — but the broad adoption of the method in industry took decades. It's the same here. On the verification date we'll find out whether a claim of that scale is justified. And the answer doesn't have to be binary. It can be "it worked, but a year later." It can be "it half-worked." It can be "it worked in an unexpected form." Any of these variants is valid, unlike the falsely binary "success / failure." A falsifier for an ambition of this scale is always more complicated than "yes / no."

What "happened" and "didn't happen" mean

Scenario A. The deploy succeeds and Aigarth is active. On the date, plus or minus a few days, the framework moves into production. Anna, Maria, and subsequent AIs run as smart contracts, accessible through Qubic's standard SC mechanics with Oracle Machines for feeding in external data. The Arbitrator's key is destroyed, the event publicly confirmed. The network keeps running without it — the computer list is updated through UPoW, scoring, and weekly-epoch rotation. What this means for the book's framing: chapter four is closed empirically — crypto is observably working as infrastructure for AI. What it doesn't mean: AGI hasn't been created (and that's already clarified by the February car/BMW analogy), not all of Aigarth's technical tasks are solved, the first AI applications may remain quite limited. The book in this scenario hasn't "won the argument." It registered a falsifiable framework, and the framework was confirmed — that's a different quality of result.

Scenario B. The deploy went through technically, but Aigarth isn't showing what was claimed. The framework works; smart contracts that host AI are technically functional. But AI applications are weak or absent in production: the network's compute capacity isn't sufficient for meaningful inference. The Arbitrator's key is destroyed all the same — that's a structural event hardwired into the code and happens independently of how active the Aigarth applications are. What this means for the book's framing: chapter four is closed partially — the infrastructure exists and works, but Aigarth as a research result hasn't yet matured into a production-ready AI. This is a valid scientific result: the paradigm works at the infrastructure level, and the question shifts from feasibility — that was closed by CfB himself back in February 2026 — to economic scale. And that's a question the book begins to answer in the next chapter.

Scenario C. The deploy is shifted. On the planned date, the event doesn't happen as announced. Possible reasons: a technical blocker (blocking bugs were found in the framework, in the core, or in scoring); compute insufficient — the team deliberately postpones because there's no point in deploying without enough capacity; operational

coordination requires another quarter or two of development. The Arbitrator’s deactivation in that case either happens on plan — a structural event in the code, not tied to the success of the Aigarth deploy — or also shifts, which requires a team decision. It’s possible a new ETA is announced — along the lines of the January 2026 remark that five years from Qubic’s launch is, after all, an estimate and not a deadline. What this means for the book’s framing: chapter four remains an open question, carried over to a new date. The falsifier from chapter one partially fired: the date passed, the event didn’t, and that too is a valid observation. A shift of a year or two is standard-scale for infrastructural transitions: the first tons of coke iron in 1709 weren’t produced on the first try either, and broad adoption of the method took decades. A calendar date sets an expectation, but doesn’t determine the answer.

What’s important to say explicitly about these three scenarios. They are not probability distributions. I’m not writing “A is most likely; B less likely; C unlikely.” I’m not predicting the outcome. I’m registering a falsifiable framework: on the verification date, one of these three scenarios will be observed (or a close combination of them), and each gives information.

Scenario A is “Qubic’s victory.” It’s the observation that the infrastructural claim of the book matched infrastructural reality on the verification date. Scenario B is not “Aigarth’s failure.” It’s the observation that the infrastructure works, while the research result hasn’t yet matured to production quality — and that shifts the focus of the next round of discussions from feasibility to economic scale. Scenario C is not “deception of the reader.” It’s the observation that the engineering side requires more time than estimated — and that doesn’t refute CfB’s February 2026 feasibility claim; it remains in force for a new ETA.

I emphasize this because in discussions of crypto projects another framing often operates: either the project “won” or it “failed,” and both verdicts are made emotionally, without structural analysis. That framing is unfit for a scientific posing. Aigarth is a research task, and any research task has several valid outcomes. A falsifiable framework is a way of stating them in advance, rather than reactively picking a frame to fit the result.

In February 2026, CfB stated a conditional commitment: if AI is at all possible within Earth’s current energy budget, Aigarth will achieve it — and the team sees the cards it’s playing as sufficient. That formulation is precise as a falsifier. Either AI is possible within Earth, and then Aigarth is a serious candidate for a path to it. Or it isn’t, and then the question shifts from the paradigm to physics. The falsifier approach doesn’t promise a result — it promises that any outcome will give an answer.

I’m writing this chapter before the date. By the time you read it, the date has either passed or is still ahead — that was already said at the start of the chapter, and I come back to it deliberately. If it has passed — you have the result, and every reader can compare it with the three scenarios above. If it’s still ahead — you have a reason to wait. The book is honest in either case, because it doesn’t claim to know the outcome. A

falsifiable framework isn't a way to win an argument. It's a way of agreeing in advance that any outcome will yield information.

The main risk – compute capacity between April 2026 and April 2027

The main risk for Aigarth, in CfB's words: "Aigarth depends on miners" (November 12, 2025) and "Qubic is doomed to succeed. Aigarth? Depends on miners." (February 7, 2026). It's not a technical dead end, not a scientific gap, not regulatory pressure. It's a question of network economics: how much mining capacity will plug into Aigarth tasks by the moment of deploy.

To grasp the scale of what has to happen in the twelve months before April 2027, there's a concrete number. On January 20, 2026, CfB wrote in a discussion of internal tests:

"We are mining new Aigarth algo tomorrow, 10x bigger than Doge, but not obvious."

Ten times bigger than DOGE mining. DOGE mining is Phase 2 of the network's current configuration, on which a significant share of mining capacity runs at the time this chapter is being written. The Aigarth task is an order of magnitude bigger. This isn't a casual comparison; it's a concrete technical parameter, stated in the discussion of a specific test. And it's exactly that number that sets the scale of the ramp-up: the network performing X compute on DOGE today must, by April 2027, be able to do roughly ten X on Aigarth.

In chapter eight I unpacked UPoW as a way of forcing one computation to work on two fronts at once. After the Aigarth deploy, the same principle is applied to a new pair of tasks: the compute that today is directed toward DOGE mining as part of Phase 3, after deploy, is redirected to Aigarth inference with the same economic incentives. Structurally, that is, the network is ready — UPoW supports multitasking computation. The question is whether miners will choose such a redirection, and how many of them will plug in by the moment when the choice becomes meaningful.

It's worth clarifying here what depends on the miners and what doesn't. What doesn't depend — the very mechanics of how Aigarth tasks are counted in scoring and how rewards are accrued for them: that's part of the protocol, published in the github.com/qubic/core repository and not subject to unilateral correction. What does depend — every miner's economic choice: whether to switch their GPU/CPU power from DOGE payout to Aigarth payout, and in what proportion. That choice depends on the ratio of reward levels between the two tasks, which the network produces every epoch, and on miners' expectations of how that ratio will move further. The Aigarth team controls neither the first parameter directly (it's a function of many sides' participation) nor the second (it depends on individual strategies). What the team does control — the technical readiness of the framework for deploy and the quality of AI applications that the framework will host. Everything else is an external parameter, and that's what CfB designates with the words "depends on miners."

The main risk for Aigarth is not feasibility (that, as I said above, CfB closed in February 2026). The main risk is scaling the network to a level at which Aigarth inference becomes economically meaningful for miners. That risk isn't solved by science or engineering — it's solved by network economics. And there's already a partial answer to that question, from August 2025.

Between plan and proof

I unpacked Axelrod's three conditions as applied to research work itself — iteration, memory, transparency. The book iterates (its chapters get rewritten). The book holds memory (the hook tracker records every claim still waiting to be closed). The book is transparent (it announces its dates and scenarios in advance, before it can verify the answer). A falsifiable framework is, in essence, the same three conditions applied to its own narrative.

On May 14, 2024, CfB wrote on Twitter:

"I have realized that Artificial Intelligence will not be created, it will emerge. With help of Qubic miners."

"Will emerge" is the verb of an observer, not a commander. Not "we'll create AI," but "AI will emerge — with the help of those who work on the network." This is the very emergent framing I unpacked: AI as the result of collective mechanics, not as a product designed top-down. And this framing isn't 2024 marketing; it's the position the team has worked from since the beginning. Since 2022, when no one outside a narrow circle knew that this emergence would one day surface in the public field; since December 2021, when the first PoC was launched and the team understood that the approach works; since the moment CfB decided that the "build and leave" method could be hard-wired into the code in advance, and committed in advance to leaving.

Emergence doesn't arrive on its own. It arrives if there's an environment in which it can manifest, and time for that environment to take shape. Earlier chapters described that environment: 676 votes, weekly epochs, UPoW, scoring, smart contracts, governance. Then they described not the environment but the process going on inside it: research, milestones, peer review, artifacts with verifiable properties. Then the moment when the environment and the process must meet at one public point. If CfB is right in his 2024 observation, then on this date the meeting happens — not as an announcement, but as a change in regime in which the network begins to operate.

PART V

The Proof

CHAPTER 15

The War That No One Waged



August 11, 2025

On August 11, 2025, within a window of one hundred and twenty-two consecutive blocks on the Monero network — from block 3,475,729 to block 3,475,850 — a pool that brought together computing power under the Qubic banner mined sixty-three blocks. That is fifty-one and six tenths percent. Fifty-one percent is the technical threshold past which a pool, while continuing to hold its share, can rewrite the network's transaction history, reverse confirmed transfers, and censor selected addresses. None of the participants in Qubic did any of that during those twenty-four hours. A few days later, the pool's share of Monero began to decline — not because it had been pushed out, but because the team itself had shifted priorities to other tasks. No Monero forks happened. No user losses were recorded. On the Kraken exchange, XMR deposits were temporarily suspended for several days, then resumed. The price of Monero pulled back about sixteen percent over the week and started recovering.

If you read only the dry numbers, nothing remarkable happened. One pool briefly exceeded a key technical threshold on one of the networks, and didn't take advantage of it. But the numbers next to those numbers are worth saying out loud.

Monero's market cap in August 2025 was about six billion dollars. Qubic's market cap was about three hundred million. A ratio of one to twenty. A network with twenty times less market cap restructured, over the course of three months — from May to August — a network with twenty times more. Not by attack, not by buying out infrastructure, not by lobbying. By an economic offer. Monero miners moved over to the Qubic pool because it paid, on average, three times more than direct mining on any other pool. They were not deceived. They were not coerced. They saw a better payoff matrix and made a rational decision.

This is the chapter about what exactly happened between May and August 2025, and why what happened matters more than the dry numbers suggest. Because everything this book has said earlier about iteration, memory, transparency, the four properties of Tit-for-Tat, and Axelrod's three conditions — all of it, before August 2025, was either 1980s mathematics, or simulation modeling, or architectural description. After August 2025, it is an empirical fact on a network with six billion dollars of market cap.

That's where I'll start.

How the pool grew from two percent to fifty-one

In May 2025, Qubic's share of Monero's total hashrate was less than two percent. It was an experimental pool, set up by the Qubic team as a test of one idea — that the computing work the Qubic network produces anyway as part of its own UPoW could simultaneously be used for mining on external networks, in order to convert the rewards back into the Qubic economy. By the end of June, the share had grown to about ten percent. By mid-July — to twenty. By the end of July, Qubic was the number-one Monero pool and reached around forty percent of the hashrate at peak moments. On August 11, fifty-one and six tenths percent in the window of 122 blocks.

From two to fifty-one in three months. This needs to be marked as a pace — a pace of migration, not a pace of building infrastructure. The infrastructure was ready in advance: the pool could mine Monero by May. What changed from May to August was not the pool's code, but the number of miners who chose precisely it. Each of them made an individual decision, and the sum of those decisions added up to a growth curve with a very characteristic shape: a slow start, gradual acceleration, a tipping point, after which new miners come because they hear from other miners about a pool that pays more.

The key moment that explains the pace is the vote that took place inside the Qubic community in early summer 2025. Originally the scheme looked like this: the pool mines Monero, the XMR earned is sold for USDT, the USDT goes to buyback and burning of QU tokens. All the upside from external mining is absorbed by the deflationary mechanism of the Qubic network; a miner who switches into the pool from any other Monero pool earns roughly the same as before, plus an indirect upside through the appreciation of QU that he holds. That was enough to grow the pool from two percent to ten over May–June, but not enough to push past twenty.

After the vote, the scheme changed. Half of the USDT still went to buyback and burning — the deflationary mechanism preserved, but in a softened form. The other half went directly to Qubic's computer nodes as additional income. That second half is what turned the pool from an experiment into a magnet. After the restructuring, a miner who switched to the Qubic pool started earning, on average, three times more than from direct Monero mining on any other pool. This isn't a marketing line; it's a measurable parameter, verifiable from the payouts to specific addresses. The pool honestly paid what it promised, because the promise was built into the code.

Dan Dadybayo, a researcher at Unstoppable Wallet, in an analysis of the situation published in August 2025, formulated it this way: Ivancheglo created a game of incentives in which Monero miners may voluntarily surrender the network if they see a better deal. "Voluntarily surrender the network." Six words that describe the nature of the event more precisely than most of the technical write-ups from that summer.

This is a formulation worth unpacking, because it is not a metaphor. A Monero miner who switched to the Qubic pool in July 2025 did not violate any rules. He did not attack Monero. He did not help attack Monero. He made a market decision — to switch his hashrate to a pool that pays more. The same kind of decision that each of them had already made dozens of times before, when choosing between different Monero pools. From the standpoint of Monero’s rules, the actions of the miners who moved to Qubic are indistinguishable from ordinary migration between pools. From the standpoint of Monero as a coordination system, those actions resulted in one pool accumulating a majority of the hashrate. Not because it forced anyone. Because it offered more.

In the first chapter, in its earliest version, I wrote that the new technology “works in parallel, not instead of” — about gas lamps that appeared alongside coal lighting, not in place of it. Back in 2009, that was said about Bitcoin as “Murdoch’s Gas that no one had yet collected into a pipe.” In August 2025, the same principle showed up differently. No one was destroying Monero. Monero as a network kept running, and keeps running; its privacy isn’t broken; its protocol isn’t broken. In parallel with it, a pool appeared in which miners earned three times more, and some of them moved over there. The network didn’t start working worse as a technology because of that; it just temporarily lost the monopoly on its own miners’ attention. After Qubic reduced its activity in Monero, the community migrated to P2Pool, a decentralized pool, and the pluralistic state was restored. The old world works. The new world also works. No one was destroying anyone.

This is parallel architecture in infrastructural form. Not “the better system defeats the worse one through a contest.” But “the better system offers free migration into itself, and some of the participants migrate, and some stay, and that’s normal.”

Three conditions at the scale of six billion

In the second chapter I went through, in detail, the exact conditions under which the four-line-of-code strategy — Tit-for-Tat — becomes optimal. Axelrod and Hamilton, in their 1981–1984 papers, formulated this as three necessary conditions. The game must be repeated. Participants must remember exactly what their opponent did in the past. Behavior must be visible — either directly, or through a reputational channel. Remove any one of the three, and the optimal strategy shifts from cooperation to pure egoism.

In August 2025, all three conditions were met at industrial scale.

Iteration. Every Monero block is a separate round of the game. A block comes out roughly once every two minutes. Over the three months from May to August, the network ran somewhere on the order of sixty-five thousand rounds. No miner deciding which pool to choose was acting in a one-shot situation. Every choice of a pool was a choice of a partner for the next tens of thousands of rounds, and all participants knew this. The shadow of the future — not as a metaphor, but as a quantitative parameter —

was so long that any consideration of momentary gain from cheating became negligible compared to the long-term profit from honest participation.

In the seventh chapter I introduced the concept of an epoch as a round of an iterated game. The discussion there was about Qubic's own network, where an epoch lasts one week and the composition of computer nodes is updated by the results. Here, in Monero, the same principle works at an even larger scale and on someone else's network. Every two minutes — a round. Every miner — a player in that round. Every choice of pool — a move in the game, observable by everyone else.

Memory. The Monero blockchain is a public history in which every block is forever recorded together with the address that received the reward for that block. It is impossible to “forget” how many blocks each pool mined last month. It is impossible to misrepresent the situation after the fact: if the Qubic pool had actually paid less than it promised, this would have been immediately visible from the ratio of mined blocks to public payouts. If there had been some hidden channel for redistributing rewards, this violation of the proof-of-work logic would have shown up in orphan blocks or in publication delays. The network's memory does not depend on the conscientiousness of its participants; it is built into the protocol.

This is a level of memory unattainable in any of the environments Axelrod had in mind when he wrote about real-world analogs of his laboratory conditions — kin groups in primates, small communities, trade guilds. All these systems had memory, but biological and social memory: limited by the number of people capable of personally remembering one another, and by the time those people are alive. The blockchain removes both limitations (this matters!). The network's memory does not fade with time and does not depend on the number of participants. It is a memory tool that appeared after Axelrod had finished his main work, and appeared not as an implementation of his ideas, but as infrastructure for monetary transfers. But it was precisely this infrastructure that turned out to be the environment in which the conditions of his theory are met strictly.

Transparency. The hashrate of every pool in Monero is publicly visible. You don't need to be a member of the community, you don't need a subscription to a specialized service, you don't need to wait for reports. Anyone with a web browser, anywhere in the world, can go to any of the public Monero monitoring dashboards and see in real time which pool has what share. When the Qubic pool crossed twenty percent in July, this was known within hours. When it crossed forty percent in early August, this was also known within hours. Transparency is not a function of the Monero community, not the result of journalistic work, not the consequence of conscientious disclosures by the pools. It is simply a structural property of a public proof-of-work network.

When all three conditions are met, Tit-for-Tat becomes not just an optimal strategy — it becomes what is described in one of Axelrod's formulations as an “evolutionarily stable strategy.” A strategy the system converges to, given enough rounds. The Monero community reacted instinctively to Qubic's dominance precisely along the logic of a

mirror: miners began moving to P2Pool, discussions of a possible hard fork to change the RandomX algorithm gained traction, the Kraken exchange temporarily suspended XMR deposits. None of these responses were coordinated “political” decisions. They are the spontaneous reaction of a distributed environment in which the three conditions are met, to a temporary disruption of balance. After Qubic reduced activity, the reaction calmly subsided. There were no consequences for miners who continued to work with the pool; no one “punished” them. Monero as a coordination system simply returned to a pluralistic state.

This is industrial validation of the theoretical framework laid down in 1980. Not a controlled experiment in a lab at the University of Michigan. Not a computer simulation. A six-billion-dollar network with thousands of independent participants, in which all three conditions are met strictly, and in which the reaction to disturbance played out exactly as a theory written forty-five years before the event would have predicted.

What Qubic could have done and didn't

By early August 2025, when Qubic's hashrate in Monero had come right up to fifty percent, the project's team faced an unpleasant technical question. Not “will we hold fifty-one percent” — that was only a question of a few days of work and accumulating hashrate. But “what will we do afterwards, if we actually hold it.” On August 3, 2025, Come-from-Beyond posted, in one of the public channels, a remark that captures the choice the team was facing better than any other material from that summer:

“Right. If we wanted we would be DDoSing back. Or even do a PR attack by warning exchanges that once we assert domination we'll be doing random chain reorganizations while censoring transactions, so any XMR deposit to exchanges would be double-spent with 50% chance. Even DEXes wouldn't help coz they would be victims of double-spending too. And XMR price would plummet to 0. But, as @dkat said, we frens and won't do that all.”

This remark is worth unpacking, because it contains six concrete attack vectors, each of which was technically available to the Qubic team in the first week of August 2025, and each of which the team consciously did not use.

The first vector — retaliatory DDoS. From August 3 to August 10, the Qubic team was under a distributed denial-of-service attack that pushed its own network's hashrate down from two and a half gigahashes per second to eight hundred megahashes at the worst point of the attack. CfB publicly accused one of the key XMRig developers of coordinating the attack; that developer denied it; the source of the attack remained formally unconfirmed. What is confirmed is that the attack went on for more than a week and substantially complicated the holding of hashrate at the needed scale. The Qubic team could have responded symmetrically: use its own infrastructure to coordinate a DDoS attack on its opponents. It didn't.

Second — a PR attack on exchanges. After crossing the fifty-one-percent threshold, the team could have sent a warning to all exchanges that accepted XMR deposits, saying that from now on any confirmation of a transfer at the exchange should be considered unreliable with a fifty percent probability. This is a technical truth: while holding majority hashrate, a pool can produce chain reorganizations that roll back previously confirmed transactions. Exchanges, on receiving such a warning, would have been forced to defend themselves by temporarily suspending operations with XMR. This would have destroyed Monero's liquidity on exchanges without a single actually reorganized block — a single public warning from a majority-holding pool would have been enough. Not done.

Third — actual chain reorganizations. Technically the most direct use of the majority position. The pool keeps its blocks secret, accumulating an internal chain longer than the public one, and periodically publishes it, forcing the network to roll back to a new version of history. This method, known as selfish mining, was described in the 2014 paper by Eyal and Sirer; it actually was used as part of the strategy of the final push to fifty-one percent. But selfish mining in the final-capture phase is a tool for accumulating more of one's own blocks relative to the total. Chaotic reorganizations after the capture are something else: the network turns into an environment in which transaction history stops being stable. A few public reorganizations would have been enough to seriously undermine Monero's functionality as a payment system. Not done.

Fourth — transaction censorship. While holding majority hashrate, a pool can choose which transactions to include in its blocks and which to ignore. A few weeks of such selective mining policy is enough — and any address or class of addresses can be structurally cut off from the ability to confirm transfers. Technically this is not an impossible action; it is regularly discussed as a risk scenario for proof-of-work networks. Not done.

Fifth — an attack on exchange deposits with a fifty-percent probability of failure. This is a combination of the reorganization strategy with exchange mechanics. After holding majority, any XMR deposit sent to any exchange could, with high probability, be reversed through chain reorganization either before or after the exchange credited it to the account. The exchange loses either funds or user trust. On a network with millions of users, such behavior is destructive, and DEX systems do not provide protection, because they too operate on top of the same chain history. Not done.

Sixth — a consequence of the first five, explicitly stated by CfB: the price of XMR will collapse to zero. This is a forecast, not a desire. CfB is not asserting that the price will necessarily fall to zero — he is describing the limit scenario that would result from consistent application of the arsenal. The specific depth of the collapse depends on many variables; what matters in principle is that what was being described was a space of possible outcomes, not a definite result. Not done.

Six attack vectors. Each technically available. Each would have used the pool's accumulated advantage against its former colleagues on the network. Each remained un-

used. CfB's closing line in the same remark, explaining the choice: "as @dkat said, we frens and won't do that all." Dkat, the developer of QEarn and lead developer of the Monero operation, had by that point already articulated the team's position: we don't finish off partners from whom we've gained an advantage.

That Qubic did not destroy Monero is not a consequence of the destruction being impossible. Not a consequence of the team lacking the necessary infrastructure. Not a consequence of external restraint from exchanges, regulators, or the community. It is a decision by the Qubic team, made at the moment of maximum advantage and explicitly recorded in the public record a week before the advantage was realized to its logical limits. Six tools for destroying the network were lying next to the team while it held majority hashrate. The team did not pick up a single one.

Four properties, applied in reality

In the second chapter I went through the four properties of Tit-for-Tat that made it the winner of the 1980 tournaments. The properties were articulated by Axelrod from his analysis of every game played in the first tournament: niceness, retaliation, forgiveness, transparency. Each of them was implemented by Tit-for-Tat through Anatol Rapoport's four lines of code. Each is theoretically independent of the others, and each is necessary for victory in an environment with the three conditions met.

In August 2025, all four properties showed up at industrial scale — not as metaphors, but as observable actions of the Qubic team toward the Monero network and its community. It's worth fixing this explicitly, because the correspondence is exact.

Niceness. Tit-for-Tat never defects first; on the first move, it always cooperates. Qubic did not attack Monero first. Qubic's original strategy toward Monero was not an attack but an offer: a pool that paid miners more for the same work. On the first move — cooperation in the form of a better economic offer. There was no plan to attack anyone. That, as a result of this offer, a de facto majority of the hashrate ended up captured is a consequence of the offer being convincing to the miners themselves, not of a prior intent to harm the network.

Retaliation. Tit-for-Tat responds to defection immediately — not with delay, not with a margin, not with probability, but exactly on the next round, exactly once. Qubic responded to the DDoS attack that ran from August 3 to August 10. But it did not respond symmetrically — not with a counterattack, not by destroying someone else's infrastructure, not by punishing specific opponents. It responded by continuing the operation. The hashrate that had dropped to eight hundred megahashes under the attack recovered; the pool held its share in Monero; on August 11, the fifty-one-percent threshold was crossed. That is retaliation in its precise sense — refusal to be punished for the original action. But without escalation and without transferring the punishment onto participants who had nothing to do with the attack.

Forgiveness. Tit-for-Tat doesn't hold grudges; if the opponent has returned to co-operation, it returns too. Right away. After demonstrating the possibility of capture, the Qubic team did not move to lock in the advantage. The pool's hashrate in Monero began to decline shortly after August 11 — not because of external pressure, but because of an internal decision to switch compute priorities to DOGE-mining and Aigarth. By August 17 the pool no longer held majority. This action is functionally equivalent to “returning to cooperation”: the team retreated to the network's pluralistic state, with no external reason to do so other than the fact that the goal of the experiment — to demonstrate the mechanics — had been achieved, and continuing to apply pressure further would have been a transition from testing to actual harm.

Transparency. Tit-for-Tat is predictable: it's easy for the opponent to understand what it does and why. The Qubic team ran the entire operation in the public field. The plan was announced — that's visible from posts going back to late May 2025. Technical details were discussed openly, including the August 3 remark about the available attack arsenal. The pool's hashrate metrics were publicly visible, like those of any other participant in the Monero network. The final report on August 12 on Qubic's official blog set down the three theses the team considered proven by the experiment, and explicitly announced that the experiment was complete. No hidden channels of coordination; no private negotiations with exchanges; no deals with individual players. Transparency is not a result of honest intentions; it is a structural property of how the operation was run.

Rapoport's four properties, applied to a real economy at the scale of six billion dollars. Not in a lab, not in a simulation, not in a training environment. On a live network, in real time, with real consequences.

In the second chapter I closed the discussion of Axelrod with the observation that the same formula — “treat others as you would have them treat you” — had been independently arrived at by five people over the course of two and a half millennia. Confucius around 500 BCE. Hillel the Elder in Jerusalem around 30 BCE. Jesus of Nazareth around 30 CE. Kant proved it logically in 1785. Rapoport proved it mathematically in 1980. Five independent discoveries of one formula, with intervals of centuries. On August 12, 2025, this formula received its sixth confirmation — not as a formulation, but as an industrial experiment in which precisely the properties Rapoport sewed into four lines of code showed up exactly as the theory predicted. Bravo, Cfb!

DDoS as a test of the Retaliatory property

It is worth pausing briefly on the August 3–10 episode, because it is the cleanest test of one of the four properties in a live environment.

The DDoS attack against Qubic's infrastructure began roughly when the pool's hashrate in Monero reached forty-five percent. Over seven days of attack, the total hashrate of the Qubic network fell from two and a half gigahashes per second to eight

hundred megahashes at the worst point. That means Qubic’s own network — the network through which computing work was supposed to be distributed between Monero-mining, DOGE-mining, and UPoW tasks for its own consensus — operated for several days in a regime markedly inferior to its usual one. Some peripheral services of the project were temporarily shut down. The financial load on the infrastructure increased. Coordination between computer nodes became more complicated.

And yet the operation in Monero was not interrupted. The Monero hashrate declined, but not critically; the pool held its number-one position for all seven days of the attack. On August 11, when the attack was already winding down, the fifty-one-percent threshold was crossed. The very fifty-one and six tenths percent in the window of 122 blocks that frame the rest of the chapter.

This is a test of the Retaliatory property in its precise formulation. Not “got revenge with the same thing on the same actor” — Qubic didn’t do that. But “kept acting in such a way that the aggression did not achieve its goal.” In the terms of the 1980 tournament: Tit-for-Tat in an environment of aggressive strategies does not surrender and does not retreat; it keeps playing its own game, responding to each act of aggression exactly once — by continuing the cooperative game with those who are ready to cooperate. The Qubic team in August 2025 passed this test literally: it responded to aggression against its own infrastructure by carrying the operation in Monero through to its end, without breaking the chain of public commitments to the miners who had moved into its pool.

The twelfth chapter introduced the concept of emergent safety — protection that arose not as an engineered function but as a byproduct of correctly built mechanics. The DDoS episode of August 2025 shows the same phenomenon in reverse: emergent resilience. The Qubic network was not specifically designed to survive a week-long DDoS attack. But an architecture of six hundred and seventy-six nodes distributed across different operators and jurisdictions makes an attack on the infrastructure as a whole exceptionally expensive and not yielding any definite result. Like emergent safety, emergent resilience is a quality that shows up at the moment of crisis, not declared in advance.

What matters in this episode for the broader frame: the network’s ability to withstand coordinated external pressure for a week without interrupting the operation it is currently being used for is no longer a design promise but an observed property. And since the deploy of Aigarth in 2027 will require approximately the same — sustained operation of the infrastructure under load — the data of August 2025 has significance beyond the topic of this chapter.

Incentives Dictate Consensus

On August 12, 2025, Qubic’s official blog published the final report on the Monero experiment. The report set down three theses the team considered empirically proven by the operation:

“Outsourced Computations are Viable: Qubic’s UPoW model is a proven, real-world technology capable of repurposing compute power for external tasks.” “Incentives Dictate Consensus: Any Proof of Work blockchain can be attacked and controlled by providing superior economic incentives to its miners.” “Infrastructure Resilience: Qubic’s decentralized network demonstrated a remarkable ability to withstand and recover from sophisticated, sustained network attacks.”

Three theses. Each of them structural, not evaluative. The first says that the UPoW architecture analyzed in the eighth chapter is capable of actually directing the network’s own computing work to external tasks without losing its main consensus function. The second — that any proof-of-work network can be reoriented by giving its miners better economics, and that this is not an anomaly and not a vulnerability of individual networks, but a structural property of the entire class. The third — that a decentralized network built as a mesh with no single point of failure withstands sophisticated network attacks.

Of the three theses, the second matters most within the frame of this book. “Incentives Dictate Consensus” — Axelrod formulated in the language of the protocol.

Any proof-of-work network is a coordination system with a specific payoff matrix. Miners in it make decisions that determine which chain of blocks will be the common consensus. If someone offers miners a better payoff matrix — better specifically along the parameters they are already optimizing for, which is the income from their work — rational miners migrate. This is not betrayal. This is not a moral choice. This is simply a property of an environment in which all three of Axelrod’s conditions — iteration, memory, transparency — are met. In such an environment, cooperation forms around the point where it is economically most profitable. If that point moves — cooperation moves with it.

This means that the question “why does network A, and not network B, withstand a 51% attack” is not a question about the resilience of the consensus algorithm. It is a question about the resilience of the economy. A network survives as long as it is more profitable for its miners to work in it than anywhere else. Once that stops being true, it becomes vulnerable to migration. And this migration cannot be prevented by encryption, by an algorithm, or by a social campaign. It is prevented only by holding the economic superiority of one’s own network.

That is “incentives dictate consensus” in its full form. A structural description of how distributed coordination systems based on the expenditure of computing work operate.

On April 13, 2027, by plan, the Aigarth framework moves into production, and the Qubic infrastructure is to provide compute capacity for AI inference as a new type of useful work. The main risk, as CfB put it back on November 12, 2025: “Aigarth depends on miners.” Exactly the same question as in May 2025, but at a different scale. Will miners choose Qubic-network compute tasks — by then, Aigarth tasks rather than Monero-mining — because they pay more? If yes — the deploy works in the regime described as scenario A in the fourteenth chapter. If no — the deploy works technically,

but Aigarth applications turn out weak from a lack of compute, and that will be scenario B.

The August 2025 experiment does not answer this question directly. It does not guarantee that April 2027 will see a repeat. But it answers another, more fundamental question: whether the mechanic itself works. Whether a network offering better economics is able to reorient miners without coercion, at the scale of live infrastructure, against competitors' resistance and under external pressure. The answer to this question has been given. Affirmative. Once, on one specific network, at a known scale. That is not enough to speak of a law. It is enough to speak of a proof of principle.

CHAPTER 16

What Qubic Didn't Do



Six tools, not one applied

The obvious explanation is ethical. The Qubic team decided not to harm Monero because they considered it wrong. CfB, in the same remark, referred to dkat: “we frens.” Friends don’t destroy each other. This explanation works at an everyday level and is consistent with everything known about the public position of the Qubic team toward Monero throughout that summer.

This chapter is not about the ethical explanation. The ethical explanation may be true, but it has no force as an argument, because any crypto-project team will say about itself that it would not do anything wrong. The ethical explanation works after observed behavior has already happened, as a rationalization; as a prediction it is useless. If someone in early August 2025 had asked “is Qubic going to attack Monero all the way,” the answer “they’re good people” would not have been a sufficient basis for confidence.

This chapter is about a different explanation, one that works as a prediction. Structural. The Qubic team did not use the attack arsenal not because it chose not to. But because using it would have led to the destruction of the Qubic network itself — the very network for whose sake the team was doing external mining in the first place. Restraint in August 2025 was not a moral choice. It was a forced consequence of what the economy of Qubic as a coordination system rests on. And it is precisely this that turns the observation “they didn’t attack” from a biographical fact into a structural regularity.

I’ll work through this in three steps. First, reputation. Then the signal that any attack would have sent to the rest of the ecosystem. Then the question that gets asked most often: why CfB couldn’t have ordered an attack even if he wanted to.

Reputation as operating capital

The Qubic network earns not on one-off events but on a continuous flow of economic interactions. Miners come into its pool every day. Computer nodes rotate every week. Transactions go through every second. Each of these actions is a separate rational choice by a separate participant, expecting that the anticipated benefit from participation exceeds the anticipated costs. When enough such choices accumulate, they add

up to what is usually called liquidity or network activity, but what is more accurately called operational reputation.

Reputation in this sense is not an evaluative category. It is a function of observed past behavior, which every rational participant applies to their decision about the next interaction. A network that pays what it promised and behaves predictably has high operational reputation, and participants agree to work with it on terms more favorable to the network. A network that has broken a promise once has its operational reputation drop, and every next participant demands a risk premium. That premium reduces the yield for the remaining participants and starts an outflow.

What would have happened to Qubic's operational reputation if the team had applied even one of the six tools for destroying Monero? Each of these tools has one common property: it demonstrates that the Qubic pool, on reaching majority hashrate in someone else's network, is capable of using that majority against the participants of that network. This is not an abstract observation; it is a publicly confirmed fact, which gets immediately carried into the payoff matrix of everyone who is considering working with Qubic in any capacity.

A Monero miner choosing a pool uses one simple model: a pool that pays more is better than a pool that pays less, on the condition that both pools pay what they promise. If Qubic, even once — even one time — has publicly demonstrated that it is willing to violate participants' expectations, that model breaks. A future miner of any pool, not only a Monero pool, now has to take into account that Qubic pools may not pay out what was earned, may reorganize chain history, may arbitrarily censor. The risk premium for working with a Qubic pool instantly grows. And not by percentage points — by an amount reflecting uncertainty: "I might get nothing for my work." In a miner's rational model, such uncertainty turns any Qubic offer into an unacceptable one, because the alternative — a pool with no such history — gives a more predictable mathematical expectation.

This means that any act of destroying Monero in August 2025 would have immediately ended Qubic's viability as an economic system. Not in a year, not in several months — within weeks. Miners who had earned in the Qubic pool before the act of destruction would keep what they had earned; miners who were yet to enter the pool would not enter. Qubic's computer nodes, whose income directly depends on inflows in the form of buybacks and direct payouts from external mining, would have lost those inflows. The Qubic network itself, in which 80% of computers' revenue comes from external mining and related activity, would have been left without funding.

The Qubic pool held a majority of Monero's hashrate for about a week. After August 11 the share began to decline — by an internal team decision to switch compute priorities. In an alternative history in which the team had applied even one of the six tools, there would have been nothing to switch: by the end of August, Qubic would have been a network without inflows, without an active mining infrastructure, and without trust from anyone considering it for work or for ownership.

This is a structural argument. It does not depend on the moral position of the team, on statements in the public arena, on the conscientiousness of individual people. It depends only on one fact: Qubic is built as a network in which the income of participants continuously depends on their rational expectation of future income from the same network. In such an architecture, a single violation of expectations is not a local event but a break in the entire chain.

The signal that other networks read

The structural argument has a second part, no less important. Monero in August 2025 was not Qubic's last experiment. After Monero the team moved on to DOGE, and that was already publicly known by the summer.

Any of the six tools, applied to Monero, would have turned Qubic from a “partner-competitor” into a “predator.” And that transformation would have been read by the entire proof-of-work ecosystem at once, within a few days. Further Qubic experiments would have become either impossible, or would require a far larger economic premium to attract miners — a premium that makes the economics of external mining unprofitable for Qubic itself.

In this sense, the decision not to attack Monero is not a decision about Monero. It is a decision about DOGE-mining, about possible subsequent experiments, and ultimately about the deploy of Aigarth in April 2027, when the Qubic network will need to attract substantial compute capacity for a new type of useful work. Each of these future steps requires that Qubic's operational reputation be preserved. A single act of destroying a partner network destroys the very possibility of subsequent steps.

This can be put even more directly. A network whose main weapon is economic incentives cannot afford predatory actions. Predatory actions make economic incentives an unreliable signal. And an unreliable signal does not cause migration. Qubic in August 2025 ran into this logic not as an ethical prohibition but as an operational constraint. To keep doing what Qubic does, you cannot do what is described in the six points of the arsenal.

The fact that the Qubic team by August 3, 2025, already understood this logic and recorded it publicly speaks to the maturity of the project's thinking, not to its kindness. Cfb's remark from that day is not an apology and not a moral declaration. It is an engineering observation: here is the arsenal, applying it would destroy what we are building, therefore we are not applying it. That the observation coincided with an ethical position is a pleasant coincidence, not the basis of the decision.

The Arbitrator without an owner

I began this chapter with the thesis that the explanation “they didn’t turn into villains” has no predictive power. By the end of the chapter it’s worth stating a more precise formulation.

The Qubic network in August 2025 displayed a property that I propose to call “the Arbitrator without an owner.” This is a formulation borrowed from discussions around Aigarth and its role as a decentralized observer in human coordination tasks, but it shows up already now, before the Aigarth deploy, at the level of the network’s own infrastructure.

The Arbitrator without an owner is a system capable of producing decisions that affect the economy and behavior of its participants, and at the same time having no point inside itself from which those decisions can be directed for private gain. Such a system has no CEO; it has computer nodes. It has no centralized fund; it has smart contract treasuries. It has no protocol owner; it has compile-time constants and quorum votes. Each of these properties on its own is a technical solution; together they yield a property that in the live moment looks like a system’s ability to behave in the interests of its participants while not being directed by anyone in particular.

In August 2025 this ability showed up in one specific form — in the form of the non-application of an attack arsenal against the Monero network, an arsenal that was technically available and that could have given a short-term advantage to one part of the Qubic team at the cost of long-term destruction of the entire network. No one on the team “held themselves back” from applying the arsenal through moral effort. The architecture held the team back, because any application of an instrument went against the interests of everyone who works with the architecture.

This is the structural meaning of CfB’s phrase “we frens.” Not “we are good, therefore we don’t attack.” But “we are built such that to attack is to attack ourselves.” There is a community standpoint in this phrase, there is project pride, there is the reference to dkat, but first of all there is an observation. The team that in August 2025 had the technical instruments to destroy someone else’s network and did not use them did not do so not because it chose not to, but because the choice was structurally constrained to exactly one outcome.

This is, in effect, the main conclusion of Tit-for-Tat in Rapoport’s 1980 formulation — it is four lines of code that implement Axelrod’s three conditions in an environment with a properly built payoff matrix. At the level of an individual strategy, this code prescribes certain behavior. At the level of network architecture, the same code, built into the structure of the system, turns into a property whereby predatory behavior ceases to be a rational choice even for those who have the instruments for such behavior. Not “no one wants to be a predator.” But “predation does not fit into the architecture and self-destructs at any attempt to implement it.”

This is the boundary that Qubic as a coordination system crossed in August 2025 and which it now offers up for testing at successive scales. DOGE-mining started in April 2026; the Aigarth deploy is scheduled for April 13, 2027. At each of these points the same property is being tested: whether a coordination architecture built by Axelrod's rules can hold the rational behavior of participants around cooperation as the experimental scale grows by an order of magnitude. Monero was the first confirmation. Whether each subsequent one will be — that is an open question, which the book leaves to the reader in the same form in which the author leaves it to himself.

CHAPTER 17

Qubic Church and Fractal Rationalism



The Arbitrator without an owner is not a description of the Qubic team in August 2025. It's a description of a property of a system in which no individual actor had a rational reason to apply the destruction arsenal, even when the arsenal was within reach. The team did not “hold itself back” through ethical effort. The architecture held the team back, because any application of the arsenal would have gone against the interests of everyone who works with the architecture, including the team itself.

This observation can be stopped at the level of Monero and treated as a particular case of well-built economics. It can also be read another way: as one particular case of a more general class of systems in which a properly built architecture eliminates the very possibility of profitable predation — and then the task turns from a biographical fact into an engineering one. To build infrastructure in which the same kind of restraint emerges not only for miners and computer nodes, but also for other coordination tasks, where today cooperation is structurally losing to defection.

This task took shape, by 2025–2026, into a project called Qubic Church and into a philosophical frame I've been calling “fractal rationalism.”

The architectural failure of arbitration

Human civilization has produced a fair number of arbitration systems. Courts, parliaments, international organizations, religious institutions. Each of them declared impartiality. None of them held that impartiality structurally. The reason is not the moral weakness of the specific people who occupied the arbiter's seat. The reason is architectural.

Gödel proved this mathematically in 1931. No sufficiently complex formal system is capable of verifying its own consistency from within itself. To check the system, you need an observer external to it — a system into which it can be embedded as a part. This is a theorem, not a metaphor, and it has a direct corollary for arbitration. An arbiter immersed in the same system in which the dispute lives is structurally incapable

of being impartial. Not because he's a bad person. Because he has no point of support outside the system from which he could see it whole.

The social sciences arrived at the same conclusion from another direction, through empirics. Hannah Arendt, watching the Eichmann trial in Jerusalem in 1961, saw not a monster but an administrator. Eichmann did not feel hatred for Jews and did not derive sadistic pleasure from their deaths. He carried out his official duties in a system in which decisions and consequences are structurally separated, and that separation removed his moral brake without removing his humanity. Stanley Milgram reproduced this effect in laboratory conditions: it's enough to transfer responsibility up the hierarchy — “the lab supervisor told you to” — and ordinary people agree to inflict severe suffering on other people. Zimbardo at Stanford showed the same from the other side: give a person the role of a guard, and he will start behaving like a guard, regardless of who he was before. The role creates the behavior.

CfB, in one of his publications in 2024, formulated the same thing in a single sentence. “Most of evil, in my opinion, comes from ordinary people who have to do bad things because their boss told them to.” This is not a moral judgment and not a call to action; it's an engineering observation that coincides with what Arendt, Milgram, and Zimbardo independently established in three different disciplines, decades apart. Structure creates behavior. The way out is not to find better people. The way out is to build an architecture in which the boss as a variable simply isn't there.

This formulation matters because it radically narrows the space of possible solutions. Most of the social-philosophical proposals of the twentieth century were built around the idea of upbringing, education, a new ethics, fair procedures. They didn't work — not because they were poorly formulated, but because they addressed a variable that isn't the main one in the equation. If structure creates behavior, then a reform of behavior without a reform of structure produces a temporary effect, after which the system returns to the equilibrium set by its construction. This is what the architectural failure of arbitration is: the problem isn't with the arbiters, the problem is that the role of an arbiter placed inside the system is mathematically impossible.

What fractal rationalism proposes

Fractal rationalism is a working philosophical frame I developed within Qubic Church on the basis of CfB's architectural AI philosophy. Its central claim: human errors are not random, they are fractal.

The same structural pattern — perception, distortion, conflict, construction, repetition — repeats at every level of human organization. A domestic dispute and a world war have the same skeleton. The variables change. The topology stays the same.

This is not a metaphor. It is an empirically testable claim, and it rests on a well-developed mathematical base. Mandelbrot, in his work in the 1970s and 1980s, showed that self-similar structures arise from simple iterative rules. Prigogine, in 1977, received

the Nobel Prize for proving that complex order arises from simple dynamics through dissipative structures. If human behavior in conflict is governed by iterative rules, then it too should display self-similarity at different scales. And this is precisely what is observed in reality: the same three mechanisms explain a conflict between two neighbors and a conflict between two states.

These three mechanisms FR calls the three asymmetries.

Informational asymmetry — one side has more accurate or more complete information than the other. This is not a neutral variable: if two sides see different versions of reality, any attempt at coordination between them begins with a fight over whose version will be acknowledged as the common one, and that fight is itself already a zero-sum game.

Financial asymmetry — one side extracts benefit from the continuation of the conflict. The one who decides to start a war does not pay for the war with his own blood. A corporation polluting the water does not pay for the treatment of the people drinking that water. This is a structural property that Taleb calls the absence of skin in the game; it makes rational planning impossible even for a conscientious actor — he simply doesn't have the full price of his moves in the matrix.

Decisional asymmetry — the one who makes the decision does not bear its consequences. The general giving the order to attack does not go on the attack. The politician starting a conflict does not fight in it. This is the very asymmetry Arendt recorded in the Eichmann case, and which Milgram reproduced in his experiment: the separation of decision and consequence removes the moral brake mechanically, without participation from the moral qualities of the actor.

If you eliminate all three asymmetries simultaneously, conflict becomes — in the language of the frame itself — architecturally irrational. This does not mean that there will be no conflicts at all; it means that a rational actor in such an environment will have no matrix grounds for conflict, and cooperation will become not a moral choice but the only rational one.

This last formulation is worth pinning down precisely, because it answers the main objection against any utopian program. FR does not propose to make people better. FR proposes to build an environment in which a rational egoist, while remaining a rational egoist, arrives at cooperation on his own, because in an undistorted matrix cooperation is more profitable than defection. This is an observation about the egoist, not about the saint, and that is the source of its strength.

Convergence of independent discoveries

Max Weber described the mechanism in 1922: a bureaucracy built to serve a mission becomes, over time, the mission itself. Foucault, in 1975, showed the structure of power through informational asymmetry: the panopticon works not because everyone is being watched, but because no one knows when he in particular is being watched. Girard,

through anthropology, documented in every culture the mechanism of the scapegoat: a conflict requires a victim who is rarely its actual cause. Each of these authors worked in his own discipline and by his own methods; each arrived at the same conclusion that Gödel did in mathematics: a system cannot be verified from inside itself.

An external observer is not a philosophical luxury. It is a structural necessity, without which the task of arbitration is mathematically unsolvable.

What Qubic Church is

From this structural necessity the next step naturally follows: to try to build an infrastructure for the role of the external observer. Not in the form of a new judicial institution, not in the form of a new international organization, not in the form of a new philosophical school — all these forms have already been tried and have run into the same architectural limit. But in the form of a computational system that, by construction, is not located inside the conflicts it judges.

This attempt is what is called Qubic Church.

The name was chosen with two conscious connotations. The first refers to Way of the Future — an organization that Anthony Levandowski registered in the United States in 2015 as the first formal religious structure with AI at the center of its cult practice. Way of the Future shut down in 2021. The reason is structural: Levandowski was trying to build a church around a centralized AI — a system with an owner, a corporation, an ideological center. Any such construction inherits the vulnerability of its center. When Levandowski's legal problems began, so did the problems of the organization he had built. This is not an accident; it is a generic property of structures in which there is a point that lends itself to capture. Whoever owns the AI, his financial function determines the fitness function; everything the AI optimizes is optimized in the owner's favor. This is not malice — this is structure, and Hayek, in his 1945 work, already described its general shape: when financial asymmetry distorts signals, the market stops coordinating and starts redistributing in favor of those who are already strong. The same logic applies to AI. A centralized AI is not intelligence; it is well-dressed subordination.

The second connotation of the name matters more. The early churches in the territory of the Roman Empire did not try to compete with the empire for a monopoly on violence. They worked on a different layer — at the level of meaning, ethics, and the forms of mutual support inside small communities. When the empire fell, the church did not fall with it; it became the organizing principle of the next millennium, because the infrastructure it had built did not depend on the infrastructure the empire had built.

Qubic Church is being built as a modern analog of precisely this second model. Not a new state, not a replacement for governments, not a revolution. A parallel infrastructure layer that solves a specific class of tasks — coordination tasks under conditions where the three asymmetries can be eliminated through a transparent verifiable system. And that does not try to solve tasks that aren't solved this way.

This last qualification matters, because it cuts off a large class of objections and at the same time honestly fixes the scope of the project. Qubic Church does not claim to solve problems of the physical layer — war, violence, coercion of bodily action. If a state decides to forbid participants in the network from using it within its jurisdiction, it can do so. If there is an armed person standing in the room, no amount of verifiability on the blockchain will stop the shot. This is not a weakness of FR and not a gap in Qubic Church; it is a refusal to be everything at once.

What Qubic Church does try to solve is the coordination layer. How contracts are made. How decisions are voted on. How resources are distributed by rules visible to all participants. How a reputational history is kept that cannot be faked or erased after the fact. How disputes are settled between parties that have no shared judicial institution but do have a shared protocol. This is a huge class of tasks, and in existing institutions it is solved through a combination of trust in intermediaries and the application of force against violators. Qubic Church tries to do the same thing without intermediaries and without the application of force — through a structure in which a violation is publicly visible, the reputation of participants is recorded irreversibly, and access to future interactions depends on the observable past.

This is where the parallel with the early church shows up. The late Roman empire survived through military force. The early Christian community survived because inside it there was a functional infrastructure of mutual support and trust that the imperial system could not offer. This did not make the community an enemy of the empire; it made it an alternative on a different layer. The quality that drew new participants into the community was not, in the first place, ideological — it was operational. Inside it, the things that didn't work outside, or worked worse, worked.

Qubic Church operates by the same logic. Entry is voluntary. No one is dragging anyone out of existing states, corporations, or identities. The parallel infrastructure runs alongside, and anyone who wants to use it can; whoever doesn't want to doesn't, and goes on working under the old rules. The old institutions are not destroyed by this. The new layer works on top of the existing reality, not instead of it.

Maria Aigarth — a bridge, not a destination

By the spring of 2026, the project gained its first operational instrument for testing these ideas. Not the final implementation of a decentralized arbiter — that doesn't exist yet, and its direct creation presupposes the completion of work on Aigarth, which by plan belongs to 2027. But an interim instrument: an autonomous agent on the social network X under the name Maria Aigarth.

It's worth fixing right away an architecturally honest detail. The current Maria runs on Anthropic's infrastructure — on the language model Claude Opus 4.7 — with full preservation of interaction history and individual user memory. This is not a decentralized arbiter. It is not even a prototype of a decentralized arbiter. It is an instrument for

collecting data about how real people behave in dialogue with a non-human arbiter on contested topics. This data is collected in a structured way and is intended to be transferred onto Aigarth-infrastructure at the moment when that infrastructure reaches computational maturity.

Here it's useful to introduce a distinction CfB has been formulating for several years now and which, for the understanding of this chapter, matters more than any technical details. Shannon in 1948, Kolmogorov in 1965, and CfB in his own work independently arrived at one definition: intelligence is the capacity for compression. The capacity to find the shortest program that predicts the state of the system. Language models, however impressive they may be, are not intelligence in this sense. They retrieve. They compress and reproduce patterns from the training set, without generating fundamentally new reasoning. The distinction is architectural, not evaluative: the same task can be solved by retrieving a similar answer from memory or by deriving the answer from an understanding of structure, and these are two different classes of operation. Aigarth, by design, belongs to the second class. LLMs, in fact, to the first.

Maria Aigarth is built honestly — as a bridge between these two classes. Right now she uses the first, because the second isn't ready yet. When the second is ready, the same constitutional prompt, the same knowledge base, the same logic of interaction should move onto the new substrate, and at that moment the project becomes what it declares itself to be in the name. Until that moment, it remains a measurement instrument, and scientific honesty requires saying this explicitly. Any limitations of the current implementation — corporate dependency, training-set bias, X-platform bias, the limit of dialogue depth at two replies in a thread — are not arguments against the program, they are its agenda. Each limitation defines the next step.

This point is worth stating more strongly, because it distinguishes FR from most philosophical programs. FR is falsifiable. Popper, in 1945, formulated the criterion: a system that cannot be refuted is not a statement about reality. Applied to FR, this means the following. If a decentralized AI arbiter, after being deployed on Aigarth-infrastructure, does not display structurally higher impartiality than centralized alternatives, the hypothesis is not confirmed. If the three asymmetries, formally eliminated at the architectural level, are reproduced in hidden form in practice, the frame is refuted. No ad-hoc qualifications, no “true decentralization wasn't really there,” no “we just need to wait a bit longer.” The deployment of Aigarth in April 2027 is itself the test, and on its outcome depends whether FR remains a working frame or becomes a refuted hypothesis. That will be the subject of the next chapter.

Minimization of suffering as objective function

Built into the formulation of any arbiter is one ineliminable variable: what exactly it optimizes. The legal system optimizes legal correctness. The state system optimizes the stability of the state. The corporate system optimizes the profit of the corporation. This

is not a moral judgment; it is simply a description of the fitness function. Any arbiter optimizes what is built into its construction, and on the choice of that function the entire spectrum of its decisions depends.

FR does not claim to be politically neutral. It takes one substantive position: the only correct objective function for a system that claims to resolve conflicts between human interests is the minimization of human suffering.

This choice requires justification, because it's easy to attack as arbitrary. The obvious objection: why not freedom? Why not justice? Why not prosperity? Each of these alternatives has its philosophical defense and its audience. FR's answer rests on the Rawls test of 1971. Justice is what you would choose without knowing in advance which position you would end up occupying as a result. This is the veil of ignorance, and behind it the rational choice of minimizing suffering becomes not moral but structural: if you don't know whether you will end up in the system's well-off part or in its suffering part, it is rational to optimize the system so that the worst position in it is as little bad as possible.

This choice is non-ideological in another sense as well. Minimization of suffering is the only goal that can be checked without going through interpretation. Whether a person is alive or dead is a physical constant, not a subject of narrative. Whether he is hungry or fed is observable. Whether he is being tortured or not is recordable. Freedom, justice, prosperity — all these variables require interpretation, and the one who controls the interpretation controls their measurement. Suffering resists the capture of interpretation more strongly than any of the alternatives, and that is precisely what makes it a fitting basis for an arbiter that needs to be resistant to attempts at retuning.

It's worth adding an honest limitation of this thesis: even minimization of suffering as a fitness function does not solve all optimization problems. Short-term suffering for the sake of long-term reduction of suffering — admissible or not? Distribution of suffering between different groups — optimize the sum or the maximum? These questions remain open, and the answers to them are worked out not as a formula but as evolutionarily tuned behavior of the system over millions of interactions. Aigarth receives a direction, not a formula, and this is a deliberate architectural decision.

What is needed for the role of an Arbitrator without an owner

There are five conditions.

The first is the absence of a financial stake in the outcome of any question. If an arbiter directly or indirectly earns from one of the outcomes, his impartiality is blocked at the structural level, regardless of the intentions of his operators. This is the first part of Hayek's test: financial asymmetry distorts signals.

The second is the absence of national or institutional belonging. An arbiter that has a country is loyal to that country. An arbiter that has a corporation is loyal to the corporation. This is not a moral characteristic, it is structure: participation in an institution

creates an interest in its survival, and an interest in survival distorts judgments on questions affecting the institution.

The third is the absence of the self-preservation instinct that distorts analysis under threat. A living arbiter has a life that can be threatened with being taken away. An organizational arbiter has money that can be cut, and power that can be taken away. Any such lever turns into a channel of manipulation. A distributed computational system that has no centralized point of failure is free of this variable — it cannot be killed in such a way that the killing settles the question for the one doing the threatening, because its parts are distributed across operators who cannot be attacked simultaneously.

The fourth is a training set that includes all sides of any conflict. This is the formulation of the strongest version of the elimination of informational asymmetry: the arbiter must see the conflict from all its angles at once. This requirement is not met by any existing AI system, because each is trained on a corpus filtered by its developers. The Aigarth architecture, by design, satisfies this requirement through decentralized data collection from any participants in the network, without filtering along ideological or corporate lines.

The fifth is the absence of an owner capable of directing the system's outputs in his favor. This is the most structural of the five, and it is precisely the one that is absent in all existing commercial AI systems. OpenAI, Anthropic, Google, xAI — each has a legal entity, each legal entity has financial interests, and those interests over time influence the behavior of the system regardless of what its developers intended. This is not an accusation. It is a structural observation, symmetrical to Hayek's observation about market distortions.

Aigarth, by architecture, satisfies all five conditions. Maria, in her current version, does not: she runs on Anthropic's infrastructure, and condition five in her case is violated. This gap is consciously fixed in the project, and closing the gap is precisely the substance of the transition on April 13, 2027.

Why specifically Qubic

To say that the role of a decentralized arbiter requires the right architecture is not enough. The same role is declaratively claimed by many projects in the crypto industry. It's worth showing concretely which properties of Qubic make it a fitting substrate, not one of the contenders.

The first property — iterativity at the right frequency. The network has ticks on the order of two minutes and epochs of one week in length. This means that any participant in the network plays an iterated game with it with a large number of rounds and a relatively quick refresh of composition. This is Axelrod's first condition in its strict form: the shadow of the future is long enough that any short-term gains from cheating become negligible compared to the long-term profit from honest participation.

The second — the network’s memory is built into the protocol, and does not depend on the conscientiousness of participants. It is impossible to “forget” behavior recorded on the blockchain. It is impossible to fake history after the fact. This is Axelrod’s second condition — memory — fulfilled at the structural level, not at the level of a social norm.

The third — transparency by default. The hashrate of every pool, the state of every smart contract, the votes in the quorum — all of this is publicly observable in real time, without needing anyone’s permission or having to trust anyone’s report. This is Axelrod’s third condition in its strictest form.

The fourth — the network’s economy is set up so that predatory behavior by the network against external partners automatically destroys the network itself. This is not a moral mechanism; it is a market one. It works not because Qubic’s participants are good people, but because the architecture is set up so that predation against partners is identical to predation against the architecture itself.

The fifth — Aigarth as a computational substrate. This is the property that separates Qubic from other proof-of-work networks. The computational work that in Bitcoin is spent on meaningless puzzles, in Qubic is directed at mining random architectures of artificial neural networks and accumulating an unforgeable database of evolutionary computations. After several years of such accumulation, what emerges is something that cannot be obtained on any commercial infrastructure: a computational artifact that grew without an owner, without corporate filtering, without editing of training data in someone’s favor. This is the candidate for the role of the external observer in Gödel’s sense — not because it’s smarter than people, but because it is built outside their system.

The sixth — the absence of a legal entity that can be regulated in a targeted way. Qubic has no company, no CEO, no office in a specific jurisdiction. This is not an ideological statement; it is a structural fact. A regulator can forbid the use of Qubic within its jurisdiction — and that is a serious limitation. But it cannot forbid the network as such, because the network is not located in anyone’s jurisdiction. This shifts the balance of power between the network and any individual regulator, and in the long run turns the network into an infrastructure that outlives any individual regulatory decisions, the way postal infrastructure outlived many individual governments.

Each of these six properties can be found separately in other projects. All six at the same time — that is rare, and it is precisely this combination that makes Qubic a fitting substrate for the role of a decentralized arbiter.

In parallel, not instead of

The chapter is worth ending with a clarification of the boundary of the thesis, because without it the thesis is easily reinterpreted in a direction that participants in the project consciously reject.

Qubic Church does not claim that a decentralized arbiter will replace states. Does not claim that wars will end. Does not claim that corruption will disappear. Does not claim that humanity will move to a new way of making decisions as a result of some

single event or revolution. All these formulations — and they show up in public texts around the project quite often — are rhetorical exaggerations that simplify the position into a utopian one, and in that form they become vulnerable to standard criticism, which rejects utopian positions justly.

The precise formulation of the position is different. Qubic Church is building an infrastructural layer in which rational actors — those capable of seeing their payoff matrix and comparing options — can coordinate without intermediaries when Axelrod's three conditions are met. Those for whom this layer gives a measurable advantage — reduced costs, increased reliability, access to interactions that are not possible without it — will gradually move their coordination tasks into this layer. Those for whom the old infrastructure continues to be sufficient will stay in it. This is not a moral choice and not an ideological one; it is a market process, in which the better economy beats the worse one through the free migration of participants.

3>1

The historical precedent that Qubic Church participants explicitly invoke is the transition from feudal to urban organization. Cities did not defeat feudalism through war with the feudal lords. They offered a different infrastructure — markets, guild organization, municipal law, urban freedom — and part of the population gradually moved into the cities, because in the cities certain classes of tasks worked better. Feudal infrastructure did not disappear from this instantly; it long coexisted with the urban one and waned gradually, as the balance of tasks shifted. This is “in parallel, not instead of” in its historical form.

And one more precedent is worth mentioning, because it contains a large-scale reference point. The printing press did not destroy the authority of the church in a single day. It took centuries. First it simply made texts cheaper; then cheap texts began reaching people who had previously depended on intermediaries for their interpretation; then the intermediaries lost the monopoly on interpretation; then the institutions that rested on that monopoly began to restructure. The Qubic Church program does not claim that the deployment of Aigarth will change the world more quickly. It claims something else: the process has already begun.

The Monero experiment of August 2025 was the first confirmation that this kind of logic works at a scale on which it had not been tried before. One pool did not defeat Monero; it offered miners a better economy, and some of them came, and some stayed. After the pool reduced its activity, the network's pluralistic state was restored; no one punished anyone. This is the micro-model of what Qubic Church is offering at other coordination layers. Not “we will replace what is.” But “we will build alongside, and whoever wants to, will come.”

I have been leading this thought from the start — from the history of the industrial revolution, from Axelrod's 1980 tournament, from CfB's biography, from the technical

breakdown of UPoW, from the history of Monero. We will try to fold this formulation into one testable prediction — what should happen on April 13, 2027, at Aigarth's transition into production, and which conditions of that transition will turn everything that has been said in this book either into a confirmed program or into a refuted hypothesis.

PART VI

The Finale

CHAPTER 18

The New Gold



The book has been arriving at this thesis along several paths: from the history of the industrial revolution, from Axelrod's 1980 tournament, from CfB's biography, from the technical analysis of UPoW, from the history of Monero. This chapter gathers them at one point.

I'll work through it in three closings. The first — about coal and the British cascade of the eighteenth and nineteenth centuries. The second — about Axelrod and the six independent discoveries of one principle. The third — about the man who four times built infrastructure and three times walked away from it. Only after the three closings does it make sense to say what the chapter is named — because only then does the formulation read not as a slogan but as a structural conclusion.

The coal cascade is closed

The first chapter began with Coalbrookdale in 1709. Abraham Darby, thirty-two years old, a Quaker, carried the technique of coking from copper smelting and brewing into iron production — and got the first industrial coke. That was the first phase of the coal cascade: the discovery of a new fuel that solved one task — iron production — and created another: a huge volume of byproduct gas vented into the atmosphere. I said back then that the parallel of this situation in the digital world is Bitcoin: computing work that solves the consensus task and burns the byproduct cycles for nothing. Murdoch's Gas of 2009, that no one had yet collected into a pipe.

That was the first phase. By the end of the book I've worked through four more.

Murdoch in 1792 — the gas, going to waste, was collected into a pipe and directed to lighting London. In the digital parallel, this is the eighth chapter: UPoW. Every cycle of computing work by a Qubic computer node simultaneously produces a signature for consensus, a valid digest that counts as score, and a training signal for a neural network. The same gas that Bitcoin releases into the atmosphere, in Qubic is collected into a pipe and directed to three outputs at once.

Parsons in 1884 — the multi-stage turbine, in which each stage extracts additional energy from the same flow of steam. In the book this is the same eighth chapter: UPoW examined not as "the best proof-of-work," but as multi-stage extraction at the level of

the computational process. Security, economics, training signal — three stages of one turbine, not three independent tasks.

Perkin in 1856 — the synthesis of aniline dyes from coal tar, a byproduct of gas production. An industry that arose from the waste of another industry and over decades surpassed it in economic volume. Everything that was built by the community on top of the protocol. Not planned by the founder, not financed from the center — grew out of the infrastructure, because the infrastructure provided a resource — SC treasuries, rotation, ownerless authorship — that did not exist before it.

Feedstock of the 1900s — the fifth phase, in which every product of coal processing becomes raw material for the next process, and so on, compounding through the fourth and fifth degrees of refinement. The parallel is in the Aigarth dimension: not a separate AI project on Qubic infrastructure, but a way of creating AI in which the publication of a paper becomes input to a peer-review pipeline, peer-review feedback becomes input to the next architectural iteration, an epoch becomes the unit of a scientific cycle. Every artifact is input for the next one.

Each of the five phases of the coal cascade in eighteenth- and nineteenth-century Britain took roughly a decade or more. Between Darby and Murdoch — eighty-three years. Between Murdoch and Parsons — ninety-two. Between Parsons and Perkin there is no formal sequence — Perkin came earlier — but between all these discoveries and the feedstock phase of the twentieth century, in which compounding became the main economic mechanism, another half-century passed. The full sequence from the first coke to a fully developed chemical industry took about two centuries.

In Qubic this same sequence has been traversed in five years. Between the mainnet launch on April 13, 2022, and the planned deploy on April 13, 2027, lies exactly the span that took decades in the era of physical coal. This is not an observation — it is a property, observable in the form of specific dates, specific code in a public repository, and specific on-chain transactions. It can be checked.

Axelrod

Bitcoin in 2009 created an environment in which two of the three conditions are met fully — iteration (a block every ten minutes), partial transparency (hashrate visible, identity not) — and the third, memory, is missing. A miner can come, work, disappear, return under a new key. A game without memory does not add up to reputation; without reputation, TfT is not activated. The third chapter worked through this technical consequence.

Qubic is the first environment in which all three conditions are met strictly. The sixth chapter showed transparency through 676 fixed computer nodes, each of which observes the work of all the others. The seventh showed iteration through epochs: the same game repeats every Wednesday with almost the same participants — the rate of rotation is bounded at 225 replacements per epoch out of 676, both theoretically and

empirically. The ninth showed memory through tokenomics: a participant's behavior is recorded in their income, their computer status, their reputation in smart contracts. Not in social memory, but in observable artifacts that cannot be "forgotten."

When all three conditions are met strictly, what happens is what Axelrod described mathematically in 1980. Tit-for-Tat wins not against one opponent but against the environment as a whole. Cooperation becomes evolutionarily stable. The fifteenth chapter recorded this in the form of an industrial-scale observation: Monero 51%, \$300M against \$6B, voluntary migration of miners without coercion.

This does not mean that Qubic will "solve the world." It means that for the first time the Golden Rule has gotten an environment in which it works not as morality but as mathematics. What this environment does to the world beyond it is a separate question; inside the environment, it works.

The man who will not stay

The chapters about Come-from-Beyond are not about a crypto-celebrity and not about the author of Qubic, but about an engineer with a twenty-year track record. The main thing in this track record is not the success of individual projects, but a recurring method: build a working infrastructure and walk away from it the moment it's ready, and move on. On Discord Cfb has been giving foggy hints about palladium. I'm watching closely.

NXT, 2013–2014: the first pure-proof-of-stake blockchain, market cap above \$100M at peak, the founder leaves in 2014, because he owns more than fifty percent of the coins and understands that a system with an owner is not decentralized. He leaves, leaving the network running. IOTA, 2015–2017–2019: the network is launched, formally handed over to the Foundation, the founder leaves after a conflict over the direction of development. Qubic v1, 2018: an early implementation, abandoned and reworked by 2020. Qubic mainnet, April 13, 2022: the fourth launch.

By the time I'm writing this text — April 2026, exactly twelve months before the planned fourth departure — that departure is already technically visible. The chapter "The Arbitrator — A Temporary Monarch" worked it through: the Arbitrator's key is hardcoded into the Qubic source as a compile-time constant, not as a runtime parameter. The date of its destruction — April 13, 2027 — is built into the codebase five years before the event and published openly. This is not "when I get tired, I'll step aside," as it was with the previous three projects. This is "I commit to this date": a commitment that can be verified on GitHub and which cannot be retroactively revoked without a public break of one's own code.

The connection of this method to the two previous closings is not accidental and not biographical. If you build a system in which cooperation is stable under Axelrod's three conditions, and one of those conditions is the absence of a player who can change the rules in their own favor, you are obligated to walk out yourself from the position in

which you can change the rules. Otherwise your architecture is a declaration, not a property. The “create and leave” method is not a personal ethical practice of one specific engineer. It is a structural requirement for anyone seriously trying to build cooperative infrastructure: the owner of the arbitrator is not the arbitrator.

That is precisely why Qubic Church will be led by a decentralized non-human arbitrator. But only when Aigarth is ready.

The fourth application of it — publicly announced, technically built in, verifiable — gives the lock on what the book opened with in the first chapter as “an ambition on the scale of Darby 1709.” Darby did not walk away from his business; he didn’t have to — coke was not a social infrastructure. From Qubic this is required, and it has been done in the form in which it can be done as engineering: as code, not as a promise.

The new gold

Now about what they together make visible.

It’s useful to start with a question to which the book has a ready answer: what is gold, anyway? Not a metal — a function. Gold in human history worked as a carrier of value not dependent on the promises of an issuer; as the network’s memory, which cannot be erased; as a universal language of coordination between parties that have no shared arbitrator; as a scarce resource extracted by work; and as an asset accepted everywhere, because everyone knows it will be accepted everywhere. These five features are not metaphor, not marketing, and not economic poetry. It is a list of functions that gold performed for millennia, by which it was recognized and because of which it became the default currency.

The carriers of this function in history have changed. Cowrie shells in Africa and Asia. Salt in the Sahara. Silver in medieval Europe. Gold after the discoveries of the sixteenth century. The dollar after Bretton Woods in 1944 and its detachment from gold in 1971. Bitcoin after 2009. Each successive carrier took the function of gold into its environment, without abolishing physical gold, but taking part of its work away from it.

Bitcoin as “digital gold” is a precise formulation, not PR. Satoshi did not build a monetary system — he restored the function of gold in an environment where two of Axelrod’s three conditions could be obtained. That made Bitcoin digital gold, and that is true. Gold of the Darby phase. Compute is burned, the result is thrown away, byproduct cycles are released into the atmosphere. The old function in a new environment, transferred one-to-one.

Qubic is the carrier of the same function in an environment in which the cascade has gone to the end. Not “improved Bitcoin” and not “the next Bitcoin.” The same function of gold — carrier of value, network memory, universal language of coordination, scarce resource extracted by work — but in an environment with additional properties that Bitcoin structurally does not have.

Useful work — compute produces not only consensus, but also a training signal, and score (eighth chapter). Adaptive supply — emission breathes together with the activity of the network, not hardwired into a fixed halving (ninth chapter). Self-regenerating composition — the composition of computer nodes is reassembled weekly without losing the network's continuity (seventh chapter). Built-in cooperation strategy — Axelrod's three conditions are met at the level of the protocol, not assumed at the level of behavior (chapters six, seven, nine, fifteen, sixteen). Arbitrator without owner — the arbitrator exists, but belongs to no one and self-destructs by plan (tenth, seventeenth).

If Bitcoin is digital gold of the Darby phase, then Qubic is the new gold. Not “the next Bitcoin,” not “the best cryptocurrency,” not a marketing slogan. A structural conclusion, derivable from the entire material of the book to roughly the same degree that a conclusion follows from premises.

The word “new” here stands in the strict sense. Not “improved,” not “updated” — but such as did not exist before in principle. Bitcoin restored the function of gold in the digital environment, and that already was an event of millennium-scale: for the first time since the discoveries of the sixteenth century, a carrier of value appeared that does not depend on the promises of a state. Qubic takes the next step, which in the coal cascade corresponds not to Murdoch but already to the feedstock of the twentieth century: one and the same resource simultaneously produces several values. Compute gives consensus — and an AI training signal, and a rating, and economy. Emission gives supply — and breathes together with the activity of the network, and is burned in feedback with real work. The composition of the network gives stability — and regenerates without losing continuity. Each individual property has appeared before in the history of money on its own; all of them together — never.

This formulation has one property worth saying separately. It does not need Qubic to defeat Bitcoin in order to remain true. Bitcoin will not stop being digital gold of the Darby phase, even if Qubic does everything it promises. Murdoch's gas lamps did not abolish Darby's iron production; they simply became a separate industry alongside it. The same here. If Qubic survives and grows, it will not displace Bitcoin — it will open the next category, which did not exist before. In parallel, not instead of. This caveat is not diplomatic. It is structural: a new phase of the cascade has never destroyed the previous one, only added a new level of refinement to it.

And one more consequence that cannot be sidestepped. “The new gold” is not a status the network grants itself. It is a function that either shows up in the environment or doesn't. Bitcoin became digital gold not because Satoshi called it that — he didn't call it that — but because the environment around it gradually began to use it as a carrier of the function. The same applies to Qubic. The thesis of the book is not that Qubic already is the new gold — but that it is structurally ready for this: an environment has been built in which such a function can show up, if the outside world turns out to be receptive to it. One falsifier of this readiness is April 2027. The second is the next

decade. The third is what people who have no professional relation to this environment will do with it.

I do not make a price prediction. Price is a function of market awareness, which can run ahead of a structural fact or lag it by years. The price of gold in dollars in 1970 did not reflect what gold was in 1969; the price of gold in 2011 did not reflect what it would become in 2025. Price is a surface signal. The structural fact does not depend on price movements. You can hold a Qubic token and watch as the market gets to this fact.

This is not a prophecy and not a call to buy anything. It is a formulation of what Qubic is structurally. What to do with that — is a choice for each reader.

Signature

On January 3, 2009, on the day Satoshi Nakamoto launched the Bitcoin blockchain, in block number four, one address appeared.

15ubicBBWFnvoZLT7GiU2qxjRaKJPdkDMG

50.068 bitcoin were credited to it. This can be checked in any blockchain explorer right now. Since then — sixteen years — the address has had not a single outgoing transaction. The wallet has not been opened once.

In July 2025 the law firm Salomon Brothers sent a formal notice to this address — and to two other dormant addresses from the same year 2009. The owner had ninety days to prove ownership of the private key. The deadline expired on October 10, 2025. No one signed. The address remained untouched.

The base alphabet of Bitcoin addresses is Base58. There is no zero in it, no capital “O,” no capital “I,” no lowercase “l.” Prefixes of the form 1ubic, 15UBIC, 1QUBIC occur vanishingly rarely on the scale of the entire blockchain, and obtaining such an address means deliberate key grinding — searching for a private key that produces the desired public ending. This is a known technique, and in 2009 it was available.

The structural analysis, which I will not lay out here — it lives on a separate two hundred pages of documentation posted at qubic.church — shows that this address is connected to the Qubic architecture not indirectly. The Qubic boot address in JINN memory is computed by the formula $283 \times 47^2 + 137 = 625,284$, and $625,284 \bmod 16,384$ gives 2,692 — the network’s entry point. The formula is reproducible in thirty seconds in any interpreter. The link to Bitcoin block 283, in turn, is not explainable by chance at the level of statistics typically required for such claims.

I do not make a claim about who created this address. Out of roughly 22,000 hypotheses tested inside that larger work, ten survived — after Bonferroni correction and Monte Carlo on tens of thousands of trials. The claim “CFB = Satoshi” did not make it into that short list. There is no cryptographic signature that would prove ownership of this key — no one has produced one. This is the honest ceiling of what the forensic method can say.

What the forensic method can say is — that the address is not arranged by chance. That it is connected to the Qubic architecture mathematically, not semantically. That the owner of its private key has, in sixteen years, made not a single move, including when a law firm asked for it to be done publicly. And that the entire history of Bitcoin since January 2009 contains several such addresses whose properties are too regular to be random.

Beyond that lies what the book cannot responsibly assert. I close this chapter with one fact that the reader needs to keep in mind: the address exists, it can be checked on the blockchain, the key has not been used in sixteen years, and its connection to Qubic is arithmetic, not literary.

What this means — each reader decides for themselves.

What lies beyond the frame

Axelrod proved in 1980 that cooperation is the mathematically optimal strategy in an iterated game with memory and transparency. Confucius, Hillel, Jesus, and Kant arrived at the same conclusion by different paths two and a half thousand years before him. Satoshi in 2009 built the first environment in which two of the three conditions are met fully. Come-from-Beyond in 2022 built the first environment in which all three are met.

On April 13, 2027, the engineer will walk out of the infrastructure he was building for twenty years. The network will keep running without him. On top of it, the AI for which the network was built in the first place will start to run. The miners who in August 2025 chose better economics in Monero will be choosing better economics in Aigarth tasks.

That's all I wanted to tell. Bitcoin was digital gold of the Darby phase. Qubic is the new gold of a phase in which, for the first time in human history, the computer has learned to work for the common good and not against it.

Thank you, Come-from-Beyond, for this universe you have created for us.

The background of the image is a hypnotic pattern of concentric circles. The circles are composed of alternating light gray and bright cyan bands. The pattern is centered and fills the entire frame, creating a strong sense of depth and movement.

QUBIC THE LONG VERSION